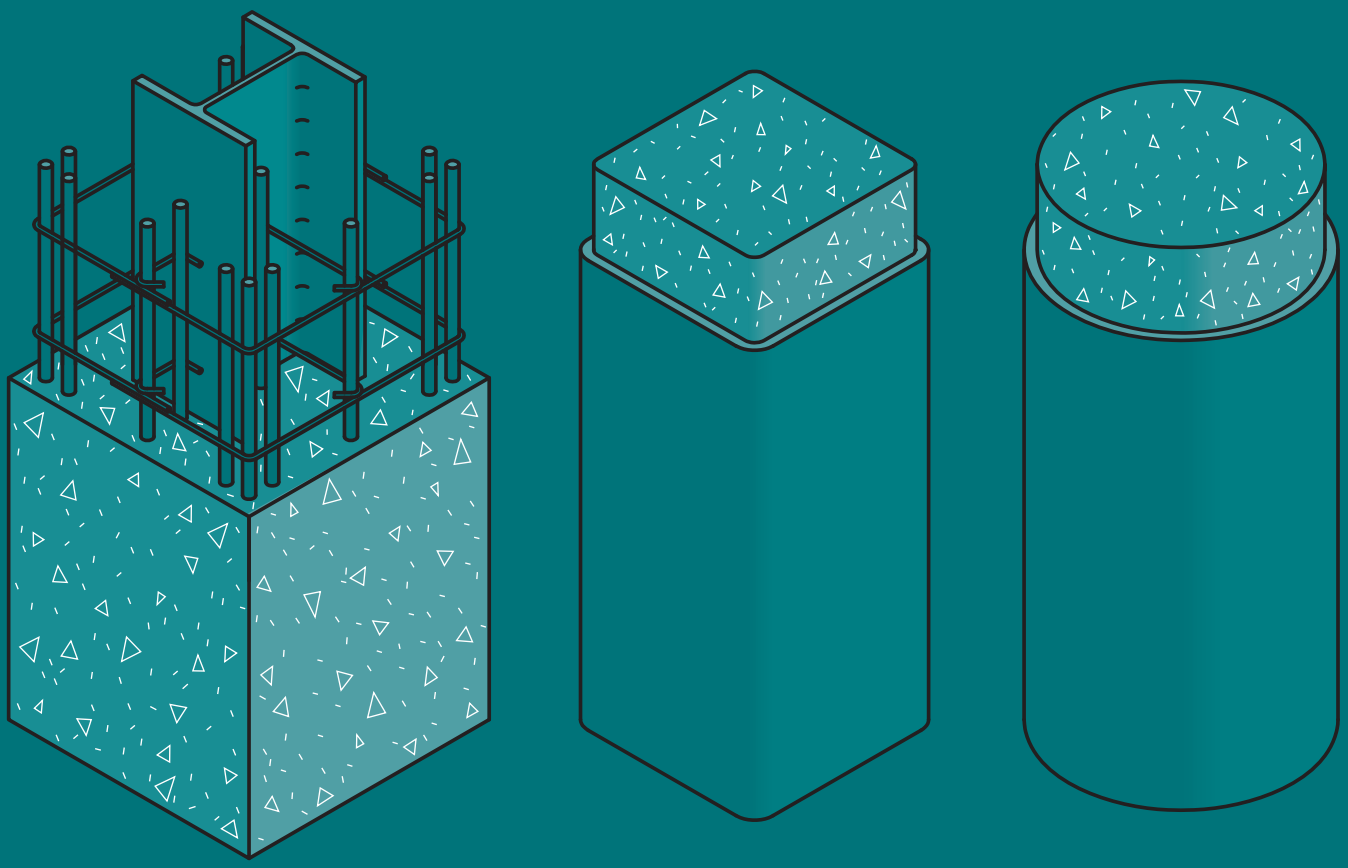


Design Guide 6

Composite Column Design

Second Edition

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Second Edition

Matthew S. Trammell, SE, PE
Mark D. Denavit, PE, PhD
Tiziano Perea, PhD
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Preface

The first edition of AISC Design Guide 6 was written by pioneering engineer Lawrence G. Griffis of Walter P. Moore and Associates, Inc., and focused on encased composite columns. Its purpose was to provide supplemental reference information, beyond what was available in the first edition of the AISC *Manual of Steel Construction for Load and Resistance Factor Design*, to practicing engineers engaged in composite frame design. It contained numerous tables of available axial and flexural strengths for composite columns with different wide-flange steel shapes, reinforcing ratios, material properties, and effective lengths.

Since the publication of the first edition of this Design Guide in 1992, the provisions for design of composite columns in the AISC *Specification* have evolved significantly. The purpose of this edition of Design Guide 6 is to provide updated and expanded guidance on the design of encased and filled steel-concrete composite members. Additionally, it introduces a more modern spreadsheet-based tool to assist with calculating available axial, flexural, and shear strengths of composite members.

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Chapter 1
Introduction

Steel-concrete composite columns are stiff, strong, and ductile structural members that can be an effective alternative to more traditional options in all steel or concrete framing systems. The behavior and design of composite columns share commonality with that of structural steel and reinforced concrete columns, while also featuring unique advantages and challenges. Provisions for steel-concrete composite columns have developed significantly over the past several decades and continue to evolve as research and use in practice worldwide advance the state of knowledge. The intent of this Design Guide is to help engineers navigate and interpret the most recent U.S. provisions included in the AISC *Specification for Structural Steel Buildings* (AISC, 2022e), hereafter referred to as the AISC *Specification*, to design efficient composite columns.

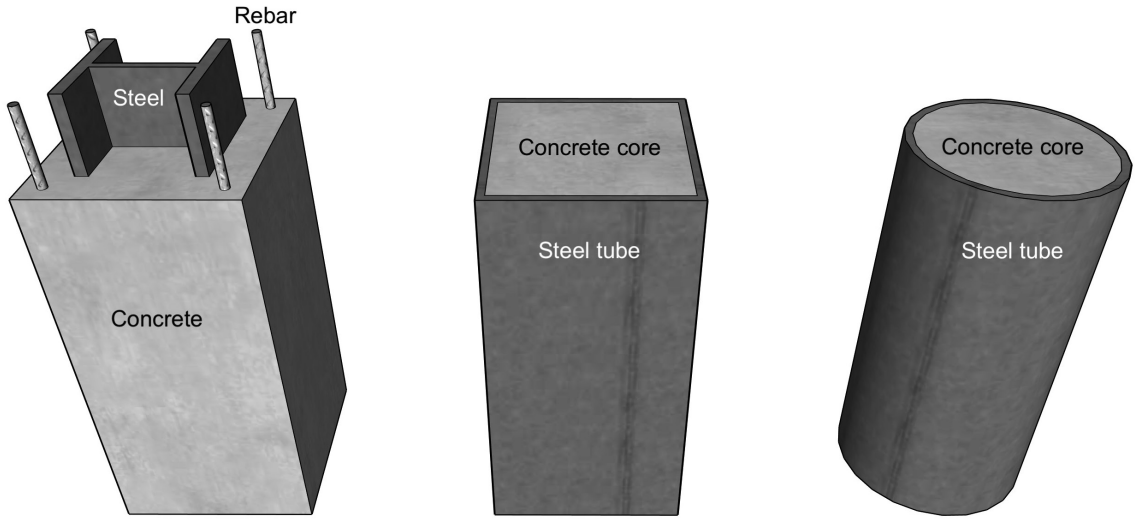
The use of the term "composite column" throughout this Design Guide is intended to indicate a scope limited to encased and filled composite members, to the exclusion of other composite members such as composite beams and composite walls. The use of the term "column" is not meant to indicate that this Design Guide is limited to members that are only subjected to axial compression. This Design Guide covers composite columns subjected to a variety of actions primarily within the context of nonseismic design. Seismic design of composite columns is not addressed comprehensively in this Design Guide as the distinct analysis, design, and detailing requirements are outside its scope. However, composite columns have been shown to be particularly ductile and robust under both seismic and blast actions (Fujikura et al., 2008; Denavit et al., 2011), and thus, engineers are encouraged to explore the application of composite columns to those design situations.

Composite columns can be constructed in many forms. The primary focus of this Design Guide is on the more traditional forms of composite columns, as shown in Figure 1-1, comprised of either:

- A single I-shaped steel section encased by a rectangular concrete section with longitudinal and transverse reinforcement.
- A filled composite member comprised of a rectangular, square, or round hollow structural section (HSS) or box column filled with concrete either with or without internal longitudinal and transverse reinforcement.

The design and detailing principles described herein can be applied, with due professional care, to other composite column cross sections.

The information contained in this Design Guide is based on the 2022 AISC *Specification* and the 2019 ACI *Building Code Requirements for Structural Concrete* (ACI, 2019),



(a) Rectangular encased composite member

(b) Rectangular filled composite member

(c) Round filled composite member

Fig. 1-1. Types of composite columns considered in this Design Guide.

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hereafter referred to as ACI 318. Provisions for the design of composite columns had been included in ACI 318 since its earliest edition; however, the provisions specific to composite columns had not been updated since the 1956 edition and were eliminated in the 2019 edition. Nonetheless, the AISC *Specification* references ACI 318 for design, detailing, and material properties related to the concrete and reinforcing steel portions of composite construction.

Chapter 2 of this Design Guide provides a summary of the pertinent provisions in the AISC *Specification* for composite column members and structural systems with composite column members. Design examples are interspersed throughout Chapter 2 to illustrate how the provisions are applied. Chapter 3 discusses connections to composite column members. Chapter 4 discusses practical design considerations. Chapter 5 provides a brief look at the potential future of composite construction. Finally, details of a composite column program implemented within a Microsoft Excel spreadsheet are presented in Appendix A. The program is included in lieu of the extensive design tables provided in the first edition of this Design Guide (Griffis, 1992b).

1.1 STEEL-CONCRETE COMPOSITE CONSTRUCTION

Structural members comprised of steel shapes in combination with plain or reinforced concrete have been utilized by engineers for many years. Early structures simply took advantage of the protection that the concrete provided to the steel shapes for resistance to fire and corrosion. Research on the strength of such members was conducted in the early 1900s (Burr, 1912; Talbot and Lord, 1912) and design provisions for composite columns were included in the first ACI *Building Code* (ACI, 1910; Furlong, 2012a; 2012b). However, it was not until the development and popularity of modern composite frame construction in the 1960s that composite columns became a common and viable structural member type (Viest et al., 1997).

Fazlur Khan, in his early discussions of structural systems for tall buildings, first proposed the concept of a composite frame system utilizing composite mega-columns as part of the overall wind and earthquake resisting systems for very tall buildings (Roeder, 1998; Ali and Moon, 2018). The term "composite frame structure" describes a building employing concrete-encased or -filled steel columns and a composite floor system (structural steel and concrete-topped steel deck). Composite frame construction has been adopted for many high-rise buildings all over the world. As of 2025, the Council on Tall Buildings and Urban Habitat lists 17 of the world's 20 tallest buildings as being steel-concrete composite construction (CTBUH, 2025).

In composite frame systems, the bare steel columns resist the initial gravity, construction, and lateral loads until the concrete is cast around them to form composite columns capable of resisting the total gravity and lateral loads of the completed structure. The combination of structural steel and reinforced concrete produces a structure having the advantages of both materials. Composite frames have the advantage of speed of construction by allowing a vertical spread of the construction activity so that numerous trades can engage simultaneously in the construction of the building. An example of this sequence of construction is shown schematically in Figure 1-2. Inherent stiffness is obtained with the reinforced concrete to control the building drift due to lateral loads and reduce the perception of motion. The greater strength and lighter weight obtained from the use of structural steel results in savings in foundation costs, while the presence of concrete aids in fire resistance, stiffness, and damping.

Traditionally, in steel-framed buildings, stability and resistance to lateral loads are developed as the structure is erected. Welded or bolted moment connections are made, or braces are connected between columns in a steel building immediately behind the erection of the steel frame to provide stability and resistance to lateral loads. Shear walls, or the monolithic casting of beams and columns, provide stability and resistance to lateral loads soon after the concrete has cured for reinforced concrete buildings. However, for composite frame structures, the full stability and resistance to design lateral loads is not achieved until the concrete around the steel frame has cured, which typically occurs anywhere from a minimum of 6 to as many as 18 floors behind the erection of the bare steel frame. Thus, temporary lateral bracing of the portion of the frame with uncured concrete will typically be required.

1.2 PRACTICAL USE OF COMPOSITE COLUMNS

Practical applications for the use of composite columns can be found predominately in high-rise structures with some uses in low-rise structures. In low-rise structures such as a covered playground area, a warehouse, a transit terminal building, or a canopy, it may be necessary or desirable to encase a steel column with concrete for aesthetic or practical reasons. For example, architectural appearance, resistance to corrosion, or protection against vehicular impact may be important. In such structures, it may be structurally advantageous to utilize the concrete encasement of the rolled steel shape that supports the steel roof structure by designing the member as a composite column resisting both gravity and lateral loads. Filled composite columns can be advantageous in industrial structures with large loads or for building structures where high stiffness is required.

In high-rise structures, composite columns are frequently used in the perimeter of "tube" buildings, where the closely spaced columns work in conjunction with deep spandrel beams (either steel or concrete) to resist the lateral loads. In

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some recent high-rise buildings, composite mega-columns placed at or near the corners of the building have been utilized as part of the lateral frame to maximize the resisting moment provided by the building's dead load. Composite shear walls with encased steel columns to carry the floor loads have also been utilized in the central core of high-rise buildings. Frequently, in high-rise structures where floor space is a valuable and income-producing commodity, the large area taken up by a concrete column can be reduced by the use of a heavy encased rolled shape to help resist the extreme loads encountered in tall building design. Sometimes—particularly at the bottom floors of a high-rise structure where large open lobbies or atriums are planned—a heavy encased rolled shape as part of a composite column is a necessity because of the large load and unbraced length. A heavy rolled shape in a composite column is often utilized where the column size is restricted architecturally and where reinforcing steel percentages would otherwise exceed the maximum code allowed values.

In general, practical use will exploit one or more of the advantages of composite columns, which include:

- Smaller cross section than required for a conventional reinforced concrete column.
- Ductility and toughness available for use in areas of intermediate and high seismic risk.
- Speed of construction when used as part of a composite frame.
- Fire resistance when compared to steel columns.
- Higher rigidity, especially when part of a lateral force-resisting system.
- Higher damping characteristics for motion perception in tall buildings when part of a lateral load-resisting system.
- Stiffening effect for resistance against buckling of the steel shape.

There are also some disadvantages and limitations. In high-rise composite frame construction, as with reinforced concrete high-rise design, engineers sometimes have difficulty in controlling the rate and magnitude of column shortening of the composite column with respect to adjacent steel columns or shear walls. These problems are exacerbated by the wide variation in construction staging often experienced in the zone between the point where the steel erection columns are first erected and the point where concrete is placed around the steel to form the composite column.

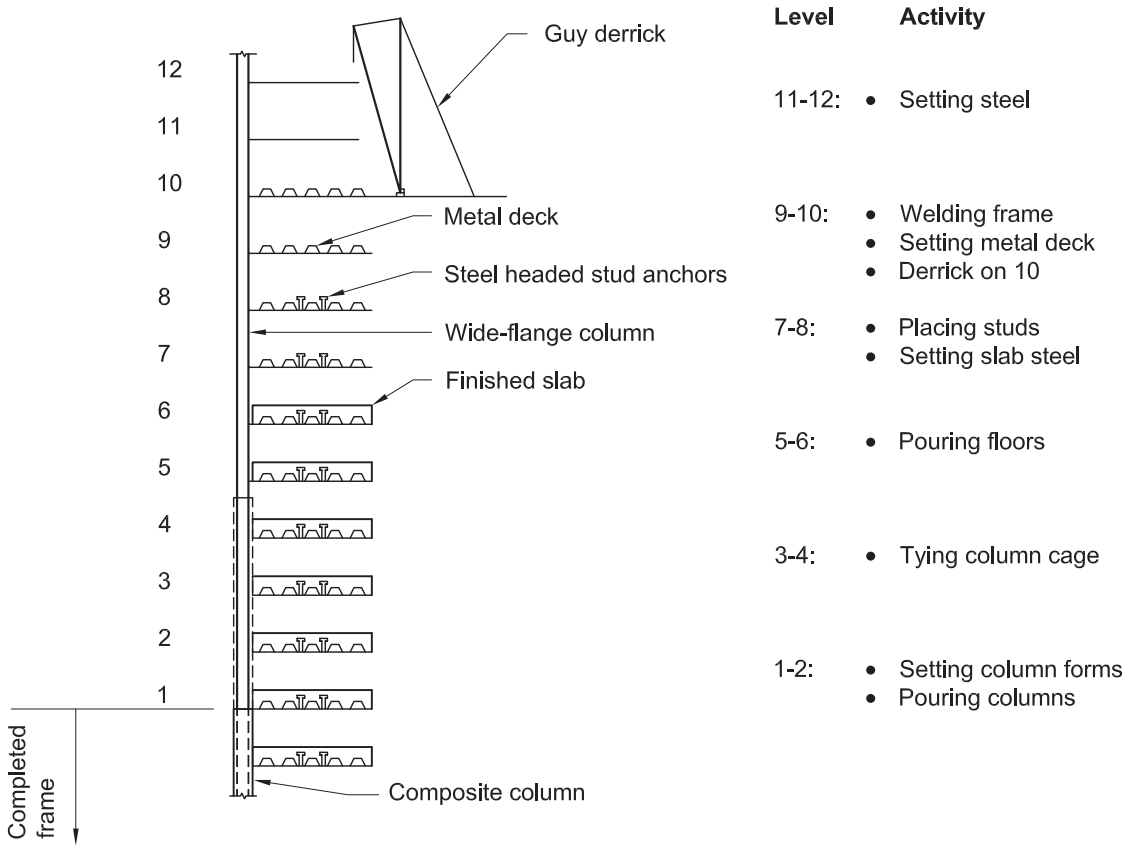


Fig. 1-2. Composite frame construction sequence (adapted from Griffis, 1992a).

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Table 1-1. Seismic Performance Factors for Systems with Composite Columns			
Seismic Force-Resisting System	**Response Modification Coefficient, *R***	**Overstrength Factor, Ω_0**	**Deflection Amplification Factor, C_d**
Steel and concrete composite eccentrically braced frames (C-EBF)	8	2½	4
Steel and concrete composite special concentrically braced frames (C-SCBF)	5	2	4½
Steel and concrete composite ordinary braced frames (C-OBF)	3	2	3
Steel and concrete composite special moment frames (C-SMF)	8	3	5½
Steel and concrete composite intermediate moment frames (C-IMF)	5	3	4½
Steel and concrete composite ordinary moment frames (C-OMF)	3	3	2½

This variation in the number of floors between construction activities has made it difficult to calculate with accuracy the effect of column shortening. Creep effects on the composite columns with respect to the all-steel core columns, or between shear walls, can also be troublesome to predict. The net effect of these problems can be floors that are not level from one point to another. The development and use of structural analysis software that can track construction stages has considerably reduced this problem, but care is still needed in accounting for time-dependent effects. Further discussion on shortening of composite columns in tall structures is presented in Section 4.5 of this Design Guide.

As with reinforced concrete columns, engineers must be aware of the potential problems related to reinforcing steel placement such as congestion, and how they affect the constructability of composites columns. This is particularly true at connections to encased composite columns where potential interference among steel beams, column reinforcing bars, connection ties, and shear connectors can all cause difficulty in the placement of reinforcement and concrete and can lead to honeycombs or voids. Careful attention to detailing can help avoid these problems.

1.3 EXISTING STRUCTURES

This Design Guide does not address the evaluation of composite members in existing structures because the procedures discussed herein have been developed from recent experimental data and analysis. The basic principles outlined in this Design Guide can be used to evaluate older columns only if the same assumptions apply. Extreme care must be used in establishing the material stress-strain relationships and the ductility of both the materials and details used.

Encasing an existing structural steel member in concrete to form a composite column can be an effective means of strengthening or repair; however, such applications are not specifically addressed in this Design Guide. In such instances, steps need to be taken to ensure that the additional steel and concrete portions can take their share of the load.

1.4 SEISMIC PROVISIONS

Seismic design and the AISC *Seismic Provisions for Structural Steel Buildings* (AISC, 2022d) are not the focus of this Design Guide. However, research has shown that steel-concrete composite frames are capable of exhibiting excellent performance during earthquakes (Kawaguchi et al., 2002; Nakashima and Chusilp, 2003; Braconi et al., 2008; Herrera et al., 2008; Tsai et al., 2008). A brief overview of the seismic force-resisting systems available for use with composite columns is presented in this section.

Six seismic force-resisting systems for frames with composite columns are described in AISC *Seismic Provisions* Chapters G and H. Additional dual systems consisting of composite braced frames with special moment frames are listed in ASCE/SEI 7 *Minimum Design Loads and Associated Criteria for Buildings and Other Structures* (ASCE, 2022). The seismic performance factors from ASCE/SEI 7 for the primary six systems are listed in Table 1-1. The seismic performance factors were largely developed based on qualitative comparisons to other structural systems; however, the values were validated for C-SMF and C-SCBF by Denavit et al. (2016a) and Denavit and Hajjar (2014), respectively.

For the three moment frame systems, the composite columns can be paired with structural steel, concrete-encased composite, or composite beams, but the connections must be fully restrained connections. For the three braced frame systems, the composite columns can be paired with either structural steel or composite beams and either structural steel or filled composite braces. As for most seismic design that relies on ductile behavior to resist lateral drifts, the systems in Table 1-1 require careful member and connection detailing beyond the strength design that this Design Guide addresses.

Provisions for seismic evaluation and retrofit of existing buildings with composite structural systems can be found in ASCE/SEI 41 *Seismic Evaluation and Retrofit of Existing Buildings* (ASCE, 2023) and AISC *Seismic Provisions for Evaluation and Retrofit of Existing Structural Steel Buildings* (AISC, 2022c).

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Chapter 2
Members and Frames

This chapter describes the strength of composite columns in the permanent condition. Strength during construction is discussed in Sections 4.3 and 4.4 of this Design Guide.

2.1 METHODS OF DESIGN

The design basis for steel-concrete composite columns is the same as that for structural steel columns. The primary objective being that no applicable strength or serviceability limit states are exceeded when the structure is subjected to applicable load combinations. Design for strength can be performed according to the provisions for load and resistance factor design (LRFD) or allowable strength design (ASD). The term "design" in this Design Guide includes analysis to determine required strength and proportioning of columns to have adequate available strength.

Design according to the LRFD method satisfies the requirements when the design strength, ϕR_n, of each structural component equals or exceeds the required strength, R_u, determined using LRFD load combinations. Design should be performed in accordance with AISC *Specification* Equation B3-1.

$$R_u \le \phi R_n \qquad \text{(}Spec.\text{ Eq. B3-1)}$$

where

R_n = nominal strength, kips
R_u = required strength using LRFD load combinations, kips
ϕ = resistance factor
ϕR_n = design strength, kips

Design according to the ASD method satisfies the requirements when the allowable strength, R_n/Ω, of each structural component equals or exceeds the required strength, R_a, determined using ASD load combinations. Design should be performed in accordance with AISC *Specification* Equation B3-2.

$$R_a \le \frac{R_n}{\Omega} \qquad \text{(}Spec.\text{ Eq. B3-2)}$$

where

R_a = required strength using ASD load combinations, kips
R_n/Ω = allowable strength, kips
Ω = safety factor

The strength criteria and associated terminology are summarized in Table 2-1.

The general stability requirements for composite columns are the same as that for structural steel, including consideration of the "big five" effects noted in AISC *Specification* Section C1: (1) flexural, shear, and axial member deformations, and all other component and connection deformations that contribute to the displacements of the structure; (2) second-order effects (including P-Δ and P-δ effects); (3) geometric imperfections; (4) stiffness reductions due to inelasticity, including the effect of partial yielding of the cross section that may be accentuated by the presence of residual stresses; and (5) uncertainty in system, member, and connection strength and stiffness. One difference, however, is that concrete cracking is a major component of stiffness reductions due to inelasticity for composite columns.

As stated in AISC *Specification* Section C1 "any rational method of design for stability that considers all of the listed effects is permitted." The direct analysis method as described in AISC *Specification* Chapter C and expanded upon for composite columns in AISC *Specification* Section I1.5 meets the general requirements, is permitted for all structures, and is recommended by the authors. The effective length method described in AISC *Specification* Appendix Section 7.2 also meets the requirements when the limitations of AISC *Specification* Appendix Section 7.2.1 are satisfied. A comparison of the direct analysis method and effective length method when applied to composite columns is presented in Table 2-2. The methods of design by advanced analysis described in AISC *Specification* Appendix 1 and the first-order analysis method described in AISC *Specification* Appendix Section 7.3 have not been specifically evaluated for composite structures and are not discussed in this Design Guide.

Methods of design generally consist of rules for the calculation of required strength and rules for the calculation of available strengths, which will be discussed individually in the following sections.

2.2 CALCULATION OF REQUIRED STRENGTHS

Both the direct analysis method and the effective length method are based on elastic analysis. Both are subject to the general analysis requirements of AISC *Specification* Section C2.1. Both must include consideration of initial imperfections, by either direct modeling or use of notional loads for the direct analysis method or only use of notional loads for the effective length method. These requirements, however, are no different than for structural steel members, and readers are referred to AISC Design Guide 28, *Stability Design of Steel Buildings* (Griffis and White, 2013), for detailed guidance.

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Table 2-1. LRFD and ASD Strength Criteria and Associated Terminology				
Method	**Strength Criteria**	**Required Strength**	**Available Strength**	**Nominal Strength**
LRFD	$R_u \le \phi R_n$	R_u computed using LRFD load combinations (e.g., $1.2D + 1.6L + 0.5L_r$)	ϕR_n also referred to as the design strength (ϕ is a resistance factor)	R_n
ASD	$R_a \le R_n/\Omega$	R_a computed using ASD load combinations (e.g., $D + L$)	R_n/Ω also referred to as the allowable strength (Ω is a safety factor)	R_n

Table 2-2. Key Aspects of Methods of Design for Composite Structures		
Method of Design	**Direct Analysis Method**	**Effective Length Method**
Analysis type[1]	Second-order elastic analysis	Second-order elastic analysis
Notional load[2]	$N_i = 0.002\alpha Y_i$ (or direct modeling of imperfections)	$N_i = 0.002\alpha Y_i$
Flexural stiffness, EI	$0.8\tau_b(EI)_{eff} = 0.64(EI)_{eff}$	$(EI)_{eff}$
Axial stiffness, EA	$0.8(E_sA_s + E_sA_{sr} + E_cA_c)$	$E_sA_s + E_sA_{sr} + E_cA_c$
Axial strength, P_n	Based on unbraced length, L	Based on effective length, KL
Interaction strength	Any set of interaction equations from AISC *Specification* Section I5 or Commentary	Only the interaction equations of AISC *Specification* Section H1.1 [other methods acceptable if $(EI)^* = 0.8(EI)_{eff}$]
Limitations	None	AISC *Specification* Appendix Section 7.2.1

[1] See requirements of AISC *Specification* Section C2.1.
[2] See exception for load combinations with lateral load in AISC *Specification* Section C2.2b(d).

The key difference between composite structures and steel structures in the determination of required strengths is the determination of the various stiffnesses. Defining the "nominal stiffness" of structural steel members is straightforward. For example, the nominal flexural stiffness, EI, of a structural steel member is the modulus of elasticity of steel multiplied by the gross moment of inertia in the plane of bending. Composite columns, however, include concrete that cracks at low levels of tensile stress and has a relatively low proportional limit in compression. As a result, it is common for the nominal stiffness to be reduced from the gross cross-sectional properties. This reduction is in addition to further adjustments to the stiffness required for the direct analysis method.

For the direct analysis method of design, AISC *Specification* Section I1.5 includes requirements on stiffnesses to be used when calculating required strengths. The nominal flexural stiffness of composite columns in compression is required to be taken as the effective stiffness of the composite section, $(EI)_{eff}$, as defined in Section I2 for the determination of axial compressive strength. The stiffness reduction parameter, τ_b, is required to be taken as 0.8. When combined with the 0.8 reduction specified in AISC *Specification* Section C2.3, the resulting flexural stiffness for use in analysis is $0.64(EI)_{eff}$.

The use of $0.64(EI)_{eff}$ has been shown through analysis to be accurate or conservative for a wide range of cases (Denavit et al., 2016b). However, $0.64(EI)_{eff}$ overestimates the stiffness when the composite column is subjected to very low axial loads and high bending moments. Concrete cracking, which reduces effective stiffness, is more pronounced under low axial loads. As a result, use of $0.64(EI)_{eff}$ may result in unconservative errors for columns that buckle under relatively low loads because of long member lengths, boundary conditions, or high leaning columns loads. Denavit et al. (2016b) suggest identifying such stability sensitive structures by computing the maximum axial load that the column can support under a gravity-only load combination and comparing the value to 15% of the nominal axial compressive strength without consideration of length effects (i.e., $0.15P_{no}$). If, when subjected to gravity-only loading, the column can support no more than 15% of its cross-sectional strength, then the column should be designed with further stiffness reduction (e.g., $\tau_b = 0.4$).

AISC *Specification* Section I1.5 also includes requirements for the axial stiffness of composite members in

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compression and for composite members in tension. The nominal axial stiffness, EA, is taken as the summation of the elastic axial stiffnesses of each component.

$$EA = E_s A_s + E_s A_{sr} + E_c A_c \quad (2\text{-}1)$$

where

A_c = cross-sectional area of concrete, in.^{2}

A_s = cross-sectional area of steel, in.^{2}

A_{sr} = cross-sectional area of longitudinal reinforcing bars, in.^{2}

E_c = modulus of elasticity of concrete, ksi

$= w_c^{1.5}\sqrt{f'_c}$

E_s = modulus of elasticity of steel

= 29,000 ksi

The axial stiffness of Equation 2-1 is subject to the 0.8 reduction specified in AISC *Specification* Section C2.3. Stiffness of members subject to net tension should be taken as the stiffness of the bare steel components (i.e., the steel shape and reinforcing).

The AISC *Specification* does not include any specific requirements for the stiffness of composite columns for the calculation of required strengths within the effective length method. The AISC *Specification* Commentary suggests one of two alternative approaches. The first is the use of the nominal stiffness equal to the effective stiffness $[EI = (EI)_{eff}]$ and the use of the interaction strength of AISC *Specification* Section H1.1. The second is the use of the reduced stiffness of 0.8 times the effective stiffness $[EI = 0.8(EI)_{eff}]$ and the use of the interaction strength with any applicable methods described in AISC *Specification* Section I5 and Commentary.

This section is focused on the calculation of required strengths, thus the stiffnesses described in this section are for that purpose, and not necessarily for use in other analyses such as those for determination of drift, deflection, vibration, or fundamental period. Section 2.8 of this Design Guide provides guidance for these situations.

2.3 CALCULATION OF AVAILABLE STRENGTHS

The available strengths of composite columns are calculated in accordance with AISC *Specification* Chapter I or Appendix 2 with the effective length, L_c, determined in accordance with the chosen method of design. Chapter I is organized with general provisions in the first section, followed by sections on different types of actions (i.e., axial force, flexure, shear, and combined flexure and axial force), then sections on load transfer, composite diaphragms and collector beams, and steel anchors. Appendix 2 contains supplemental provisions to compute the available strength of rectangular filled composite members with material strengths that exceed the limits set forth in Chapter I.

For ease of use, this Design Guide will be organized by member type. This section describes aspects of available strength applicable to all member types. Section 2.4 describes the calculation of available member strength for encased composite columns. Section 2.5 covers rectangular or square filled composite columns. Section 2.6 covers rectangular composite columns with high-strength materials. Section 2.7 covers round filled composite columns.

2.3.1 Nominal Strength of Composite Sections

AISC *Specification* Section I1.2 describes four methods of determining the nominal strength of composite sections: the plastic stress distribution method, the strain compatibility method, the elastic stress distribution method, and the effective stress-strain method. Explicit use of one of these methods is not always necessary. For example, for a composite column subjected to axial compression only, the provisions of AISC *Specification* Section I2 are sufficient to determine the available strength. The methods of AISC *Specification* Section I1.2 are primarily used when determining available strengths in flexure or combined flexure and axial force. However, when addressing combined flexure and axial force, the nominal strength of the composite section cannot be used directly because member length effects must be considered.

Plastic Stress Distribution Method

The plastic stress distribution method provides a general method for calculating the cross-sectional strength of composite columns. It is a relatively simple and convenient calculation method for the most common design situations. For the plastic stress distribution method, the nominal strength is computed assuming that steel components have reached a stress of F_y in either tension or compression, and concrete components in compression have reached a stress of $0.85f'_c$. For round HSS filled with concrete, a stress of $0.95f'_c$ is permitted to be used for concrete components in compression to account for the beneficial effects of concrete confinement. The concrete is assumed to have no tensile strength.

Points on the interaction diagram are computed by assuming a location of the plastic neutral axis, assigning stresses to the various components following the assumptions described above, and integrating the stresses to obtain an axial load and bending moment. With many points computed, an essentially continuous interaction diagram is formed. However, in practice, it is common to compute only a handful of points, resulting in a multilinear interaction diagram. AISC *Steel Construction Manual,* hereafter referred to as the AISC *Manual,* Tables 6-2a, 6-2b, 6-3, and 6-4 (AISC, 2023) provide closed-form expressions for computing anchor points identified as A, B, C, D, and E for standard composite cross sections. The difference between a multilinear interaction diagram computed using the AISC *Manual* tables and

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a continuous interaction diagram computed numerically by dividing the composite cross section into many small sections—as is done in the composite column program described in Appendix A—is shown in Figure 2-1.

The plastic stress distribution method assumes that sufficient strains have developed in the steel and concrete for both to reach their yield strength regardless of the bond condition at their interface, and that local buckling is delayed until yielding and concrete crushing have taken place. Accordingly, the plastic stress distribution method is only allowed for compact composite cross sections. The accuracy of the plastic stress distribution method has been confirmed for a range of composite cross sections. However, the method results in unconservative errors for encased composite members and rectangular filled composite members with relatively high steel ratios or relatively high steel strengths where the assumption of the steel and concrete achieving their peak strengths simultaneously breaks down (Behnam and Denavit, 2020).

Strain Compatibility Method

The strain compatibility method is an alternative method for calculating the cross-sectional strength of composite columns. It is recommended for use with irregular sections and for cases where the steel does not exhibit elastic-plastic behavior.

The strain compatibility method is identical to the method of determining moment and axial strength described in ACI 318. A linear distribution is assumed with the maximum concrete compressive strain equal to 0.003. The stress-strain relationships for steel and concrete should be obtained from tests or from published results. It is common to assume an elastic-perfectly plastic response for the steel and to use the rectangular stress block defined in ACI 318, Section 22.2.2.4. With these stress-strain relationships, the interaction strength obtained from the strain compatibility will always be less than or equal to that obtained from the plastic stress distribution method.

The strain compatibility method was found to be accurate or conservative in comparison to analysis results over a wide range of composite cross sections with the explicit consideration of strain compatibility avoiding the errors observed with the plastic stress distribution method (Behnam and Denavit, 2020).

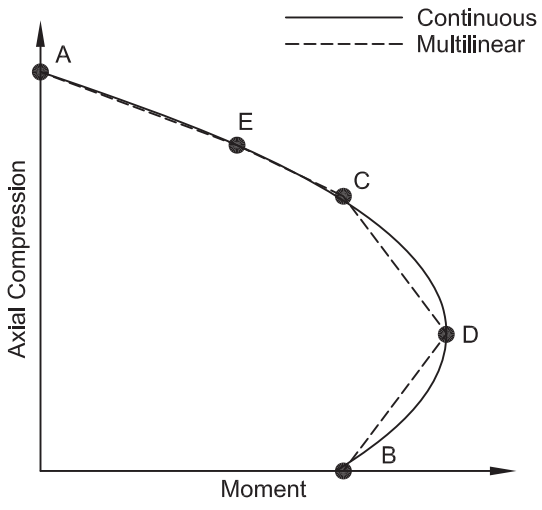


Fig. 2-1. Cross-sectional interaction diagram based on the plastic stress distribution method.

Elastic Stress Distribution Method

In the elastic stress distribution method, stresses are proportional to strains with the exception of concrete in tension, which is assumed to have zero stress. Stress in the steel is limited to F_y in tension or compression and the stress of concrete in compression is limited to $0.85f'_c$. Use of this method is among the options permitted for calculating the nominal flexural strength of encased composite members without steel anchors. It may also be appropriate for irregular cross sections or for evaluation of existing structures when the material properties are unknown.

Effective Stress-Strain Method

Neither the plastic stress distribution method nor the strain compatibility method explicitly consider the effects of local buckling because it is not a material response. The effective stress-strain method enables calculation of the nominal strength of composite cross sections with consideration of local buckling and other effects that significantly impact the strength of the cross section, such as yielding, residual stresses, concrete cracking, concrete crushing, and confinement. The method is intended to be used within a fiber-based approach for calculating the cross-section axial force-moment strength interaction where strain compatibility is enforced. Effective stress-strain relationships are published in literature. AISC *Specification* Section I1.2d Commentary cites four references that describe effective stress-strain relationships (Sakino et al., 2004; Han et al., 2005; Liang, 2009; Lai and Varma, 2016). However, it is the responsibility of the engineer to select a stress-strain relationship that is appropriate for the situation.

2.3.2 Available Strength of Composite Members Subjected to Combined Flexure and Axial Force

The methods described in AISC *Specification* Section I1.2 for determining the nominal strength of composite sections do not consider member length effects or resistance or safety

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factors. Member length effects need to be considered, and resistance or safety factors need to be applied before evaluating the strength of a composite member.

The available strength of composite members for individual actions is defined in AISC *Specification* Sections I2, I3, and I4 for axial force, flexure, and shear, respectively. The available strength of composite members subjected to combined flexure and axial force is less well defined. AISC *Specification* Section I5 provides some requirements with specific methods described in the Commentary.

This section describes three methods for assessing the interaction strength of composite members. The interaction equations of AISC *Specification* Sections H1 and I5 are presented first because they are defined in the AISC *Specification*. The transformed cross-sectional strength method is presented third and is based on a method defined in the AISC *Specification* Section I5 Commentary. Despite not being in the main body of the AISC *Specification*, transformed cross-sectional strength is the primary approach used for encased composite members and compact composite filled composite members.

The methods presented in this section are applicable to more than one of the member types covered in the following sections. Additional methods applicable only to specific member types are presented in the following sections along with other details.

Interaction Equations of AISC Specification Section H1

The simplest method of assessing interaction strength for composite members is to use the interaction equations of AISC *Specification* Section H1. These equations were developed and are primarily intended for structural steel members but can be applied conservatively to composite members.

The interaction equations of AISC *Specification* Section H1 require only the available compression strength, available tension strength, and the available flexure strengths about the axes of the member. The shape of the interaction surface is governed by the form of the equations and their constant coefficients. For the case of axial compression and uniaxial flexure, the interaction diagram is as shown in Figure 2-2. Member length effects are considered in this method by using an available axial compressive strength computed as described in AISC *Specification* Section I2.

The degree of conservatism in using the interaction equations of AISC *Specification* Section H1 generally depends on the magnitude of the concrete's contribution to the overall strength relative to the steel's contribution. The greater the contribution from the steel section, the less conservative the strength prediction of the interaction equations from AISC *Specification* Section H1. This approach is generally more conservative for members with high concrete compressive strength as compared to members with low concrete compressive strength.

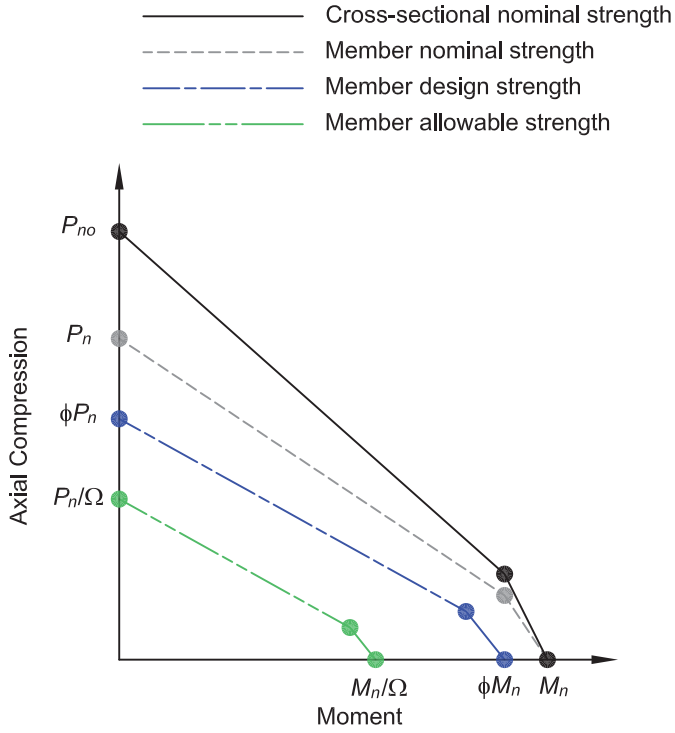


Fig. 2-2. Interaction diagram based on the equations of AISC Specification *Section H1.*

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Interaction Equations of AISC Specification *Section I5*

AISC *Specification* Section I5 includes interaction equations that are applicable to filled composite members with noncompact composite or slender-element composite sections. Based on the form of the equations, they are also only applicable to members subjected to combined uniaxial flexure and axial compression.

These equations are similar in form to the interaction equations of AISC *Specification* Section H1, but the intermediate point is shifted based on cross-sectional properties. As shown in Figure 2-3, the location of the intermediate point is based on coefficients c_p and c_m. These coefficients were empirically curve fit to the results of nonlinear finite element analyses of composite members (Lai et al., 2016).

Use of the interaction equations of AISC *Specification* Section I5 can result in a significant reduction in conservative error in comparison to the interaction equations of AISC *Specification* Section H1. Filled composite members with noncompact composite or slender-element composite sections, by definition, have relatively thin steel sections, and thus, the concrete contribution to the overall strength is relatively large. The shifted intermediate point, shown in Figure 2-4, better captures the shape of the interaction strength for these members.

Transformed Cross-Sectional Strength

The interaction equations of AISC *Specification* Sections H1 and I5 provide a direct means of assessing the strength of composite members with consideration of length effects. However, the interaction equations of AISC *Specification* Section H1 are conservative, and the interaction equations of AISC *Specification* Section I5 are limited in scope based on the range of calibration of their empirical shape parameters. Another method of assessing the strength of composite members is by using equations based on transformed cross-sectional strength.

This method is described in AISC *Specification* Section I5 Commentary as "Method 2–Interaction Curves from the Plastic Stress Distribution Method."

The first step of Method 2 is to compute anchor points on the cross-sectional nominal strength interaction diagram using the plastic stress distribution method, as shown in Figure 2-5. Closed-form equations for the axial compression and bending moment at each of these points are available in AISC *Manual* Tables 6-2a, 6-2b, 6-3, and 6-4. Points A, B, and C are required. Points D and E are not required. Point D is the point of maximum flexural strength. Point E, which is only defined for some cross-section types, lies between points A and C and can help better characterize the shape of the cross-sectional interaction diagram. Neither of these points is used in this method of assessing interaction strength for simplicity and to avoid potential unconservative error.

Points A, B, and C on the nominal strength interaction diagram of the cross section need to be transformed by applying a reduction for length effects and resistance or safety factors to generate the available strength interaction diagram of the member. The stability reduction for length effects is done by multiplying the axial force of each point by $\chi = P_n/P_{no}$, where P_n is the nominal compressive strength of the member and P_{no} is the nominal axial compressive strength without consideration of length effects. The resistance factors are

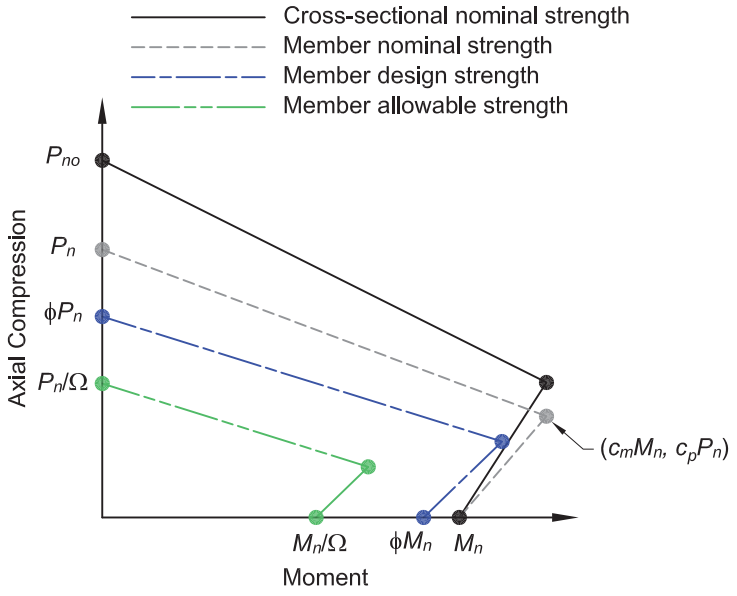


Fig. 2-3. Interaction diagram based on the equations of AISC Specification *Section I5.*

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Table 2-3. Construction of Interaction Diagram						
	Nominal Strength of Cross Section		Design Strength of Member (LRFD)		Allowable Strength of Member (ASD)	
Point	Flexure	Axial Force	Flexure	Axial Force	Flexure	Axial Force
A	0	P_{no}	0	$P_c = \phi P_n$	0	$P_c = P_n/\Omega$
C	M_n	P_C	ϕM_n	$P_{cc} = \phi\chi P_C$	M_n/Ω	$P_{cc} = \chi P_C/\Omega$
B	M_n	0	ϕM_n	0	M_n/Ω	0

ϕ = 0.75 for axial compression and 0.90 for flexure
Ω = 2.00 for axial compression and 1.67 for flexure
$\chi = P_n/P_{no}$

applied by multiplying the flexure of each point by $\phi = 0.90$ and the axial force of each point by $\phi = 0.75$. The safety factors are applied by dividing the flexure of each point by $\Omega = 1.67$ and the axial force of each point by $\Omega = 2.00$. The reductions are summarized in Table 2-3 and shown in Figure 2-5.

These simple transformations for length effects and from nominal strength to available strength (i.e., the application of resistance or safety factors; see also Table 2-1) are the source of the potential unconservative error that led to exclusion of point D from the interaction diagram. Applying these transformations to point D results in a nominal member strength that is outside of the nominal cross-sectional strength and available strength that is insufficiently reduced near the balance point. Alternative transformations that include point D and avoid some of this potential error have been proposed (Denavit, 2021) but have not been fully validated over the range of composite columns permitted by the AISC *Specification*. Interaction diagrams computed using the provisions of AISC *Specification* Section I5 can also show member nominal strength (with consideration of length effects) that is outside of the cross-sectional nominal strength (without consideration of length effects), as shown in Figure 2-3. However, the interaction equations of AISC *Specification* Section I5 were calibrated directly to the results of beam-column analyses (Lai et al., 2016).

Once the available strengths are determined, they can be used in AISC *Specification* Equations C-I5-1a and C-I5-1b to assess interaction strength.

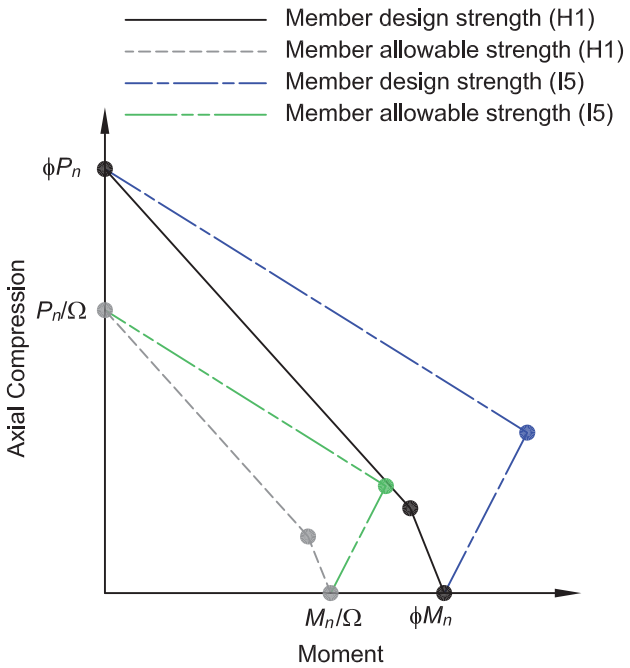


Fig. 2-4. Comparison of interaction diagrams based on the equations of AISC Specification *Section I5 to those based on the equations of AISC* Specification *Section H1.*

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If $P_r < P_{cc}$

$$\frac{M_{rx}}{M_{cx}} + \frac{M_{ry}}{M_{cy}} \le 1$$

(from *Spec*. Eq. C-I5-1a)

If $P_r \ge P_{cc}$

$$\frac{P_r - P_{cc}}{P_c - P_{cc}} + \frac{M_{rx}}{M_{cx}} + \frac{M_{ry}}{M_{cy}} \le 1$$

(from *Spec*. Eq. C-I5-1b)

where

M_c = available flexural strength, kip-in.

M_r = required flexural strength, kip-in.

P_C = nominal axial compressive strength at point C without member length effects, kips

P_c = available axial compressive strength ($P_c = \phi P_n$ for LRFD, $P_c = P_n/\Omega$ for ASD), kips

P_{cc} = available axial compressive strength at point C ($P_{cc} = \phi\chi P_C$ for LRFD, $P_{cc} = \chi P_C/\Omega$ for ASD), kips

P_r = required axial compressive strength, kips

x = subscript relating symbol to major-axis bending

y = subscript relating symbol to minor-axis bending

Note, for improved clarity and consistency, some of the symbols used here differ slightly from those used in the AISC *Specification* Commentary.

When using the plastic stress distribution method, the value of P_C does not depend on the bending axis, thus P_{cc} also does not depend on the bending axis.

Use of AISC *Specification* Equations C-I5-1a and C-I5-1b can be conservative for members subjected to biaxial bending. Additional strength can be achieved by adding a shape factor, α, as shown in Equations 2-2a and 2-2b. Behnam and Denavit (2021) recommend α = 1.25 for encased composite members, α = 1.5 for rectangular filled composite members, and α = 2 for round filled composite members. Larger values of α correspond to larger biaxial moment strength. 3D interaction diagrams as defined by Equations 2-2a and 2-2b with various values of α are shown in Figure 2-6. The shape factor does not affect the strength of members subjected to uniaxial bending (i.e., when $M_{rx} = 0$ or $M_{ry} = 0$).

If $P_r < P_{cc}$

$$\left(\frac{M_{rx}}{M_{cx}}\right)^{\alpha} + \left(\frac{M_{ry}}{M_{cy}}\right)^{\alpha} \le 1 \qquad (2\text{-}2a)$$

If $P_r \ge P_{cc}$

$$\frac{P_r - P_{cc}}{P_c - P_{cc}} + \left[\left(\frac{M_{rx}}{M_{cx}}\right)^{\alpha} + \left(\frac{M_{ry}}{M_{cy}}\right)^{\alpha}\right]^{1/\alpha} \le 1 \qquad (2\text{-}2b)$$

In the AISC *Specification* Commentary, the transformed cross-sectional strength method is referred to as "Method 2–Interaction Curves from the Plastic Stress Distribution Method" and is described only for use with the plastic stress distribution method. However, the approach can be used for other methods of determining cross-sectional strength as well. Points A, B, and C can be computed using any of the methods described in AISC *Specification* Section I1.2

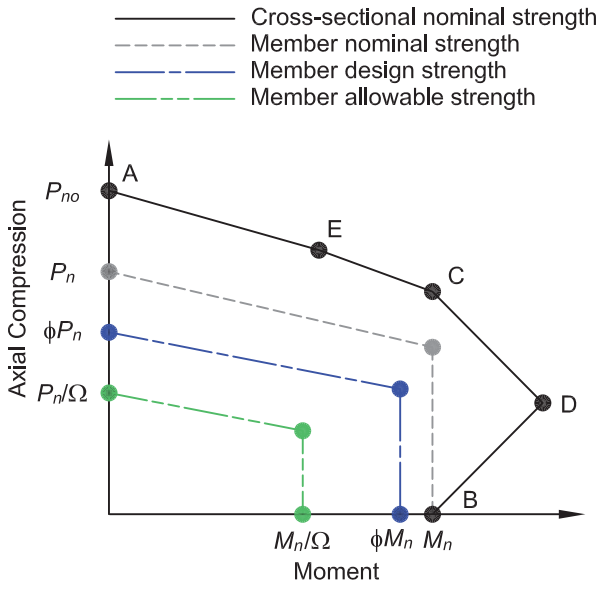


Fig. 2-5. Interaction diagram based on the transformed cross-sectional strength method.

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when defined as follows. Point A is the axial strength—that is, the intersection of the interaction diagram and the *y*-axis of a typical interaction diagram (i.e., where axial compression is plotted on the *y*-axis and bending moment is plotted on the *x*-axis). Point B is the flexural strength—that is, the intersection of the typical interaction diagram and the *x*-axis. Point C has combined flexure and axial force and is defined as the point with same flexure as Point B—that is, it is located directly above Point B in a typical interaction diagram.

Once computed, the nominal cross-sectional strengths at points A, B, and C can be transformed as described in Table 2-3 and used in AISC *Specification* Equations C-I5-1a and C-I5-1b or Equations 2-2a and 2-2b to assess strength. Note that for methods other than the plastic stress distribution method, the value of P_C when computed for bending about the major axis can be different than the value of P_C when computed for bending about the minor axis. The smaller value of P_C and thus P_{cc} should be used in the interaction equations.

2.3.3 Material Limitations

AISC *Specification* Section I1.3 specifies limits on the strength of materials used in composite columns. These limits are summarized in Table 2-4. AISC *Specification*

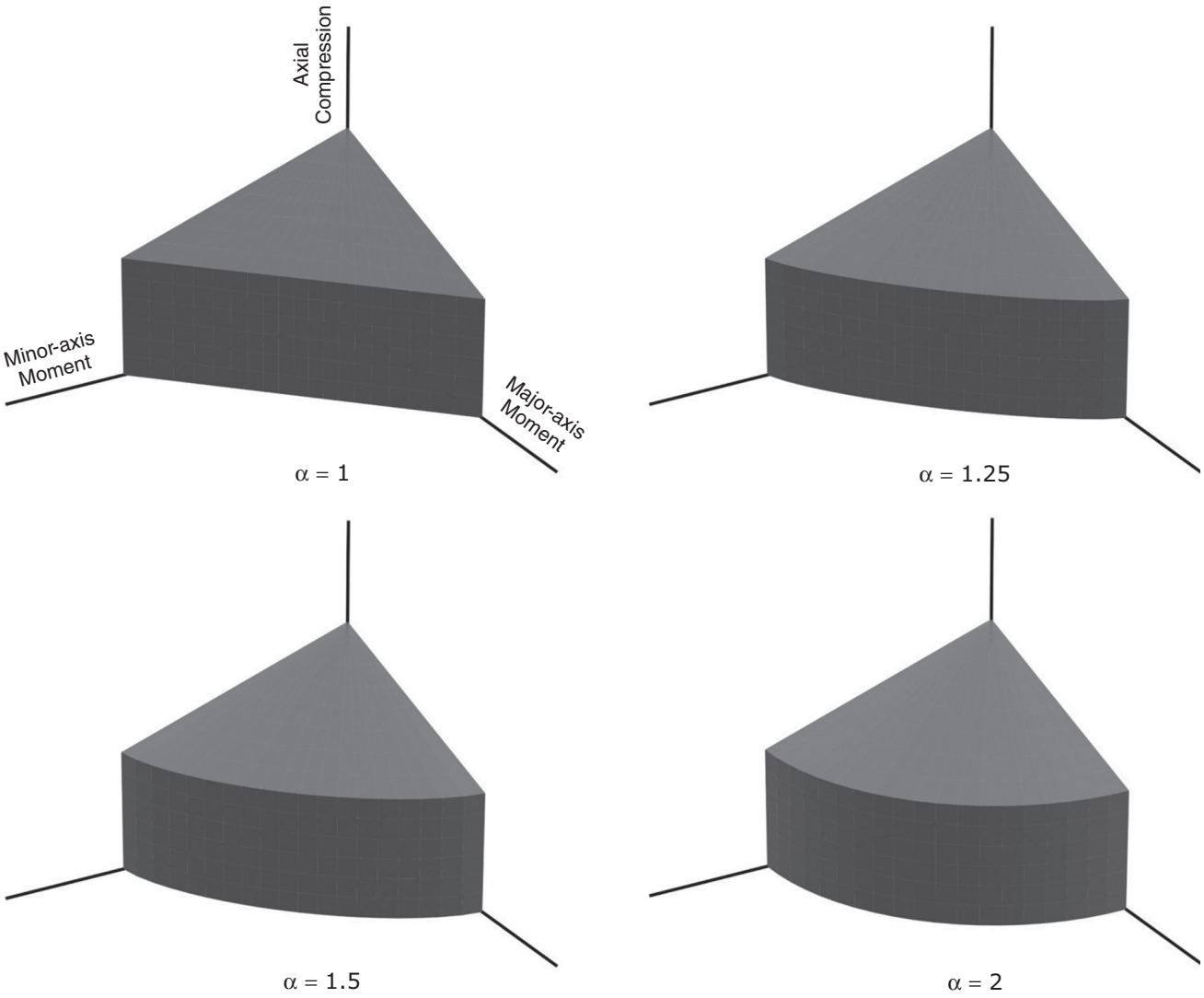


Fig. 2-6. 3D interaction diagrams showing effect of shape factor, α.

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Table 2-4. Material Limitations for AISC *Specification* Chapter I		
Material	**Minimum Limit, ksi (MPa)**	**Maximum Limit, ksi (MPa)**
Normal weight concrete	3 (21)	10 (69)
Lightweight concrete	3 (21)	6 (41)
Structural steel	N/A	75 (525)
Reinforcing steel	N/A	80 (550)

Appendix 2 has provisions for the strength of rectangular filled composite members with materials that exceed these limitations. These provisions, and the material limitations associated with them, are described in Section 2.6 of this Design Guide.

2.3.4 Singly Symmetric and Unsymmetric Members

The focus of this Design Guide is on doubly symmetric members because the vast majority of the research on composite columns has been on those with doubly symmetric sections, and the wide variety of possible singly symmetric and unsymmetric composite members makes the development of specific guidance difficult. Nonetheless, there are many practical uses of singly symmetric and unsymmetric members. For example, a corner column in a high-rise building may be L-shaped.

For singly symmetric and unsymmetric members, the strain compatibility method should be used in lieu of the plastic stress distribution method. Also, the determination of the centroidal axes should be carefully considered for consistency with the analytical model used to determine required strengths.

2.4 AVAILABLE STRENGTH OF ENCASED COMPOSITE MEMBERS

This section describes the calculation of available strength of doubly symmetric encased composite members with an I-shaped steel member, reinforcing bars, and a rectangular concrete cross section. For these members, the concrete is assumed to brace the web and flange such that local buckling of the steel shape will not affect the strength of the member regardless of the width-to-thickness ratios of the web and flanges.

2.4.1 Limitations

Limitations for encased composite members appear in AISC *Specification* Section I2.1a for members subjected to axial force and AISC *Specification* Section I3.3a for members subjected to flexure. Members subjected to combined flexure and axial force must meet both sets of limitations.

Most of the limitations are common to members subjected to axial force and members subjected to flexure and can be regarded as general limitations. These include:

- A minimum required cross-sectional area of the steel shape—that is, 1% of the total composite cross section.
- Requirement for longitudinal and transverse reinforcement and associated detailing requirements.
- Minimum longitudinal reinforcement ratios—that is, 0.4% of the total composite cross section.

Additionally, for members subjected to axial force, longitudinal reinforcement must comply with the maximum reinforcement ratio for continuous longitudinal reinforcement requirement in ACI 318.

Encased composite members serving as beams subjected to low axial loads must be classified as tension controlled as defined in ACI 318. Beams are defined in the AISC *Specification* glossary as nominally horizontal structural members that have the primary function of resisting bending moments. Low axial loads are defined in this limitation as $P_u < 0.10P_n$. The symbol P_u is typically defined as the required axial strength in compression using LRFD load combinations. For consistency, when using ASD load combinations, P_u should be taken as αP_r, where $\alpha = 1.0$ for LRFD and $\alpha = 1.5$ for ASD. To assess this limitation precisely, the location of the neutral axis must be determined using assumptions consistent with the strain compatibility method described in ACI 318, Section 22.2. Commercial software may be helpful for making this assessment. The location of the neutral axis computed according to the strain compatibility method may be different than when computed using the plastic stress distribution method. Use of the neural axis determined according to the plastic stress distribution method may be a helpful approximation.

The longitudinal and transverse reinforcing requirements are in addition to the requirements of ACI 318, as stated in AISC *Specification* Section I1.1.

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2.4.2 Axial Compressive Strength

The axial compressive strength of encased composite members is defined in AISC *Specification* Section I2.1b. The nominal axial compressive strength without consideration of length effects, P_{no}, is computed by summing the contributions of each of the components of the cross section as shown in AISC *Specification* Equation I2-7.

$$P_{no} = F_y A_s + F_{ysr} A_{sr} + 0.85 f'_c A_c \qquad (Spec.\ \text{Eq. I2-7})$$

The nominal axial compressive strength (with consideration of member length effects) for the limit state of flexural buckling, P_n, is computed as shown in AISC *Specification* Equations I2-2 and I2-3. Note that the column curve of AISC *Specification* Equations I2-2 and I2-3 is identical to the column curve for structural steel members defined in AISC *Specification* Section E3 if $P_{no} = A_s F_y$ and $(EI)_{eff} = E_s I_s$.

When $\dfrac{P_{no}}{P_e} \le 2.25$

$$P_n = P_{no}\left(0.658^{\frac{P_{no}}{P_e}}\right) \qquad (Spec.\ \text{Eq. I2-2})$$

When $\dfrac{P_{no}}{P_e} > 2.25$

$$P_n = 0.877 P_e \qquad (Spec.\ \text{Eq. I2-3})$$

where P_e is the elastic critical buckling load determined from AISC *Specification* Equation I2-4; as specified in AISC *Specification* Appendix 7, Section 7.2.3(b), when using the effective length method; or through an elastic buckling analysis.

$$P_e = \frac{\pi^2 (EI)_{eff}}{L_c^2} \qquad (Spec.\ \text{Eq. I2-4})$$

where $(EI)_{eff}$ is the effective stiffness of the composite section determined from AISC *Specification* Equation I2-5 and L_c is the effective length of the member.

The effective stiffness of the composite section, $(EI)_{eff}$, is computed by summing the contributions of each of the components of the cross section, but with a reduction factor, C_1, applied to the concrete contribution to account for the level of cracking expected at the axial compressive strength of the member.

$$(EI)_{eff} = E_s I_s + E_s I_{sr} + C_1 E_c I_c \qquad (Spec.\ \text{Eq. I2-5})$$

$$C_1 = 0.25 + 3\left(\frac{A_s + A_{sr}}{A_g}\right) \le 0.7 \qquad (Spec.\ \text{Eq. I2-6})$$

The critical buckling load, P_e, should be computed for each coordinate axis and the least value used to compute P_n. Torsional buckling and local buckling apply to some doubly symmetric structural steel members but not to doubly symmetric encased composite members because the concrete provides significant torsional stiffness and bracing to the web and flange of the steel shape.

The design compressive strength is determined using a resistance factor of $\phi_c = 0.75$ for LRFD. The allowable compressive strength is determined with a safety factor of $\Omega_c = 2.00$ for ASD. The design compressive strength need not be less than the design compressive strength of the bare steel member. The design compressive strength of the bare steel member can be greater than that of the composite member when the steel ratio is high, and the contribution of the concrete is not enough to overcome the lower resistance factor or higher safety factor used for composite members.

2.4.3 Axial Tensile Strength

The axial tensile strength of encased composite members is defined in AISC *Specification* Section I2.1c. The nominal tensile strength, P_n, is for the limit state of yielding as shown in AISC *Specification* Equation I2-8. Only the structural steel and longitudinal reinforcement contributions are included in this equation because the concrete tension strength is negligible.

$$P_n = F_y A_s + F_{ysr} A_{sr} \qquad (Spec.\ \text{Eq. I2-8})$$

The design tensile strength is determined using a resistance factor of $\phi_t = 0.90$ for LRFD. The allowable tensile strength is determined using a safety factor of $\Omega_t = 1.67$ for ASD.

2.4.4 Flexural Strength

The flexural strength of encased composite members is defined in AISC *Specification* Section I3.3a. Several methods are listed as permitted for determining the nominal flexural strength with the primary distinction being whether steel anchors are provided to aid in load transfer between the steel and concrete. Direct bond interaction between the steel and concrete has been shown experimentally with statically determinate specimens to be insufficient to develop the composite action necessary to achieve the full plastic moment capacity of the composite section.

If steel anchors are not provided to aid in load transfer between the steel and concrete, the nominal flexural strength must be determined using either the superposition of elastic stresses on the composite section for the limit state of yielding, or as the plastic moment of the steel section alone. If using superposition of elastic stresses, the stresses on the steel section from permanent loads applied to unshored beams before the concrete has hardened must be superimposed with

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the stresses on the composite section from loads applied to the beams after hardening of the concrete. For shored beams, all loads may be assumed to be resisted by the composite section.

If steel anchors are provided to aid in load transfer between the steel and concrete, the nominal flexural strength can be determined using the plastic stress distribution method or the strain compatibility method.

Closed form equations for the flexural strength of encased composite members are provided in AISC *Manual* Table 6-2a for encased members bent about the major axis of the steel shape and Table 6-2b for encased members bent about the minor axis of the steel shape. The flexural strength is denoted as M_B in these tables. The tables are reproduced in this Design Guide as Figure 2-7 and Figure 2-8, respectively. The tables are only applicable to certain reinforcing configurations. The composite column program that accompanies this Design Guide can compute the flexural strength for more general configurations of reinforcement.

The plastic stress distribution has been shown to be accurate or conservative over a wide range of cross sections, but exhibits unconservative error in the determination of flexural strength for encased shapes with high steel ratio (e.g., greater than 10%) or high steel yield stress (e.g., greater than 65 ksi) (Behnam and Denavit, 2020), especially for bending about the minor axis of the steel cross section. For these cases, the strain compatibility method or the adjustment factor described by Behnam and Denavit (2020) should be used.

The design flexural strength is determined using a resistance factor of $\phi_b = 0.90$ for LRFD. The allowable flexural strength is determined using a safety factor of $\Omega_b = 1.67$ for ASD.

2.4.5 Interaction Strength

Provisions for the interaction between flexural and axial forces for encased composite members are included in AISC *Specification* Section I5, but this section provides limited specific guidance. The AISC *Specification* Commentary provides more specific guidance, including three methods that are applicable to encased composite members. The methods in the Commentary include "Method 1—Interaction Equations of Section H1" and "Method 2—Interaction Curves from the Plastic Stress Distribution Method," both of which are described in Section 2.3.2 of this Design Guide. In this Design Guide, Method 2 is generalized for interaction diagrams from any cross-sectional strength approach as the "transformed cross-sectional strength" method. The AISC *Specification* Commentary also describes "Method 3—Design Guide 6." The first edition of this Design Guide (Griffis, 1992b) described a method for computing interaction strength and included many tables of composite member strength. While the method is obsolete, the design strength of encased composite members can conservatively be determined directly from the tables.

Assessing interaction strength by transforming cross-sectional strength from the plastic stress distribution method will often be the best approach. Note that the method requires steel anchors be provided to aid in load transfer between the steel and concrete in accordance with AISC *Specification* Section I3.3a. Cross-sectional strength can be efficiently computed using the closed form equations in AISC *Manual* Tables 6-2a and 6-2b (reproduced in this Design Guide as Figure 2-7 and Figure 2-8, respectively) or with the composite column program that accompanies this Design Guide. Cross-sectional strength should be transformed to member strength as described in Section 2.3.2 of this Design Guide.

2.4.6 Shear Strength

The shear strength of encased composite members is defined in AISC *Specification* Section I4.1. Due to uncertainty in how effectively the shear strength of the steel and the concrete in encased composite members combine, only a portion of the cross section can be considered when calculating the available strength. Three options are provided for computing the available strength:

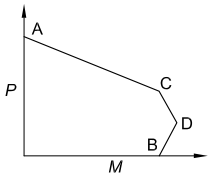
1. Calculate the available strength based on the steel section alone using the provisions of AISC *Specification* Chapter G. The potential for shear buckling (i.e., $C_{v1} < 1.0$) should be evaluated as specified in AISC *Specification* Chapter G, neglecting the bracing provided by the concrete.
2. Calculate the available strength based on the reinforced concrete portion (i.e., the concrete and the reinforcing) using the provisions of ACI 318. For this calculation it may be assumed that the steel section is concrete and not a hollow in the section.
3. Calculate the nominal strength as the summation of the nominal strengths of the steel section using the provisions of AISC *Specification* Chapter G and the transverse reinforcing using the provisions of ACI 318; then apply a resistance factor of 0.75 or safety factor of 2.00 to the combined strength.

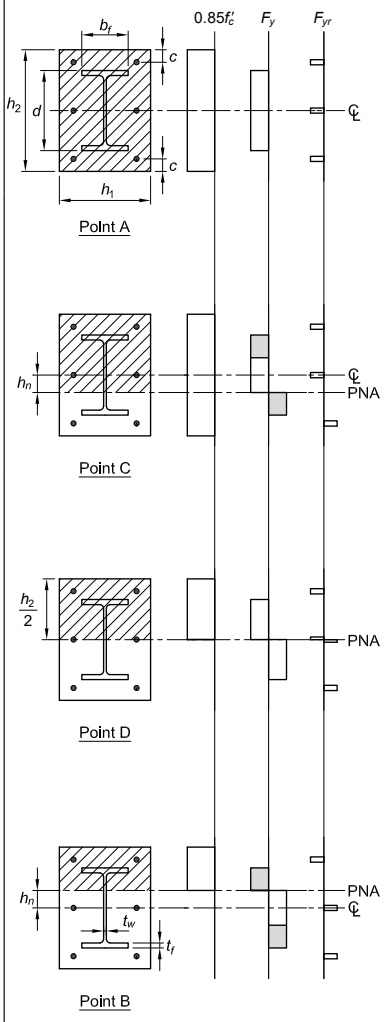
For the first option, the resistance factor and safety factor should be as defined in AISC *Specification* Chapter G. For the second and third options, the design shear strength is determined using a resistance factor of $\phi_v = 0.75$ for LRFD, and the allowable shear strength is determined using a safety factor of $\Omega_v = 2.00$ for ASD.

Future research may justify provisions for the available strength that combine the contributions from the steel section and the reinforced concrete.

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Table 6-2a
Cross-Section Strength for Rectangular Encased W-Shapes
Subject to Flexure about the Major Axis



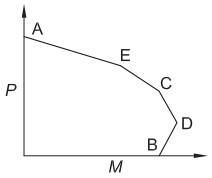
Section	Stress Distribution	Pt.	Defining Equation
		A	$P_A = F_y A_s + F_{yr} A_{sr} + 0.85 f'_c A_c$ $M_A = 0$ A_s = area of steel shape, in.^{2} A_{sr} = area of all continuous reinforcing bars, in.^{2} $A_c = h_1 h_2 - A_s - A_{sr}$
		C	$P_C = 0.85 f'_c A_c$ $M_C = M_B$
		D	$P_D = \dfrac{0.85 f'_c A_c}{2}$ $M_D = F_y Z_s + F_{yr} Z_r + 0.85 f'_c \left(\dfrac{Z_c}{2}\right)$ Z_s = full x-axis plastic section modulus of steel shape, in.^{3} A_{srs} = area of continuous reinforcing bars at the centerline, in.^{2} $Z_r = \left(A_{sr} - A_{srs}\right)\left(\dfrac{h_2}{2} - c\right)$ $Z_c = \dfrac{h_1 h_2^2}{4} - Z_s - Z_r$
		B	$P_B = 0$ $M_B = M_D - F_y Z_{sn} - 0.85 f'_c \left(\dfrac{Z_{cn}}{2}\right)$ $Z_{cn} = h_1 h_n^2 - Z_{sn}$ For h_n below the flange $\left(h_n \le \dfrac{d}{2} - t_f\right)$ $h_n = \dfrac{0.85 f'_c \left(A_c + A_{srs}\right) - 2 F_{yr} A_{srs}}{2\left[0.85 f'_c \left(h_1 - t_w\right) + 2 F_y t_w\right]}$ $Z_{sn} = t_w h_n^2$ For h_n within the flange $\left(\dfrac{d}{2} - t_f < h_n \le \dfrac{d}{2}\right)$ $h_n = \dfrac{0.85 f'_c \left(A_c + A_s - d b_f + A_{srs}\right) - 2 F_y \left(A_s - d b_f\right) - 2 F_{yr} A_{srs}}{2\left[0.85 f'_c \left(h_1 - b_f\right) + 2 F_y b_f\right]}$ $Z_{sn} = Z_s - b_f \left(\dfrac{d}{2} - h_n\right)\left(\dfrac{d}{2} + h_n\right)$ For h_n above the flange $\left(h_n > \dfrac{d}{2}\right)$ $h_n = \dfrac{0.85 f'_c \left(A_c + A_s + A_{srs}\right) - 2 F_y A_s - 2 F_{yr} A_{srs}}{2\left(0.85 f'_c h_1\right)}$ $Z_{sn} = Z_s$

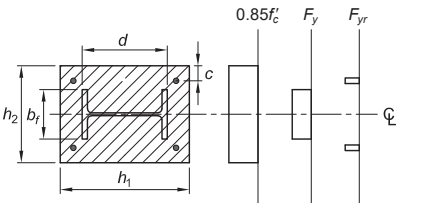
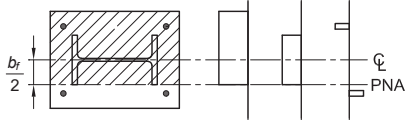
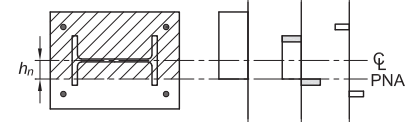
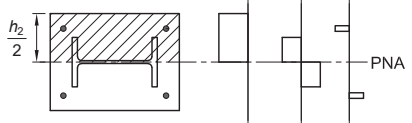
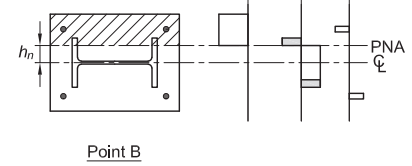
F_y = specified minimum yield stress of steel shape, ksi
F_{yr} = specified minimum yield stress of reinforcing steel, ksi

Fig. 2-7. AISC Manual *Table 6-2a—Axial and flexural interaction strength of encased composite members about the major axis of the steel shape based on the plastic stress distribution method.*

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Table 6-2b
Cross-Section Strength for Rectangular Encased W-Shapes
Subject to Flexure about the Minor Axis



Section	Stress Distribution	Pt.	Defining Equation
Point A		A	$P_A = F_y A_s + F_{yr} A_{sr} + 0.85 f'_c A_c$ $M_A = 0$ A_s = area of steel shape, in.^{2} A_{sr} = area of all continuous reinforcing bars, in.^{2} $A_c = h_1 h_2 - A_s - A_{sr}$
Point E		E	$P_E = F_y A_s + 0.85 f'_c \left[A_c - \frac{h_1}{2}(h_2 - b_f) + \frac{A_{sr}}{2} \right]$ $M_E = M_D - Z_{sE} F_y - 0.85 f'_c \left(\frac{Z_{cE}}{2} \right)$ $Z_{sE} = Z_s$ = plastic section modulus of steel shape about the y-axis, in.^{3} $Z_{cE} = \frac{h_1 b_f^2}{4} - Z_{sE}$
Point C		C	$P_C = 0.85 f'_c A_c$ $M_C = M_B$
Point D		D	$P_D = \frac{0.85 f'_c A_c}{2}$ $M_D = F_y Z_s + F_{yr} Z_r + 0.85 f'_c \left(\frac{Z_c}{2} \right)$ $Z_r = A_{sr} \left(\frac{h_2}{2} - c \right)$ $Z_c = \frac{h_1 h_2^2}{4} - Z_s - Z_r$
Point B		B	$P_B = 0$ $M_B = M_D - F_y Z_{sn} - 0.85 f'_c \left(\frac{Z_{cn}}{2} \right)$ $Z_{cn} = h_1 h_n^2 - Z_{sn}$ For h_n within the flange $\left(\frac{t_w}{2} < h_n \le \frac{b_f}{2} \right)$ $h_n = \frac{0.85 f'_c (A_c + A_s - 2 t_f b_f) - 2 F_y (A_s - 2 t_f b_f)}{2 \left[4 F_y t_f + 0.85 f'_c (h_1 - 2 t_f) \right]}$ $Z_{sn} = Z_s - 2 t_f \left(\frac{b_f}{2} + h_n \right) \left(\frac{b_f}{2} - h_n \right)$ For h_n above the flange $\left(h_n > \frac{b_f}{2} \right)$ $h_n = \frac{0.85 f'_c (A_c + A_s) - 2 F_y A_s}{2 (0.85 f'_c h_1)}$ $Z_{sn} = Z_s$

F_y = specified minimum yield stress of steel shape, ksi
F_{yr} = specified minimum yield stress of reinforcing steel, ksi

Fig. 2-8. AISC Manual *Table 6-2b—Axial and flexural interaction strength of encased composite members about the minor axis of the steel shape based on the plastic stress distribution method.*

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DESIGN EXAMPLE 2.4—Encased Composite Member Subjected to Combined Axial Compression and Flexure

Given:

Assess the adequacy of an encased composite member constructed with the cross section shown in Figure 2-9. The member is a column in the simple frame shown in Figure 2-10.

The encased composite member consists of a W14×211 ASTM A992/A992M (ASTM, 2022) steel member encased by a 36 in. by 36 in. concrete column with a specified concrete strength of $f'_c = 10$ ksi. The member has eight No. 8 longitudinal reinforcing bars and No. 4 lateral ties spaced at 16 in. on center. All reinforcing bars are ASTM A615/A615M, Grade 60 (ASTM, 2024a). The column is oriented to bend about the minor axis of the steel shape (i.e., the *y*-axis). Assume that enough steel anchors are provided to transfer load between the steel and concrete.

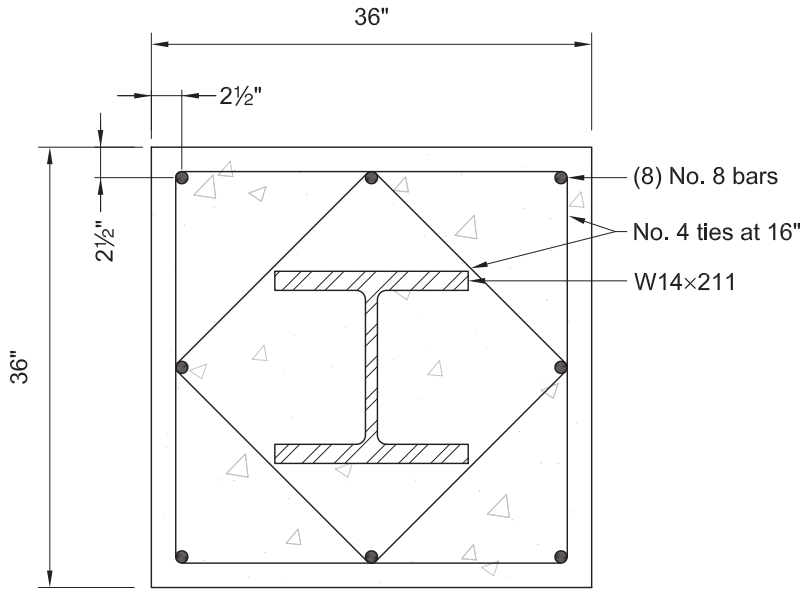


Fig. 2-9. Encased composite member for Design Example 2.4.

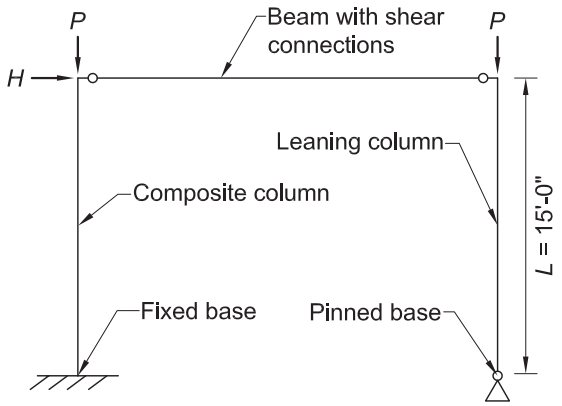


Fig. 2-10. Frame for Design Example 2.4.

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The applied loads that include the leaning column loads are as follows:

LRFD	ASD
$P_u = 1{,}000$ kips $H_u = 190$ kips	$P_a = 650$ kips $H_a = 130$ kips

Solution:

From AISC *Manual* Table 2-4, the material properties are as follows:

ASTM A992/A992M
$F_y = 50$ ksi

From ASTM A615/A615M, the material properties of the reinforcing steel are as follows:

ASTM A615/A615M Grade 60
$F_{ysr} = 60$ ksi

From AISC *Manual* Table 1-1, cross-sectional properties of the steel shape are as follows:

$A_s = 62.0$ in.^{2} $b_f = 15.8$ in. $t_f = 1.56$ in. $t_w = 0.980$ in.
$I_{sx} = 2{,}660$ in.^{4} $I_{sy} = 1{,}030$ in.^{4} $Z_{sy} = 198$ in.^{3}

From ACI 318, Appendix 2, the area of a single No. 8 reinforcing bar is 0.79 in.^{2}.

Calculate cross-sectional properties

The area and moment of inertia of the reinforcing are:

$$A_{sr} = 8(0.79 \text{ in.}^2)$$
$$= 6.32 \text{ in.}^2$$

$$I_{srx} = I_{sry}$$
$$= 3\left(0.79 \text{ in.}^2\right)\left(\frac{36 \text{ in.}}{2} - 2\tfrac{1}{2} \text{ in.}\right)^2 + 3\left(0.79 \text{ in.}^2\right)\left(\frac{36 \text{ in.}}{2} - 2\tfrac{1}{2} \text{ in.}\right)^2$$
$$= 1{,}140 \text{ in.}^4$$

The gross area of the composite member is:

$$A_g = (36 \text{ in.})(36 \text{ in.})$$
$$= 1{,}300 \text{ in.}^2$$

The area of concrete is:

$$A_c = A_g - A_s - A_{sr}$$
$$= 1{,}300 \text{ in.}^2 - 62.0 \text{ in.}^2 - 6.32 \text{ in.}^2$$
$$= 1{,}230 \text{ in.}^2$$

The moment of inertia of the concrete section about the elastic neutral axis of the composite section is:

$$I_{cx} = I_{gx} - I_{sx} - I_{srx}$$
$$= \frac{(36 \text{ in.})(36 \text{ in.})^3}{12} - 2{,}660 \text{ in.}^4 - 1{,}140 \text{ in.}^4$$
$$= 136{,}000 \text{ in.}^4$$

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$$I_{cy} = I_{gy} - I_{sy} - I_{sry}$$
$$= \frac{(36\text{ in.})(36\text{ in.})^3}{12} - 1{,}030\text{ in.}^4 - 1{,}140\text{ in.}^4$$
$$= 138{,}000\text{ in.}^4$$

Determine the required strengths

The required strengths are determined using the direct analysis method of design. Second-order analysis is performed using the approximate methods of AISC *Specification* Appendix 8.

The flexural stiffness used for calculating the required strength for bending about the *y*-axis, EI^*, is determined according to AISC *Specification* Section I1.5.

Calculate the modulus of elasticity of concrete.

$$E_c = w_c^{1.5}\sqrt{f'_c}$$
$$= (145\text{ lb/ft}^3)^{1.5}\sqrt{10\text{ ksi}}$$
$$= 5{,}520\text{ ksi}$$

Calculate the effective stiffness of the composite section according to AISC *Specification* Section I2.1b.

$$C_1 = 0.25 + 3\left(\frac{A_s + A_{sr}}{A_g}\right) \le 0.7 \qquad \text{(}Spec.\text{ Eq. I2-6)}$$
$$= 0.25 + 3\left(\frac{62.0\text{ in.}^2 + 6.32\text{ in.}^2}{1{,}300\text{ in.}^2}\right) \le 0.7$$
$$= 0.408 < 0.7$$
$$= 0.408$$

$$(EI)_{eff} = E_s I_{sy} + E_s I_{sry} + C_1 E_c I_{cy} \qquad \text{(from } Spec.\text{ Eq. I2-5)}$$
$$= (29{,}000\text{ ksi})(1{,}030\text{ in.}^4) + (29{,}000\text{ ksi})(1{,}140\text{ in.}^4)$$
$$+ (0.408)(5{,}520\text{ ksi})(138{,}000\text{ in.}^4)$$
$$= 374{,}000{,}000\text{ kip-in.}^2$$

A stiffness reduction factor of $0.8\tau_b$ is applied to the flexural stiffnesses of the composite column according to AISC *Specification* Section C2.3. The stiffness reduction parameter, τ_b, is taken as 0.8 according to AISC *Specification* Section I1.5.

$$EI^* = 0.8\tau_b (EI)_{eff}$$
$$= 0.8(0.8)(374{,}000{,}000\text{ kip-in.}^2)$$
$$= 239{,}000{,}000\text{ kip-in.}^2$$

The first-order axial force and first-order moment of the composite column with the structure restrained against lateral translation are:

LRFD	ASD
$P_{nt} = 1{,}000$ kips $M_{nt} = 0$ kip-ft	$P_{nt} = 650$ kips $M_{nt} = 0$ kip-ft

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The first-order axial force and first-order moment of the composite column due to lateral translation are:

LRFD	ASD
$P_{lt} = 0$ kips $M_{lt} = H_u L$ $= (190 \text{ kips})(15 \text{ ft})$ $= 2{,}850$ kip-ft	$P_{lt} = 0$ kips $M_{lt} = H_a L$ $= (130 \text{ kips})(15 \text{ ft})$ $= 1{,}950$ kip-ft

The second-order axial and flexural strengths are determined using the approximate second-order elastic analysis procedure from AISC *Specification* Appendix Section 8.1.

Calculate the B_1 multiplier according to AISC Appendix *Specification* Appendix Section 8.1.2.

$$B_1 = \frac{C_m}{1 - \alpha P_r / P_{e1}} \geq 1 \qquad \text{(}Spec\text{. Eq. A-8-3)}$$

Because the moment at the top of the composite column is zero, the term M_1/M_2 is zero.

$$C_m = 0.6 - 0.4(M_1/M_2) \qquad \text{(}Spec\text{. Eq. A-8-4)}$$
$$= 0.6 - 0.4(0)$$
$$= 0.6$$

$$P_{e1} = \frac{\pi^2 EI^*}{(L_{c1})^2} \qquad \text{(}Spec\text{. Eq. A-8-5)}$$
$$= \frac{\pi^2 (239{,}000{,}000 \text{ kip-in.}^2)}{(180 \text{ in.})^2}$$
$$= 72{,}800 \text{ kips}$$

Calculate B_1 using AISC *Specification* Equation A-8-3.

LRFD	ASD
$\alpha = 1.0$ $B_1 = \frac{C_m}{1 - \alpha P_u / P_{e1}} \geq 1$ $= \frac{0.6}{1 - (1.0)(1{,}000 \text{ kips})/(72{,}800 \text{ kips})} \geq 1$ $= 0.608 < 1$ $= 1.00$	$\alpha = 1.6$ $B_1 = \frac{C_m}{1 - \alpha P_a / P_{e1}} \geq 1$ $= \frac{0.6}{1 - (1.6)(650 \text{ kips})/(72{,}800 \text{ kips})} \geq 1$ $= 0.609 < 1$ $= 1.00$

Calculate the B_2 multiplier according to AISC *Specification* Appendix Section 8.1.3.

$$B_2 = \frac{1}{1 - \frac{\alpha P_{story}}{P_{e\,story}}} \geq 1 \qquad \text{(}Spec\text{. Eq. A-8-6)}$$

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From Figure 2-10, the total vertical load supported by the story is $2P$.

LRFD	ASD
$P_{story} = 2P_u$ $= 2(1{,}000 \text{ kips})$ $= 2{,}000 \text{ kips}$	$P_{story} = 2P_a$ $= 2(650 \text{ kips})$ $= 1{,}300 \text{ kips}$

From Figure 2-10, the total vertical load in moment-resisting columns, P_{mf}, is P.

LRFD	ASD
$P_{mf} = P_u$ $= 1{,}000 \text{ kips}$	$P_{mf} = P_a$ $= 650 \text{ kips}$

Calculate R_M using AISC *Specification* Equation A-8-8.

LRFD	ASD
$R_M = 1 - 0.15(P_{mf}/P_{story})$ $= 1 - 0.15(1{,}000 \text{ kips}/2{,}000 \text{ kips})$ $= 0.925$	$R_M = 1 - 0.15(P_{mf}/P_{story})$ $= 1 - 0.15(650 \text{ kips}/1{,}300 \text{ kips})$ $= 0.925$

From Figure 2-10, the composite column is fixed at the base and free at the top; therefore, the first order interstory drift can be computed as:

LRFD	ASD
$H_u = 190 \text{ kips}$ $L = (15 \text{ ft})(12 \text{ in./ft})$ $= 180 \text{ in.}$ $\Delta_H = \dfrac{H_u L^3}{3EI^*}$ $= \dfrac{(190 \text{ kips})(180 \text{ in.})^3}{3(239{,}000{,}000 \text{ kip-in.}^2)}$ $= 1.55 \text{ in.}$	$H_a = 130 \text{ kips}$ $L = (15 \text{ ft})(12 \text{ in./ft})$ $= 180 \text{ in.}$ $\Delta_H = \dfrac{H_a L^3}{3EI^*}$ $= \dfrac{(130 \text{ kips})(180 \text{ in.})^3}{3(239{,}000{,}000 \text{ kip-in.}^2)}$ $= 1.06 \text{ in.}$

Calculate the elastic critical buckling strength for the story, $P_{e\ story}$, using AISC *Specification* Equation A-8-7.

LRFD	ASD
$P_{e\ story} = R_M \dfrac{H_u L}{\Delta_H}$ $= (0.925)\dfrac{(190 \text{ kips})(180 \text{ in.})}{1.55 \text{ in.}}$ $= 21{,}400 \text{ kips}$	$P_{e\ story} = R_M \dfrac{H_a L}{\Delta_H}$ $= (0.925)\dfrac{(130 \text{ kips})(180 \text{ in.})}{1.06 \text{ in.}}$ $= 20{,}400 \text{ kips}$

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Calculate B_2 using AISC *Specification* Equation A-8-6.

LRFD	ASD
$\alpha = 1.0$	$\alpha = 1.6$
$B_2 = \dfrac{1}{1-\dfrac{\alpha P_{story}}{P_{e\,story}}} \geq 1$	$B_2 = \dfrac{1}{1-\dfrac{\alpha P_{story}}{P_{e\,story}}} \geq 1$
$= \dfrac{1}{1-\dfrac{1.0(2{,}000\text{ kips})}{21{,}400\text{ kips}}} \geq 1.0$	$= \dfrac{1}{1-\dfrac{1.6(1{,}300\text{ kips})}{20{,}400\text{ kips}}} \geq 1.0$
$= 1.11 > 1.0$	$= 1.11 > 1.0$
$= 1.11$	$= 1.11$

Because the load combination under consideration is not a gravity-only load combination and $B_2 \leq 1.7$, the notional load is not required according to AISC *Specification* Section C2.2b(d).

The required flexural strength is calculated using AISC *Specification* Equation A-8-1.

LRFD	ASD
$M_{uy} = B_1 M_{nt} + B_2 M_{lt}$	$M_{ay} = B_1 M_{nt} + B_2 M_{lt}$
$= 1.00(0\text{ kip-ft}) + 1.11(2{,}850\text{ kip-ft})$	$= 1.00(0\text{ kip-ft}) + 1.11(1{,}950\text{ kip-ft})$
$= 3{,}160\text{ kip-ft}$	$= 2{,}160\text{ kip-ft}$

The required axial strength is calculated using AISC *Specification* Equation A-8-2.

LRFD	ASD
$P_u = P_{nt} + B_2 P_{lt}$	$P_a = P_{nt} + B_2 P_{lt}$
$= (1{,}000\text{ kips}) + 1.11(0\text{ kips})$	$= (650\text{ kips}) + 1.11(0\text{ kips})$
$= 1{,}000\text{ kips}$	$= 650\text{ kips}$

Assess the limitations of AISC Specification *Section I2.1a*

AISC *Specification* Section I2.1a(a) requires that the cross-sectional area of the steel core comprise at least 1% of the total composite cross section.

$$\frac{A_s}{A_g} = \frac{62.0\text{ in.}^2}{1{,}300\text{ in.}^2}$$
$$= 4.77\% > 1\% \quad \textbf{o.k.}$$

The area of the steel core exceeds the minimum requirement of 1%, and the requirement of AISC *Specification* Section I2.1a(a) is met.

AISC *Specification* Section I2.1a(b) requires that (1) the concrete encasement of the steel core be reinforced with continuous longitudinal bars and transverse reinforcement; (2) the detailing and placement of longitudinal reinforcement conforms to the requirements of ACI 318; (3) transverse reinforcement consist of a minimum of either a No. 3 bar spaced at a maximum of 12 in. on center or a No. 4 bar or larger spaced at a maximum of 16 in. on center; and (4) the maximum spacing of transverse reinforcement not exceed 0.5 times the smaller column dimension.

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The problem statement notes that the member has eight No. 8 longitudinal reinforcing bars and No. 4 lateral ties spaced at 16 in. on center. The detail shown in Figure 2-9 illustrates the bar placement, cover, tie layout, and tie spacing.

The section meets the requirements of AISC *Specification* Section I2.1a(b).

AISC *Specification* Section I2.1a(c) requires that the reinforcement ratio for continuous longitudinal reinforcement, ρ_{sr}, exceed 0.4%.

$$\rho_{sr} = \frac{A_{sr}}{A_g} \qquad \text{(Spec. Eq. I2-1)}$$

$$= \frac{6.32 \text{ in.}^2}{1{,}300 \text{ in.}^2}$$

$$= 0.486\% > 0.4\% \quad \textbf{o.k.}$$

This satisfies the minimum required reinforcement percentage of 0.4%.

AISC *Specification* Section I2.1a(d) requires that the reinforcement ratio for continuous longitudinal reinforcement, ρ_{sr}, not exceed 8%, in accordance with ACI 318, Section 10.6.1.1.

$$\rho_{sr} = 0.486\% < 8\% \quad \textbf{o.k.}$$

This satisfies the maximum permitted reinforcement ratio of ACI 318.

All the requirements of AISC *Specification* Section I2.1a have been met.

Determine the available axial compressive strength

Calculate the nominal axial compressive strength without consideration of length effects.

$$P_{no} = F_y A_s + F_{ysr} A_{sr} + 0.85 f'_c A_c \qquad \text{(from Spec. Eq. I2-7)}$$

$$= (50 \text{ ksi})(62.0 \text{ in.}^2) + (60 \text{ ksi})(6.32 \text{ in.}^2) + 0.85(10 \text{ ksi})(1{,}230 \text{ in.}^2)$$

$$= 13{,}900 \text{ kips}$$

Calculate the nominal compressive strength. Buckling about the *y*-axis will control.

$$P_e = \frac{\pi^2 (EI)_{eff}}{L_c^2} \qquad \text{(Spec. Eq. I2-4)}$$

$$= \frac{\pi^2 (374{,}000{,}000 \text{ kip-in.}^2)}{(180 \text{ in.})^2}$$

$$= 114{,}000 \text{ kips}$$

$$\frac{P_{no}}{P_e} = \frac{13{,}900 \text{ kips}}{114{,}000 \text{ kips}}$$

$$= 0.122$$

Because $P_{no}/P_e \le 2.25$, use AISC *Specification* Equation I2-2.

$$P_n = P_{no}\left(0.658^{\frac{P_{no}}{P_e}}\right) \qquad \text{(Spec. Eq. I2-2)}$$

$$= (13{,}900 \text{ kips})(0.658^{0.122})$$

$$= 13{,}200 \text{ kips}$$

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The available compressive strength is:

LRFD	ASD
$\phi_c = 0.75$ $\phi_c P_n = 0.75(13{,}200 \text{ kips})$ $= 9{,}900 \text{ kips}$	$\Omega_c = 2.00$ $\frac{P_n}{\Omega_c} = \frac{13{,}200 \text{ kips}}{2.00}$ $= 6{,}600 \text{ kips}$

Assess the limitations of AISC Specification *Section I3.3a*

AISC *Specification* Section I3.3a(a) is not a limitation, but rather requirements for calculating the available flexural strength.

AISC *Specification* Sections I3.3a(b), I3.3a(c), and I3.3a(d) were previously checked when calculating the available flexural strength.

AISC *Specification* Sections I3.3a(e) does not apply to this member because it is a column. However, the limitation will be checked after M_n has been calculated as an example of how to evaluate if an encased composite member is tension controlled.

The requirements of AISC *Specification* Section I3.3a have been met.

Determine the available flexural strength

The composite member has steel anchors to transfer load between the steel and concrete; therefore, the flexural strength will be determined using the plastic stress distribution method on the composite section. This moment is determined as M_B using AISC *Manual* Table 6-2b:

$$M_B = M_D - F_y Z_{sn} - 0.85 f_c' \left(\frac{Z_{cn}}{2}\right)$$

First, calculate M_D using the equation from AISC *Manual* Table 6-2b.

$$M_D = F_y Z_s + F_{ysr} Z_r + 0.85 f_c' \left(\frac{Z_c}{2}\right)$$

Calculate Z_r and Z_c.

AISC *Manual* Table 6-2b provides equations for encased composite members with reinforcing bars located a distance $h_2/2 - c$ in the x-direction from the centerline of the cross section (i.e., the y-axis). The reinforcing bars located on the y-axis will be neglected for this calculation. Therefore, for this calculation, A_{sr} is taken as the area of six No. 8 bars.

$$A_{sr} = 6(0.79 \text{ in.}^2)$$
$$= 4.74 \text{ in.}^2$$

Note, the composite column program described in Appendix A can account for more general bar configurations, including all reinforcing bars in this example.

$$Z_r = A_{sr}\left(\frac{h_2}{2} - c\right)$$
$$= (4.74 \text{ in.}^2)\left(\frac{36 \text{ in.}}{2} - 2\text{½ in.}\right)$$
$$= 73.5 \text{ in.}^3$$

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$$Z_c = \frac{h_1 h_2^2}{4} - Z_s - Z_r$$
$$= \frac{(36\text{ in.})(36\text{ in.})^2}{4} - 198\text{ in.}^3 - 73.5\text{ in.}^3$$
$$= 11{,}400\text{ in.}^3$$

Calculate M_D.

$$M_D = F_y Z_s + F_{ysr} Z_r + 0.85 f_c' \left(\frac{Z_c}{2}\right)$$
$$= (50\text{ ksi})(198\text{ in.}^3) + (60\text{ ksi})(73.5\text{ in.}^3) + 0.85(10\text{ ksi})\left(\frac{11{,}400\text{ in.}^3}{2}\right)$$
$$= 62{,}800\text{ kip-in.}$$

Next, compute h_n assuming that the plastic neutral axis (PNA) is above the flange of the steel shape (i.e., $h_n > b_f/2$).

$$h_n = \frac{0.85 f_c'(A_c + A_s) - 2F_y A_s}{2(0.85 f_c' h_1)}$$
$$= \frac{0.85(10\text{ ksi})(1{,}230\text{ in.}^2 + 62.0\text{ in.}^2) - 2(50\text{ ksi})(62.0\text{ in.}^2)}{2\left[0.85(10\text{ ksi})(36\text{ in.})\right]}$$
$$= 7.81\text{ in.}$$

$$\frac{b_f}{2} = \frac{15.8\text{ in.}}{2}$$
$$= 7.90\text{ in.}$$

The assumption that the PNA is above the flange of the steel shape is incorrect because $h_n < b_f/2$. Compute h_n assuming that the PNA is within the flange of the steel shape (i.e., $t_w/2 < h_n \le b_f/2$).

$$h_n = \frac{0.85 f_c'(A_c + A_s - 2t_f b_f) - 2F_y(A_s - 2t_f b_f)}{2\left[4F_y t_f + 0.85 f_c'(h_1 - 2t_f)\right]}$$
$$= \frac{0.85(10\text{ ksi})\left[1{,}230\text{ in.}^2 + 62.0\text{ in.}^2 - 2(1.56\text{ in.})(15.8\text{ in.})\right] - 2(50\text{ ksi})\left[62.0\text{ in.}^2 - 2(1.56\text{ in.})(15.8\text{ in.})\right]}{2\left\{4(50\text{ ksi})(1.56\text{ in.}) + 0.85(10\text{ ksi})\left[36\text{ in.} - 2(1.56\text{ in.})\right]\right\}}$$
$$= 7.86\text{ in.}$$

$$\frac{t_w}{2} = \frac{0.980\text{ in.}}{2}$$
$$= 0.490\text{ in.}$$

0.490 in. < 7.86 in. < 7.90 in. **o.k.**

The assumption that the PNA is within the flange of the steel shape is correct because $t_w/2 < h_n \le b_f/2$.

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Calculate Z_{sn} and Z_{cn}.

$$Z_{sn} = Z_s - 2t_f\left(\frac{b_f}{2} + h_n\right)\left(\frac{b_f}{2} - h_n\right)$$

$$= 198 \text{ in.}^3 - 2(1.56 \text{ in.})\left(\frac{15.8 \text{ in.}}{2} + 7.86 \text{ in.}\right)\left(\frac{15.8 \text{ in.}}{2} - 7.86 \text{ in.}\right)$$

$$= 196 \text{ in.}^3$$

$$Z_{cn} = h_1 h_n^2 - Z_{sn}$$

$$= (36 \text{ in.})(7.86 \text{ in.})^2 - 196 \text{ in.}^3$$

$$= 2{,}030 \text{ in.}^3$$

Finally, calculate M_B.

$$M_B = M_D - F_y Z_{sn} - 0.85 f_c'\left(\frac{Z_{cn}}{2}\right)$$

$$= 62{,}800 \text{ kip-in.} - (50 \text{ ksi})(196 \text{ in.}^3) - 0.85(10 \text{ ksi})\left(\frac{2{,}030 \text{ in.}^3}{2}\right)$$

$$= 44{,}400 \text{ kip-in.}$$

$$M_n = M_B$$

$$= (44{,}400 \text{ kip-in.})\left(\frac{1 \text{ ft}}{12 \text{ in.}}\right)$$

$$= 3{,}700 \text{ kip-ft}$$

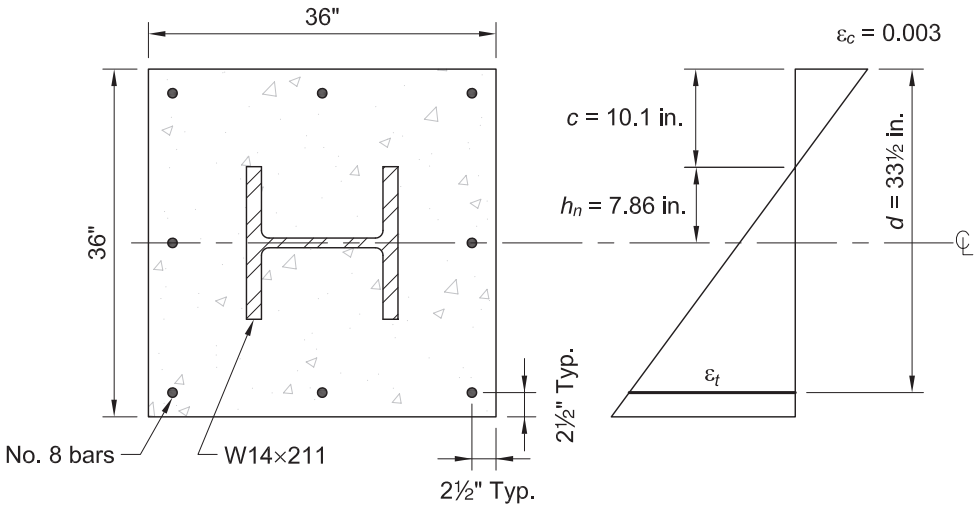


Fig. 2-11. Strain diagram for evaluating if the encased composite member is tension controlled.

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The available flexural strength is:

LRFD	ASD
$\phi_b = 0.90$ $\phi_b M_n = 0.90(3{,}700 \text{ kip-ft})$ $= 3{,}300 \text{ kip-ft}$	$\Omega_b = 1.67$ $\frac{M_n}{\Omega_b} = \frac{3{,}700 \text{ kip-ft}}{1.67}$ $= 2{,}200 \text{ kip-ft}$

AISC *Specification* Section I3.3a(e) does not apply to this member because it is a column, but the evaluation of whether the member is tension controlled as defined in ACI 318 will still be made as an example of this determination. A member is tension controlled if the tensile strain in the steel is greater than or equal to the yield strain plus 0.003. The tensile strain is computed using the neutral axis location, h_n, as shown in Figure 2-11.

$$\varepsilon_t = \frac{d-c}{c}\varepsilon_c$$
$$= \frac{(33\text{½ in.}) - (10.1 \text{ in.})}{(10.1 \text{ in.})}(0.003)$$
$$= 0.00695$$

$$\varepsilon_{ty} + 0.003 = \frac{F_{ysr}}{E_s} + 0.003$$
$$= \frac{60 \text{ ksi}}{29{,}000 \text{ ksi}} + 0.003$$
$$= 0.00507$$

Because $\varepsilon_t \geq \varepsilon_{ty} + 0.003$, the member is tension controlled.

Check combined flexure and axial compression strength

The interaction strength is determined using the transformed cross-sectional strength method described in Section 2.3.2 of this Design Guide.

The axial compression at Point C needs to be determined according to Table 2-3. The cross-sectional axial compression strength at Point C, P_C, is determined using AISC *Manual* Table 6-2b.

$$P_C = 0.85 f'_c A_c$$
$$= 0.85(10 \text{ ksi})(1{,}230 \text{ in.}^2)$$
$$= 10{,}500 \text{ kips}$$

From Table 2-3, the available compressive strength at Point C is:

LRFD	ASD
$\phi_c = 0.75$ $\phi_c \chi P_C = \phi_c \frac{P_n}{P_{no}} P_C$ $= 0.75\left(\frac{13{,}200 \text{ kips}}{13{,}900 \text{ kips}}\right)(10{,}500 \text{ kips})$ $= 7{,}480 \text{ kips}$	$\Omega_c = 2.00$ $\frac{\chi P_C}{\Omega_c} = \frac{P_n}{P_{no}}\frac{P_C}{\Omega_c}$ $= \left(\frac{13{,}200 \text{ kips}}{13{,}900 \text{ kips}}\right)\left(\frac{10{,}500 \text{ kips}}{2.00}\right)$ $= 4{,}990 \text{ kips}$

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The available strengths for anchor points A, B, and C are:

LRFD	ASD
$P_c = \phi_c P_n$ $= 9{,}900$ kips $P_{cc} = \phi_c \chi P_C$ $= 7{,}480$ kips $M_{cy} = \phi_b M_n$ $= 3{,}330$ kip-ft	$P_c = P_n/\Omega_c$ $= 6{,}600$ kips $P_{cc} = \chi P_C/\Omega_c$ $= 4{,}990$ kips $M_{cy} = M_n/\Omega_b$ $= 2{,}220$ kip-ft

Evaluate the interaction equation.

LRFD	ASD
Compare P_u and P_{cc} $P_u = 1{,}000 \text{ kips} < P_{cc} = 7{,}480 \text{ kips}$ Therefore, use Equation 2-2a, which, simplified for uniaxial bending, is: $\frac{M_{uy}}{M_{cy}} = \frac{3{,}160 \text{ kip-ft}}{3{,}330 \text{ kip-ft}}$ $= 0.949 \le 1$ **o.k.**	Compare P_a and P_{cc} $P_a = 650 \text{ kips} < P_{cc} = 4{,}990 \text{ kips}$ Therefore, use Equation 2-2a, which, simplified for uniaxial bending, is: $\frac{M_{ay}}{M_{cy}} = \frac{2{,}160 \text{ kip-ft}}{2{,}220 \text{ kip-ft}}$ $= 0.973 \le 1$ **o.k.**

Determine the available shear strength

Three options for computing the available shear strength are presented in AISC *Specification* Section I4.1. This example uses the option where the available shear strength is computed as that of the steel section alone as specified in AISC *Specification* Chapter G.

Calculate the nominal shear strength using $C_{v2} = 1.0$ according to the User Note in AISC *Specification* Section G6.

$$V_n = 0.6 F_y b_f t_f C_{v2} \qquad \text{(Spec. Eq. G6-1)}$$
$$= 0.6(50 \text{ ksi})(15.8 \text{ in.})(1.56 \text{ in.})(1.0)$$
$$= 739 \text{ kips/flange}$$

The available shear strength is:

LRFD	ASD
$\phi_v = 0.90$ $\phi_v V_n = 0.90(2 \text{ flanges})(739 \text{ kips/flange})$ $= 1{,}330 \text{ kips} > 190 \text{ kips}$ **o.k.**	$\Omega_v = 1.67$ $\frac{V_n}{\Omega_v} = \frac{(2 \text{ flanges})(739 \text{ kips/flange})}{1.67}$ $= 885 \text{ kips} > 130 \text{ kips}$ **o.k.**

Conclusion

The encased composite column is found to be adequate for the applied loads.

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2.5 AVAILABLE STRENGTH OF RECTANGULAR AND SQUARE FILLED COMPOSITE MEMBERS

2.5.1 Limitations

Limitations for filled composite members appear in AISC *Specification* Section I2.2a for members subjected to axial force and AISC *Specification* Section I3.4a for members subjected to flexure. Members subjected to combined flexure and axial force must meet both sets of limitations.

The two sets of limitations are similar. Both stipulate a minimum cross-sectional area of the steel shape (i.e., 1% of the total composite cross section). Both also state that internal reinforcement is not required. However, if longitudinal reinforcement is provided, a minimum amount of transverse reinforcement must be provided.

For members subjected to axial force, if longitudinal reinforcement is provided, it must comply with the maximum reinforcement ratio for continuous longitudinal reinforcement requirement in ACI 318.

For members subjected to flexure, if longitudinal reinforcement is provided, the reinforcing ratio must be at least 0.4%. Also, filled composite members serving as beams and subjected to low axial loads (i.e., $P_u < 0.10P_n$) must be classified as tension controlled as defined in ACI 318. The requirement that flexural members be tension controlled is to prevent brittle failure caused by concrete crushing. The concrete of filled composite members is well confined, and brittle failures like those observed in over-reinforced concrete beams have not been observed for filled composite members. See Section 2.4.1 of this Design Guide for practical guidance on how to assess this limitation.

The longitudinal and transverse reinforcing requirements are in addition to the requirements in ACI 318, as stated in AISC *Specification* Section I1.1.

2.5.2 Classification of Composite Sections

Filled composite members are required to be classified for local buckling according to AISC *Specification* Section I1.4. Sections are classified as compact composite, noncompact composite, or slender composite. For a section to be compact composite, the width-to-thickness ratio, λ, of all elements in the section must be less than or equal to λ_p as determined from Table 2-5 (based on AISC *Specification* Tables I1.1a and I1.1b). For a section to be noncompact composite, it must not be classified as compact composite, and the width-to-thickness ratio, λ, of all elements in the section must be less than or equal to λ_r as determined from Table 2-5. All other sections are classified as slender composite. However, note that if any width-to-thickness ratio exceeds the maximum permitted width-to-thickness ratio from Table 2-5, then the section is not permitted.

For members subjected to combined flexure and axial load, the local buckling classifications can be different for flexure and axial load.

For HSS, the dimension b is the clear distance between webs less the inside corner radius on each side and the dimension h is the clear distance between the flanges less the inside corner radius on each side. Values of b/t and h/t are listed in AISC *Manual* Tables 1-11 and 1-12 . For box sections, the dimensions b and h are the clear distance between the elements providing stiffening. Zhou et al. (2023) confirm that stiffeners such as those shown in Figure 2-12 mitigate local buckling and suggest limits for dimensions of stiffening plates.

The use of the terms "compact composite," "noncompact composite," and "slender composite" in the AISC *Specification* is intended to distinguish the classification of composite members from that of structural steel members. The behavior assumed in design for a noncompact composite section, for example, is different from that of a noncompact element of a structural steel member.

2.5.3 Axial Compressive Strength

The axial compressive strength of filled composite members is defined in AISC *Specification* Section I2.2b. The nominal axial compressive strength without consideration of length effects, P_{no}, is computed based on the classification of the section. For compact composite sections, P_{no} is equal to the plastic axial compressive strength, P_p, as shown in AISC *Specification* Equation I2-9a.

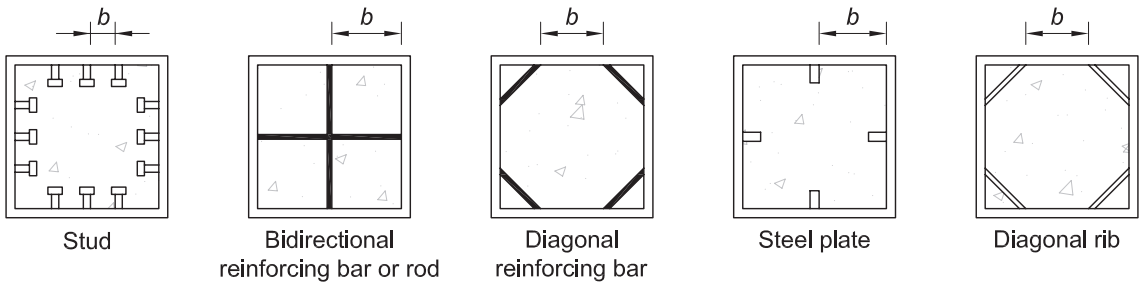


Fig. 2-12. Examples of stiffened rectangular filled composite members.

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Table 2-5. Limiting Width-to-Thickness Ratios for Compression Steel Elements in Rectangular Filled Composite Members				
Description of Element	**Width-to-Thickness Ratio**	**λ_p Compact Composite/ Noncompact Composite**	**λ_r Noncompact Composite/ Slender Composite**	**Maximum Permitted**
Walls of rectangular HSS and box sections in members subjected to axial compression	b/t	$2.26\sqrt{\frac{E}{F_y}}$	$3.00\sqrt{\frac{E}{F_y}}$	$5.00\sqrt{\frac{E}{F_y}}$
Flanges of rectangular HSS and box sections in members subjected to flexure				
Webs of rectangular HSS and box sections in members subjected to flexure	h/t	$3.00\sqrt{\frac{E}{F_y}}$	$5.70\sqrt{\frac{E}{F_y}}$	$5.70\sqrt{\frac{E}{F_y}}$

$$P_{no} = P_p \qquad \text{(}Spec\text{. Eq. I2-9a)}$$

$$P_p = F_y A_s + 0.85 f_c' \left(A_c + A_{sr} \frac{E_s}{E_c} \right)$$

(from *Spec*. Eq. I2-9b)

This equation is based on AISC *Specification* Equation I2-9b but with C_2 replaced with the value of 0.85 for rectangular sections.

For noncompact composite sections, P_{no} varies between P_p and P_y as shown in AISC *Specification* Equation I2-9c.

$$P_{no} = P_p - \frac{P_p - P_y}{(\lambda_r - \lambda_p)^2} (\lambda - \lambda_p)^2$$

(*Spec*. Eq. I2-9c)

where

P_y = yield axial compressive strength, kips

λ = the largest width-to-thickness ratio of the section

$\lambda_p = 2.26\sqrt{E/F_y}$ from Table 2-5 and AISC *Specification* Table I1.1a

$\lambda_r = 3.00\sqrt{E/F_y}$ from Table 2-5 and AISC *Specification* Table I1.1a

$$P_y = F_y A_s + 0.7 f_c' \left(A_c + A_{sr} \frac{E_s}{E_c} \right)$$

(*Spec*. Eq. I2-9d)

For slender composite sections, P_{no} is computed as shown in AISC *Specification* Equation I2-9e.

$$P_{no} = F_n A_s + 0.7 f_c' \left(A_c + A_{sr} \frac{E_s}{E_c} \right)$$

(*Spec*. Eq. I2-9e)

$$F_n = \frac{9 E_s}{\lambda^2} \qquad \text{(}Spec\text{. Eq. I2-10)}$$

The nominal axial compressive strength (with consideration of member length effects) for the limit state of flexural buckling, P_n, is computed as shown in AISC *Specification* Equations I2-2 and I2-3. Note that the column curve defined in AISC *Specification* Equations I2-2 and I2-3 is identical to the column curve for structural steel members defined in AISC *Specification* Section E3 if $P_{no} = A_s F_y$ and $(EI)_{eff} = E_s I_s$.

When $\frac{P_{no}}{P_e} \le 2.25$

$$P_n = P_{no} \left(0.658^{\frac{P_{no}}{P_e}} \right) \qquad \text{(}Spec\text{. Eq. I2-2)}$$

When $\frac{P_{no}}{P_e} > 2.25$

$$P_n = 0.877 P_e \qquad \text{(}Spec\text{. Eq. I2-3)}$$

where P_e is the elastic critical buckling load determined from AISC *Specification* Equation I2-4; as specified in AISC *Specification* Appendix 7, Section 7.2.3(b), when using the effective length method; or through an elastic buckling analysis.

$$P_e = \frac{\pi^2 (EI)_{eff}}{L_c^2} \qquad \text{(}Spec\text{. Eq. I2-4)}$$

where $(EI)_{eff}$ is the effective stiffness of the composite section determined from AISC *Specification* Equation I2-12 and L_c is the effective length of the member.

The effective stiffness of the composite section, $(EI)_{eff}$, is computed by summing the contributions of each of the components of the cross section, but with a reduction factor, C_3, applied to the concrete contribution to account for the level

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of cracking expected at the axial compressive strength of the member.

$$(EI)_{eff} = E_s I_s + E_s I_{sr} + C_3 E_c I_c$$

(*Spec*. Eq. I2-12)

$$C_3 = 0.45 + 3\left(\frac{A_s + A_{sr}}{A_g}\right) \le 0.9$$

(*Spec*. Eq. I2-13)

The critical buckling load, P_e, should be computed for each coordinate axis and the least value used to compute P_n. Torsional buckling does not apply to filled composite members because of their high torsional stiffness.

The design compressive strength is determined using a resistance factor of $\phi_c = 0.75$ for LRFD. The allowable compressive strength is determined using a safety factor of $\Omega_c = 2.00$ for ASD. The design compressive strength need not be less than the design compressive strength of the bare steel member. The design compressive strength of the bare steel member can be greater than that of the composite member when the steel ratio is high and the contribution of the concrete is not enough to overcome the lower resistance factor or higher safety factor used for composite members.

2.5.4 Axial Tensile Strength

The axial tension strength of filled composite members is defined in AISC *Specification* Section I2.2c. The nominal tensile strength, P_n, is for the limit state of yielding as shown in AISC *Specification* Equation I2-14. Only the structural steel and longitudinal reinforcement contributions are included in this equation because the concrete tension strength is negligible.

$$P_n = A_s F_y + A_{sr} F_{ysr} \qquad (Spec. \text{ Eq. I2-14})$$

The design tensile strength is determined using a resistance factor of $\phi_t = 0.90$ for LRFD. The allowable tensile strength is determined using a safety factor of $\Omega_t = 1.67$ for ASD.

2.5.5 Flexural Strength

The flexural strength of filled composite members is defined in AISC *Specification* Section I3.4b. Unlike encased composite members, direct bond interaction between the steel and concrete is sufficient to develop the composite action necessary to achieve the full flexural strength of the composite section; therefore, the determination of the flexural strength is not dependent on whether steel anchors or other supplementary shear transfer mechanisms are provided. However, due to the possibility of local buckling, the determination of flexural strength is dependent on the classification of the composite section. Classification of rectangular filled composite members is discussed in Section 2.5.2 of this Design Guide.

For compact composite sections, the nominal flexural strength, M_n, is equal to the plastic moment strength, M_p, determined as the moment corresponding to a plastic stress distribution over the composite cross section. Closed-form equations for the flexural strength of rectangular filled composite members are provided in AISC *Manual* Table 6-3. The flexural strength is denoted as M_B in this table. The table is reproduced in this Design Guide as Figure 2-13. The table does not consider internal reinforcement. The composite column program that accompanies this Design Guide can compute the flexural strength considering internal reinforcement. The strain compatibility method and elastic stress distribution methods can conservatively be used in lieu of the plastic stress distribution moment for determining M_n. The effective stress-strain method may also be used with appropriately calibrated effective stress-strain relationships (e.g., Sakino et al., 2004; Han et al., 2005; Liang, 2009; Lai and Varma, 2016).

For noncompact composite sections, the nominal flexural strength, M_n, is determined as shown in AISC *Specification* Equation I3-5b.

$$M_n = M_p - (M_p - M_y)\left(\frac{\lambda - \lambda_p}{\lambda_r - \lambda_p}\right)$$

(*Spec*. Eq. I3-5b)

In AISC *Specification* Equation I3-5b, M_y is the yield moment corresponding to yielding of the tension flange and first yield of the compression flange. The capacity at first yield is calculated assuming a linear elastic stress distribution with the maximum concrete compressive stress limited to $0.7f_c'$ and the maximum steel stress limited to F_y. The width-to-thickness ratio, λ, and corresponding limits, λ_p and λ_r, are listed in AISC *Specification* Table I1.1b (Table 2-5 of this Design Guide). The value of the combined term $(\lambda - \lambda_p)/(\lambda_r - \lambda_p)$ should be computed for both the web and flange and the larger value should be used.

For slender composite sections, the nominal flexural strength, M_n, is computed based on assumed stress distributions. The stress in the steel is assumed to vary linearly from F_n (AISC *Specification* Equation I2-10) at the compression flange to F_y at the tension flange. The stress in the concrete varies linearly from zero at the neutral axis (as defined by steel stress distribution) to $0.7f_c'$ at the extreme compression fiber of the concrete.

These assumed stress distributions for calculating M_y or M_n for slender composite members cannot be determined from an elastic stress distribution and can only arise from an elastic stress distribution if the curvatures in the steel and concrete are different. No guidance is provided in the AISC

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Specification or Commentary on how to consider internal reinforcement in these calculations. A logical approach that is consistent with the axial strength provisions would be to back calculate the curvature in the concrete assuming elastic behavior to determine the strains in each reinforcing bar and assign a stress to each bar using the transformed modulus, $0.7f_c'E_s/E_c$.

AISC *Specification* Commentary Figure C-I3.8 shows the stress distributions for a rectangular shape with sharp corners and no internal reinforcement. The composite column program that accompanies this Design Guide can compute the flexural strength considering the rounded corners of HSS shapes and considering internal reinforcement.

The design flexural strength is determined using a resistance factor of $\phi_b = 0.90$ for LRFD. The allowable flexural strength is determined using a safety factor of $\Omega_b = 1.67$ for ASD.

2.5.6 Interaction Strength

Provisions for the interaction between flexural and axial forces for filled composite members are included in AISC *Specification* Section I5. Several methods for assessing interaction strength are applicable, including those described in Section 2.3.2 of this Design Guide.

For compact composite cross sections, assessing interaction strength by transformed cross-sectional strength from the plastic stress distribution method will often be the best approach. Cross-sectional strength can be efficiently computed using closed-form equations in AISC *Manual* Table 6-3 (reproduced in this Design Guide as Figure 2-13) or with the composite column program that accompanies this Design Guide. Cross-sectional strength should be transformed to member strength as described in Section 2.3.2 of this Design Guide.

For noncompact composite and slender composite cross sections, the interaction equations of AISC *Specification* Section I5, described in Section 2.3.2 of this Design Guide, will often be the best approach.

2.5.7 Shear Strength

New in the 2022 edition of the AISC *Specification* are provisions for the available shear strength of filled composite members that combine the contributions of steel and concrete. Based on the results of recent research, AISC *Specification* Section I4.2 permits the calculation of nominal strength of rectangular filled composite members as:

$$V_n = 0.6A_vF_y + 0.06K_cA_c\sqrt{f_c'} \quad \text{(Spec. Eq. I4-1)}$$

where

A_c = area of concrete infill, in.^{2}

A_v = shear area of the steel portion of the composite member equal to the sum of the area of webs in the direction of in-plane shear, in.^{2}

The coefficient $K_c = 1$ if the cross section is not compact composite (AISC *Specification* Tables I1.1a and I1.1b) or if the shear span-to-depth ratio, $(M_r/V_r)/d$, is greater than or equal to 0.7. For the shear span-to-depth ratio, M_r and V_r are the maximum required flexural and shear strengths, respectively, along the member length (they are shown as M_u and V_u in the AISC *Specification* despite applying to both LRFD and ASD), and d is equal to the member depth in the direction of bending. If the cross section is compact composite and the shear span-to-depth ratio is less than 0.5, then $K_c = 10$. For members with compact composite cross sections and a shear span-to-depth ratio between 0.5 and 0.7, K_c can be determined from linear interpolation.

In most cases, $K_c = 1$. The greater value of K_c may be applicable for panel zones or other special situations where the shear span is small enough for a diagonal compression strut to form in the concrete.

Note that the coefficient 0.06 is used in AISC *Specification* Equation I4-1 instead of the more familiar coefficient 2 from ACI 318 because f_c' is in units of ksi instead of psi. Internal transverse reinforcement has been found to have minimal impact on the shear strength of filled composite members (Bruneau et al., 2018).

DESIGN EXAMPLE 2.5—Square Filled Composite Member Subjected to Combined Axial Compression and Flexure—Normal Strength Materials

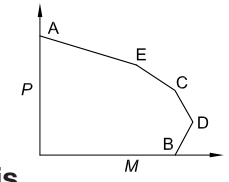
Given:

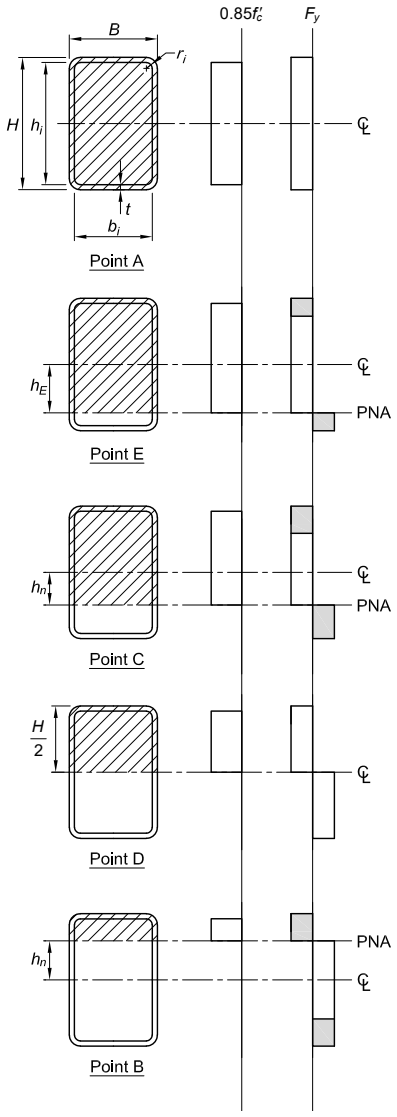
Assess the adequacy of a filled composite column member constructed with the cross section shown in Figure 2-14 and having an effective length $L_{cx} = L_{cy} = 30$ ft for the following required strengths:

LRFD	ASD
$P_u = 1{,}500$ kips	$P_a = 1{,}000$ kips
$M_{ux} = 1{,}800$ kip-ft	$M_{ax} = 1{,}200$ kip-ft
$V_u = 90$ kips	$V_a = 60$ kips

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Table 6-3
Cross-Section Strength for Composite Filled Rectangular HSS
Subject to Flexure about Either Principal Axis



Section	Stress Distribution	Pt.	Defining Equation
		A	$P_A = F_y A_s + 0.85 f'_c A_c$ $M_A = 0$ A_s = area of steel shape, in.^{2} $A_c = b_i h_i - 0.858 r_i^2$ $b_i = B - 2t$ $h_i = H - 2t$ $r_i = t$
		E	$P_E = \frac{0.85 f'_c A_c}{2} + 0.85 f'_c b_i h_E + 4 F_y t h_E$ $M_E = M_D - F_y Z_{sE} - 0.85 f'_c \left(\frac{Z_{cE}}{2}\right)$ $Z_{cE} = b_i h_E^2$ $Z_{sE} = 2 t h_E^2$ $h_E = \frac{h_n}{2} + \frac{H}{4}$
		C	$P_C = 0.85 f'_c A_c$ $M_C = M_B$
		D	$P_D = \frac{0.85 f'_c A_c}{2}$ $M_D = F_y Z_s + \frac{0.85 f'_c Z_c}{2}$ Z_s = full plastic section modulus of HSS, in.^{3} $Z_c = \frac{b_i h_i^2}{4} - 0.429 r_i^2 h_i + 0.192 r_i^3$
		B	$P_B = 0$ $M_B = M_D - F_y Z_{sn} - 0.85 f'_c \left(\frac{Z_{cn}}{2}\right)$ $Z_{sn} = 2 t h_n^2$ $Z_{cn} = b_i h_n^2$ $h_n = \frac{0.85 f'_c A_c}{2\left(0.85 f'_c b_i + 4 F_y t\right)} \le \frac{h_i}{2}$

F_y = specified minimum yield stress of steel shape, ksi
Note: Equations in this table are applicable to single-axis bending of the shape about its *x-x* axis (when $H \ge B$) or about its *y-y* axis (when $B > H$).

Fig. 2-13. AISC Manual *Table 6-3—Axial and flexural interaction strength of rectangular filled composite members based on the plastic stress distribution method.*

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The composite member consists of ASTM A572/A572M Grade 50 (ASTM, 2021) plates welded to form a box section and normal weight ($w_c = 145$ lb/ft^3) concrete fill with a specified compressive strength, $f'_c = 6$ ksi. The member does not include internal reinforcement.

Design of the welds for the box section and assessment of strength during construction when the concrete is wet are excluded from this example.

Solution:

From AISC *Manual* Table 2-5, the material properties are as follows:

ASTM A572/A572M Grade 50
$F_y = 50$ ksi

Calculate cross-sectional properties

Calculate cross-sectional properties of the gross section, the steel shape, and the concrete fill. The short protrusions of steel at each corner are to enable fabrication of the box section and will be neglected in strength calculations.

The gross area of the composite member is:

$$A_g = BH$$
$$= (25 \text{ in.})(25 \text{ in.})$$
$$= 625 \text{ in.}^2$$

The area of concrete is:

$$A_c = b_i h_i$$
$$= (24 \text{ in.})(24 \text{ in.})$$
$$= 576 \text{ in.}^2$$

The cross-sectional area of the structural steel section is:

$$A_s = A_g - A_c$$
$$= 625 \text{ in.}^2 - 576 \text{ in.}^2$$
$$= 49.0 \text{ in.}^2$$

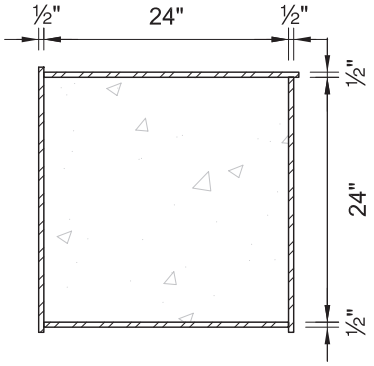


Fig. 2-14. Filled composite member for Examples 2.5 and 2.6.

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The moment of inertia of the concrete section about the elastic neutral axis of the composite section is:

$$I_c = \frac{b_i h_i^3}{12}$$
$$= \frac{(24 \text{ in.})(24 \text{ in.})^3}{12}$$
$$= 27{,}600 \text{ in.}^4$$

The moment of inertia of the steel shape about the elastic neutral axis of the composite section is:

$$I_s = \frac{BH^3}{12} - I_c$$
$$= \frac{(25 \text{ in.})(25 \text{ in.})^3}{12} - 27{,}600 \text{ in.}^4$$
$$= 4{,}950 \text{ in.}^4$$

Assess the limitations of AISC Specification *Section I2.2a*

AISC *Specification* Section I2.2a(a) requires that the cross-sectional area of the structural steel section comprise at least 1% of the total composite cross section.

$$\frac{A_s}{A_g} = \frac{49.0 \text{ in.}^2}{625 \text{ in.}^2}$$
$$= 7.84\% > 1\% \quad \textbf{o.k.}$$

This satisfies the minimum required steel percentage of 1%.

AISC *Specification* Section I2.2a(b) requires that the filled composite section be classified for local buckling according to AISC *Specification* Section I1.4. All four walls of the box section are identical.

$$\lambda = \frac{b}{t}$$
$$= \frac{24 \text{ in.}}{\frac{1}{2} \text{ in.}}$$
$$= 48.0$$

From AISC *Specification* Table I1.1a:

$$\lambda_p = 2.26\sqrt{\frac{E}{F_y}}$$
$$= 2.26\sqrt{\frac{29{,}000 \text{ ksi}}{50 \text{ ksi}}}$$
$$= 54.4$$

Because $\lambda < \lambda_p$, the section is compact composite for compression.

AISC *Specification* Sections I2.2a(c) and I2.2a(d) do not apply to this member because there is no longitudinal reinforcement.

The requirements of AISC *Specification* Section I2.2a have been met.

Determine the available axial compressive strength

Calculate the nominal axial compressive strength without consideration of length effects. For a compact composite section, $P_{no} = P_p$.

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$$P_p = F_y A_s + 0.85 f_c' \left(A_c + A_{sr} \frac{E_s}{E_c} \right) \quad \text{(from } \textit{Spec.} \text{ Eq. I2-9b)}$$

$$= (50 \text{ ksi})(49.0 \text{ in.}^2) + 0.85(6 \text{ ksi})(576 \text{ in.}^2 + 0)$$

$$= 5{,}390 \text{ kips}$$

$$P_{no} = P_p \quad \text{(}\textit{Spec.}\text{ Eq. I2-9a)}$$

$$= 5{,}390 \text{ kips}$$

Calculate the modulus of elasticity of concrete.

$$E_c = w_c^{1.5} \sqrt{f_c'}$$

$$= (145 \text{ lb/ft}^3)^{1.5} \sqrt{6 \text{ ksi}}$$

$$= 4{,}280 \text{ ksi}$$

Calculate the effective stiffness of the composite section.

$$C_3 = 0.45 + 3 \left(\frac{A_s + A_{sr}}{A_g} \right) \le 0.9 \quad \text{(}\textit{Spec.}\text{ Eq. I2-13)}$$

$$= 0.45 + 3 \left(\frac{49.0 \text{ in.}^2 + 0}{625 \text{ in.}^2} \right) \le 0.9$$

$$= 0.685 < 0.9$$

$$= 0.685$$

$$(EI)_{eff} = E_s I_s + E_s I_{sr} + C_3 E_c I_c \quad \text{(}\textit{Spec.}\text{ Eq. I2-12)}$$

$$= (29{,}000 \text{ ksi})(4{,}950 \text{ in.}^4) + 0 + 0.685(4{,}280 \text{ ksi})(27{,}600 \text{ in.}^4)$$

$$= 224{,}000{,}000 \text{ kip-in.}^2$$

Calculate the nominal compressive strength.

$$P_e = \frac{\pi^2 (EI)_{eff}}{L_c^2} \quad \text{(}\textit{Spec.}\text{ Eq. I2-4)}$$

$$= \frac{\pi^2 (224{,}000{,}000 \text{ kip-in.}^2)}{\left[(30 \text{ ft})(12 \text{ in./ft}) \right]^2}$$

$$= 17{,}100 \text{ kips}$$

$$\frac{P_{no}}{P_e} = \frac{5{,}390 \text{ kips}}{17{,}100 \text{ kips}}$$

$$= 0.315$$

Because $P_{no}/P_e \le 2.25$, use AISC *Specification* Equation I2-2.

$$P_n = P_{no} \left(0.658^{\frac{P_{no}}{P_e}} \right) \quad \text{(}\textit{Spec.}\text{ Eq. I2-2)}$$

$$= (5{,}390 \text{ kips})(0.658^{0.315})$$

$$= 4{,}720 \text{ kips}$$

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The available compressive strength is:

LRFD	ASD
$\phi_c = 0.75$ $\phi_c P_n = 0.75(4{,}720 \text{ kips})$ $= 3{,}540 \text{ kips}$	$\Omega_c = 2.00$ $\frac{P_n}{\Omega_c} = \frac{4{,}720 \text{ kips}}{2.00}$ $= 2{,}360 \text{ kips}$

Assess the limitations of AISC Specification *Section I3.4a*

AISC *Specification* Section I3.4a(a) requires that filled composite sections be classified for local buckling according to AISC *Specification* Section I1.4. All four walls of the box section are identical.

$$\lambda = \frac{b}{t}$$
$$= \frac{24 \text{ in.}}{\frac{1}{2} \text{ in.}}$$
$$= 48.0$$

From AISC *Specification* Table I1.1b, the limiting width-to-thickness ratio for the flanges of the box section is:

$$\lambda_p = 2.26\sqrt{\frac{E}{F_y}}$$
$$= 2.26\sqrt{\frac{29{,}000 \text{ ksi}}{50 \text{ ksi}}}$$
$$= 54.4$$

From AISC *Specification* Table I1.1b, the limiting width-to-thickness ratio for the webs of the box section is:

$$\lambda_p = 3.00\sqrt{\frac{E}{F_y}}$$
$$= 3.00\sqrt{\frac{29{,}000 \text{ ksi}}{50 \text{ ksi}}}$$
$$= 72.2$$

Because $\lambda \le \lambda_p$ for the flanges and the webs, the section is compact composite for flexure.

AISC *Specification* Section I3.4a(b) requires that the total cross-sectional area of the structural steel section comprise at least 1% of the total composite cross section.

The structural steel section was previously confirmed to comprise at least 1% of the total composite cross section.

AISC *Specification* Section I3.4a(c) does not apply to this member because there is no longitudinal reinforcement.

AISC *Specification* Section I3.4a(d) does not apply to this member because it is a column.

The requirements of AISC *Specification* Section I3.4a have been met.

Determine the available flexural strength

For compact composite sections, the nominal flexural strength is equal to the moment corresponding to plastic stress distribution over the composite cross section. This moment is determined as M_B using AISC *Manual* Table 6-4. Because the section is a box section, the corner radius, r_i, is taken as zero.

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First, calculate Z_s and Z_c.

$$Z_s = \frac{BH^2}{4} - \frac{b_i h_i^2}{4}$$
$$= \frac{(25\text{ in.})(25\text{ in.})^2}{4} - \frac{(24\text{ in.})(24\text{ in.})^2}{4}$$
$$= 450\text{ in.}^3$$

$$Z_c = \frac{b_i h_i^2}{4}$$
$$= \frac{(24\text{ in.})(24\text{ in.})^2}{4}$$
$$= 3{,}460\text{ in.}^3$$

Calculate M_D using the equation from AISC *Manual* Table 6-4.

$$M_D = F_y Z_s + \frac{0.85 f_c' Z_c}{2}$$
$$= (50\text{ ksi})(450\text{ in.}^3) + \frac{0.85(6\text{ ksi})(3{,}460\text{ in.}^3)}{2}$$
$$= 31{,}300\text{ kip-in.}$$

Calculate h_n, Z_{sn}, *and* Z_{cn} using the equations from AISC *Manual* Table 6-4.

$$h_n = \frac{0.85 f_c' A_c}{2(0.85 f_c' b_i + 4 F_y t)} \le \frac{h_i}{2}$$
$$= \frac{0.85(6\text{ ksi})(576\text{ in.}^2)}{2\left[0.85(6\text{ ksi})(24\text{ in.}) + 4(50\text{ ksi})(½\text{ in.})\right]} \le \frac{24\text{ in.}}{2}$$
$$= 6.60\text{ in.} < 12\text{ in.}$$
$$= 6.60\text{ in.}$$

$$Z_{sn} = 2 t h_n^2$$
$$= 2(½\text{ in.})(6.60\text{ in.})^2$$
$$= 43.6\text{ in.}^3$$

$$Z_{cn} = b_i h_n^2$$
$$= (24\text{ in.})(6.60\text{ in.})^2$$
$$= 1{,}050\text{ in.}^3$$

Calculate M_B using the equation from AISC *Manual* Table 6-4.

$$M_B = M_D - F_y Z_{sn} - 0.85 f_c' \left(\frac{Z_{cn}}{2}\right)$$
$$= 31{,}300\text{ kip-in.} - (50\text{ ksi})(43.6\text{ in.}^3) - 0.85(6\text{ ksi})\left(\frac{1{,}050\text{ in.}^3}{2}\right)$$
$$= 26{,}400\text{ kip-in.}$$

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$$M_n = M_B$$
$$= (26{,}400 \text{ kip-in.})\left(\frac{1 \text{ ft}}{12 \text{ in.}}\right)$$
$$= 2{,}200 \text{ kip-ft}$$

The available flexural strength is:

LRFD	ASD
$\phi_b = 0.90$ $\phi_b M_n = 0.90(2{,}200 \text{ kip-ft})$ $= 1{,}980 \text{ kip-ft}$	$\Omega_b = 1.67$ $\frac{M_n}{\Omega_b} = \frac{2{,}200 \text{ kip-ft}}{1.67}$ $= 1{,}320 \text{ kip-ft}$

Check combined flexure and axial compression strength

Because the section is compact composite for both axial compression and flexure, the interaction strength will be determined using the transformed cross-sectional strength method described in Section 2.3.2 of this Design Guide.

The axial compression at Point C needs to be determined according to Table 2-3. The cross-sectional axial compression strength at Point C, P_C, is determined using AISC *Manual* Table 6-4.

$$P_C = 0.85 f_c' A_c$$
$$= 0.85(6 \text{ ksi})(576 \text{ in.}^2)$$
$$= 2{,}940 \text{ kips}$$

From Table 2-3, the available compressive strength at Point C is:

LRFD	ASD
$\phi_c = 0.75$ $\phi_c \chi P_C = \phi_c \frac{P_n}{P_{no}} P_C$ $= 0.75\left(\frac{4{,}720 \text{ kips}}{5{,}390 \text{ kips}}\right)(2{,}940 \text{ kips})$ $= 1{,}930 \text{ kips}$	$\Omega_c = 2.00$ $\frac{\chi P_C}{\Omega_c} = \frac{P_n}{P_{no}} \frac{P_C}{\Omega_c}$ $= \left(\frac{4{,}720 \text{ kips}}{5{,}390 \text{ kips}}\right)\left(\frac{2{,}940 \text{ kips}}{2.00}\right)$ $= 1{,}290 \text{ kips}$

The available strengths for anchor points A, B, and C are:

LRFD	ASD
$P_c = \phi_c P_n$ $= 3{,}540 \text{ kips}$ $P_{cc} = \phi_c \chi P_C$ $= 1{,}930 \text{ kips}$ $M_{cx} = \phi_b M_n$ $= 1{,}980 \text{ kip-ft}$	$P_c = P_n/\Omega_c$ $= 2{,}360 \text{ kips}$ $P_{cc} = \chi P_C/\Omega_c$ $= 1{,}290 \text{ kips}$ $M_{cx} = M_n/\Omega_b$ $= 1{,}320 \text{ kip-ft}$

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Evaluate the interaction equation.

LRFD	ASD
Compare P_u and P_{cc} $P_u = 1{,}500 \text{ kips} < P_{cc} = 1{,}930 \text{ kips}$ Therefore, use Equation 2-2a, which, simplified for uniaxial bending, is: $\frac{M_{ux}}{M_{cx}} = \frac{1{,}800 \text{ kip-ft}}{1{,}980 \text{ kip-ft}}$ $= 0.909 \le 1$ **o.k.**	Compare P_a and P_{cc} $P_a = 1{,}000 \text{ kips} < P_{cc} = 1{,}290 \text{ kips}$ Therefore, use Equation 2-2a, which, simplified for uniaxial bending, is: $\frac{M_{ax}}{M_{cx}} = \frac{1{,}200 \text{ kip-ft}}{1{,}320 \text{ kip-ft}}$ $= 0.909 \le 1$ **o.k.**

Determine the available shear strength

Determine the shear area of the steel section as the sum of the area of webs.

$$A_v = 2(25 \text{ in.})(½ \text{ in.})$$
$$= 25.0 \text{ in.}^2$$

The span-to-depth ratio is computed as:

LRFD	ASD
$\frac{(M_u/V_u)}{d} = \frac{(1{,}800 \text{ kip-ft})(12 \text{ in./ft})/90 \text{ kips}}{25 \text{ in.}}$ $= 9.60$	$\frac{(M_a/V_a)}{d} = \frac{(1{,}200 \text{ kip-ft})(12 \text{ in./ft})/60 \text{ kips}}{25 \text{ in.}}$ $= 9.60$

Because $(M_r/V_r)/d \ge 0.7$, $K_c = 1$.

Calculate the nominal shear strength.

$$V_n = 0.6A_vF_y + 0.06K_cA_c\sqrt{f'_c} \qquad \text{(}Spec\text{. Eq. I4-1)}$$
$$= 0.6(25.0 \text{ in.}^2)(50 \text{ ksi}) + 0.06(1)(576 \text{ in.}^2)\sqrt{6 \text{ ksi}}$$
$$= 835 \text{ kips}$$

The available shear strength is:

LRFD	ASD
$\phi_v = 0.90$ $\phi_v V_n = 0.90(835 \text{ kips})$ $= 751 \text{ kips} > 90 \text{ kips}$ **o.k.**	$\Omega_v = 1.67$ $\frac{V_n}{\Omega_v} = \frac{835 \text{ kips}}{1.67}$ $= 500 \text{ kips} > 60 \text{ kips}$ **o.k.**

Conclusion

The square filled composite column is found to be adequate for the required moment and forces.

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2.6 AVAILABLE STRENGTH OF RECTANGULAR AND SQUARE FILLED COMPOSITE MEMBERS (HIGH STRENGTH)

The provisions of AISC *Specification* Chapter I include limitations on material strengths including upper limits of 75 ksi on the yield strength of steel shapes and 10 ksi on the compressive strength of normal weight concrete (see Section 2.3.3 of this Design Guide for additional details). However, experimental research has shown that composite columns constructed with materials that exceed these limits perform well. The provisions for high-strength rectangular filled composite members in AISC *Specification* Appendix 2 were added in the 2022 edition. Future editions of the AISC *Specification* are likely to expand options for high-strength composite members.

2.6.1 Limitations

The provisions of AISC *Specification* Appendix 2 apply only to rectangular filled composite members constructed with normal weight concrete. They also only apply to members where either the steel, or the concrete, or both exceed the strength limitations of AISC *Specification* Section I1.3 (i.e., if all material limitations in Chapter I are met, then Appendix 2 cannot be used). While the material strengths can exceed the limits of AISC *Specification* Section I1.3, the steel yield stress cannot exceed 100 ksi and the concrete compressive strength cannot exceed 15 ksi.

The strength equations in AISC *Specification* Appendix 2 do not require classification of the cross section for local buckling. However, the width-to-thickness ratio for compression steel elements must not exceed $5.00\sqrt{E_s/F_y}$.

Additionally, longitudinal reinforcement is not required, and if it is provided, it is not considered in the calculation of strength.

2.6.2 Axial Compressive Strength

The axial compressive strength of filled composite members with high strength materials is defined in AISC *Specification* Appendix Section 2.1.2. The nominal axial compressive strength without consideration of length effects, P_{no}, is computed as:

$$P_{no} = F_n A_s + 0.85 f_c' A_c \quad (Spec.\ \text{Eq. A-2-1})$$

where

F_n = critical buckling stress for steel section of filled composite members, ksi

$$F_n = (1.0 - 0.075\lambda) F_y \quad (Spec.\ \text{Eq. A-2-2})$$

where

λ = the largest width-to-thickness ratio of the section multiplied by $\sqrt{F_y/E}$

The critical buckling stress, F_n, was calibrated by Lai and Varma (2018) and accounts for local buckling and other effects.

The nominal axial compressive strength (with consideration of member length effects) for the limit state of flexural buckling, P_n, is computed as:

When $\frac{P_{no}}{P_e} \le 2.25$

$$P_n = P_{no}\left(0.658^{\frac{P_{no}}{P_e}}\right) \quad (Spec.\ \text{Eq. I2-2})$$

When $\frac{P_{no}}{P_e} > 2.25$

$$P_n = 0.877 P_e \quad (Spec.\ \text{Eq. I2-3})$$

where P_e is the elastic critical buckling load determined from AISC *Specification* Equation I2-4; as specified in AISC *Specification* Appendix 7, Section 7.2.3(b), when using the effective length method; or through an elastic buckling analysis.

$$P_e = \frac{\pi^2 (EI)_{eff}}{L_c^2} \quad (Spec.\ \text{Eq. I2-4})$$

where $(EI)_{eff}$ is the effective stiffness of the composite section, determined from AISC *Specification* Equation I2-12 and L_c is the effective length of the member.

The effective stiffness of the composite section, $(EI)_{eff}$, is computed by summing the contributions of each of the components of the cross section, but with a reduction factor, C_3, applied to the concrete contribution to account for the level of cracking expected at the axial compressive strength of the member.

$$(EI)_{eff} = E_s I_s + E_s I_{sr} + C_3 E_c I_c \quad (Spec.\ \text{Eq. I2-12})$$

$$C_3 = 0.45 + 3\left(\frac{A_s + A_{sr}}{A_g}\right) \le 0.9 \quad (Spec.\ \text{Eq. I2-13})$$

The critical buckling load, P_e, should be computed for each coordinate axis and the least value used to compute P_n. Torsional buckling does not apply to filled composite members because of their high torsional stiffness.

The design compressive strength is determined using a resistance factor of $\phi_c = 0.75$ for LRFD. The allowable compressive strength is determined using a safety factor of $\Omega_c = 2.00$ for ASD. The design compressive strength need not be less than the design compressive strength of the bare steel member. The design compressive strength of the bare steel member can be greater than that of the composite member

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when the steel ratio is high and the contribution of the concrete is not enough to overcome the lower resistance factor or higher safety factor used for composite members.

2.6.3 Axial Tensile Strength

AISC *Specification* Appendix 2 does not include any provisions for tensile strength, thus the provisions of AISC *Specification* Chapter I apply. The axial tension strength of filled composite members is defined in AISC *Specification* Section I2.2c. The nominal tensile strength, P_n, is for the limit state of yielding as shown in Equation 2-3 based on AISC *Specification* Equation I2-14. Only the structural steel contribution is included in this equation because the concrete tensile strength is negligible and any internal reinforcement must be neglected.

$$P_n = A_s F_y \tag{2-3}$$

The design tensile strength is determined using a resistance factor of $\phi_t = 0.90$ for LRFD. The allowable tensile strength is determined using a safety factor of $\Omega_t = 1.67$ for ASD.

2.6.4 Flexural Strength

The flexural strength of filled composite members with high-strength materials is defined in AISC *Specification* Appendix Section 2.1.3. The strength is calculated using an approach like the plastic stress distribution method. For steel components, the stress is assumed to be F_y in tension and F_n in compression, where F_n is computed using AISC *Specification* Equation A-2-2. For concrete components, the stress is assumed to be zero in tension and $0.85f_c'$ in compression. The resulting stress distribution is shown in Figure 2-15. The location of the neutral axis is determined such that the resulting force in the composite cross section is zero. The nominal flexural strength, M_n, is determined as 90% of the moment resulting from the stress distribution. The 10% reduction in flexural strength is intended to reduce potential unconservative error in the assessment of interaction strength.

Figure 2-15 shows the stress blocks for a filled composite member made with a box section. This figure is similar to AISC *Specification* Figure C-I3.8(a), but with the stress in the steel in compression equal to F_n.

The location of the neutral axis, defined in Figure 2-15 by the variable a_p, can be determined based on equilibrium as:

$$a_p = \left[\frac{2F_y H t_w + (F_y - F_n) b_i t_f + 0.85 f_c' b_i t_f}{2t_w F_y + 2t_w F_n + 0.85 f_c' b_i}\right] \tag{2-4}$$

The moment corresponding to the stress distribution can be calculated by summing the moments of the compression and tensile forces about the neutral axis.

$$M_{sd} = F_n t_f b_i \left(a_p - \frac{t_f}{2}\right) + 2F_n t_w \frac{a_p^2}{2} + 0.85 f_c' b_i \frac{(a_p - t_f)^2}{2} + 2F_y t_w \frac{(H - a_p)^2}{2} + F_y t_f b_i \left(H - a_p - \frac{t_f}{2}\right) \tag{2-5}$$

The nominal flexural strength is calculated as:

$$M_n = 0.9 M_{sd} \tag{2-6}$$

The composite column program that accompanies this Design Guide can compute the flexural strength for members constructed with box sections and cold formed sections with rounded corners.

The design flexural strength is determined using a resistance factor of $\phi_b = 0.90$ for LRFD. The allowable flexural strength is determined using a safety factor of $\Omega_b = 1.67$ for ASD.

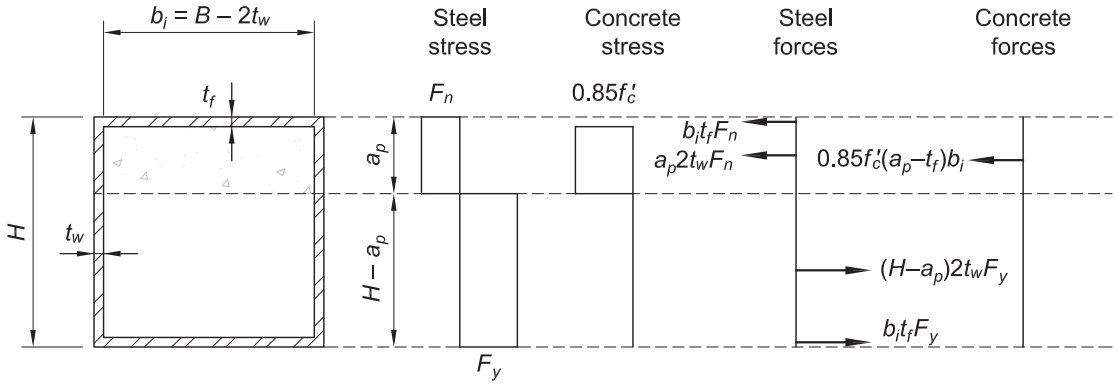


Fig. 2-15. Stress blocks for calculating nominal flexural strengths of rectangular filled composite members (high strength).

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2.6.5 Interaction Strength

The interaction strength of rectangular and square filled composite members with high strength materials is defined in AISC *Specification* Appendix Section 2.1.4. The provisions require the use of the interaction diagram defined in AISC *Specification* Section I5 but with c_p and c_m modified as:

$$c_p = 0.175 - \frac{0.075}{B/H} + \lambda \left(\frac{0.3}{P_n/P_{no}} \right) \left(\frac{f_c'}{F_y} \right) \quad \text{(Spec. Eq. A-2-3)}$$

$$c_m = 0.6 + 0.3 \left(\frac{P_n}{P_{no}} \right)^2 + 0.6\lambda \left(\frac{B}{H} \right) \left(\frac{F_{y,max}}{F_y} \right) \left(\frac{f_c'}{F_y} \right) \quad \text{(Spec. Eq. A-2-4)}$$

where

B = flange width of rectangular cross section, in.
H = web depth of rectangular cross section, in.
$F_{y,max}$ = maximum permitted yield stress of steel
= 100 ksi

The expressions for c_p and c_m were calibrated to the results of second-order inelastic analyses by Alghossoon and Varma (2020).

2.6.6 Shear Strength

AISC *Specification* Appendix 2 does not include any provisions for shear strength, thus the provisions of AISC *Specification* Chapter I apply. The provisions of AISC *Specification* Section I4.2 may be used with the higher material strengths permitted in Appendix 2 to determine shear strength. See Section 2.5.7 of this Design Guide for more information.

DESIGN EXAMPLE 2.6—Square Filled Composite Member Subjected to Combined Axial Compression and Flexure—High-Strength Materials

Given:

Assess the adequacy of a filled composite column member constructed with the cross section shown in Figure 2-14 and having an effective length $L_{cx} = L_{cy} = 30$ ft for the following required strengths:

LRFD	ASD
P_u = 2,750 kips	P_a = 1,750 kips
M_{ux} = 740 kip-ft	M_{ax} = 500 kip-ft
V_u = 90 kips	V_a = 60 kips

The composite member consists of ASTM A572/A572M Grade 50 plates welded to form a box section and normal weight ($w_c = 145$ lb/ft^3) concrete fill with a specified compressive strength, $f_c' = 12$ ksi. The member does not include internal reinforcing.

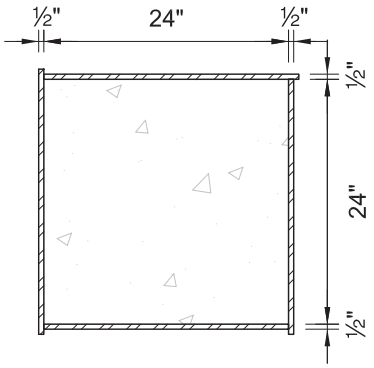


Fig. 2-14. Filled composite member for Examples 2.5 and 2.6 (repeated here for convenience).

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Solution:

From AISC *Manual* Table 2-5, the material properties are as follows:

ASTM A572/A572M Grade 50
$F_y = 50$ ksi

The geometric cross-sectional properties are the same as for the member evaluated in Design Example 2.5.

$A_g = 625 \text{ in.}^2$ $A_c = 576 \text{ in.}^2$ $A_s = 49.0 \text{ in.}^2$ $I_c = 27{,}600 \text{ in.}^4$ $I_s = 4{,}950 \text{ in.}^4$

Assess the limitations of AISC Specification *Appendix Section 2.1.1*

AISC *Specification* Appendix Section 2.1.1(a) requires that the area of the steel section comprise at least 1% of the total composite cross section.

$$\frac{A_s}{A_g} = \frac{49.0 \text{ in.}^2}{625 \text{ in.}^2}$$
$$= 7.84\% > 1\% \quad \textbf{o.k.}$$

This satisfies the minimum required steel percentage of 1%.

AISC *Specification* Appendix Section 2.1.1(b) requires that concrete be normal weight and that $f'_c \le 15$ ksi.

The concrete is normal weight, and the specified compressive strength of concrete is $f'_c = 12$ ksi, which does not exceed the maximum limitation of 15 ksi.

AISC *Specification* Appendix Section 2.1.1(c) requires that $F_y \le 100$ ksi.

The specified minimum yield stress of steel is $F_y = 50$ ksi, which does not exceed the maximum limitation of 100 ksi.

AISC *Specification* Appendix Section 2.1.1(d) requires that the maximum permitted width-to-thickness ratio for compression steel elements be limited to $5.00\sqrt{E/F_y}$.

$$\frac{b}{t} = \frac{24 \text{ in.}}{½ \text{ in.}}$$
$$= 48.0$$

$$\left(\frac{b}{t}\right)_{max} = 5.00\sqrt{\frac{E}{F_y}}$$
$$= 5.00\sqrt{\frac{29{,}000 \text{ ksi}}{50 \text{ ksi}}}$$
$$= 120 > 48.0 \quad \textbf{o.k.}$$

This satisfies the maximum permitted width-to-thickness ratio.

AISC *Specification* Appendix Section 2.1.1(e) does not apply to this member because there is no longitudinal reinforcement.

The requirements of AISC *Specification* Appendix Section 2.1.1 have been met.

Determine the available axial compressive strength

Calculate the nominal axial compressive strength without consideration of length effects.

$$\lambda = \frac{b}{t}\sqrt{\frac{F_y}{E}}$$
$$= \frac{24 \text{ in.}}{½ \text{ in.}}\sqrt{\frac{50 \text{ ksi}}{29{,}000 \text{ ksi}}}$$
$$= 1.99$$

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$$F_n = (1.0 - 0.075\lambda) F_y \qquad \text{(Spec. Eq. A-2-2)}$$
$$= \left[1.0 - 0.075(1.99)\right](50 \text{ ksi})$$
$$= 42.5 \text{ ksi}$$

$$P_{no} = F_n A_s + 0.85 f_c' A_c \qquad \text{(Spec. Eq. A-2-1)}$$
$$= (42.5 \text{ ksi})(49.0 \text{ in.}^2) + 0.85(12 \text{ ksi})(576 \text{ in.}^2)$$
$$= 7{,}960 \text{ kips}$$

Calculate the modulus of elasticity of concrete.

$$E_c = w_c^{1.5}\sqrt{f_c'}$$
$$= (145 \text{ lb/ft}^3)^{1.5}\sqrt{12 \text{ ksi}}$$
$$= 6{,}050 \text{ ksi}$$

Calculate the effective stiffness of the composite section.

$$C_3 = 0.45 + 3\left(\frac{A_s + A_{sr}}{A_g}\right) \le 0.9 \qquad \text{(Spec. Eq. I2-13)}$$
$$= 0.45 + 3\left(\frac{49.0 \text{ in.}^2 + 0}{625 \text{ in.}^2}\right) \le 0.9$$
$$= 0.685 < 0.9$$
$$= 0.685$$

$$(EI)_{eff} = E_s I_s + E_s I_{sr} + C_3 E_c I_c \qquad \text{(Spec. Eq. I2-12)}$$
$$= (29{,}000 \text{ ksi})(4{,}950 \text{ in.}^4) + 0 + 0.685(6{,}050 \text{ ksi})(27{,}600 \text{ in.}^4)$$
$$= 258{,}000{,}000 \text{ kip-in.}^2$$

Calculate the nominal compressive strength.

$$P_e = \frac{\pi^2 (EI)_{eff}}{L_c^2} \qquad \text{(Spec. Eq. I2-4)}$$
$$= \frac{\pi^2 (258{,}000{,}000 \text{ kip-in.}^2)}{\left[(30 \text{ ft})(12 \text{ in./ft})\right]^2}$$
$$= 19{,}600 \text{ kips}$$

$$\frac{P_{no}}{P_e} = \frac{7{,}960 \text{ kips}}{19{,}600 \text{ kips}}$$
$$= 0.406$$

Because $P_{no}/P_e \le 2.25$, use AISC *Specification* Equation I2-2.

$$P_n = P_{no}\left(0.658^{\frac{P_{no}}{P_e}}\right) \qquad \text{(Spec. Eq. I2-2)}$$
$$= (7{,}960 \text{ kips})(0.658^{0.406})$$
$$= 6{,}720 \text{ kips}$$

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The available compressive strength is:

LRFD	ASD
$\phi_c = 0.75$ $P_c = \phi_c P_n$ $= 0.75(6{,}710 \text{ kips})$ $= 5{,}030 \text{ kips}$	$\Omega_c = 2.00$ $P_c = P_n/\Omega_c$ $= (6{,}710 \text{ kips})/2.00$ $= 3{,}360 \text{ kips}$

Determine the available flexural strength

Use the equations presented in Section 2.6.4.

$b_i = 24$ in.

$H = 25$ in.

$t_f = t_w$

$= ½$ in.

$$a_p = \left[\frac{2F_yHt_w + (F_y - F_n)b_it_f + 0.85f_c'b_it_f}{2t_wF_y + 2t_wF_n + 0.85f_c'b_i}\right] \quad (2\text{-}4)$$

$$= \left[\frac{2(50 \text{ ksi})(25 \text{ in.})(½ \text{ in.}) + (50 \text{ ksi} - 42.5 \text{ ksi})(24 \text{ in.})(½ \text{ in.}) + 0.85(12 \text{ ksi})(24 \text{ in.})(½ \text{ in.})}{2(½ \text{ in.})(50 \text{ ksi}) + 2(½ \text{ in.})(42.5 \text{ ksi}) + 0.85(12 \text{ ksi})(24 \text{ in.})}\right]$$

$= 4.34$ in.

$$M_{sd} = F_nt_fb_i\left(a_p - \frac{t_f}{2}\right) + 2F_nt_w\frac{a_p^2}{2} + 0.85f_c'b_i\frac{(a_p - t_f)^2}{2} + 2F_yt_w\frac{(H - a_p)^2}{2} + F_yt_fb_i\left(H - a_p - \frac{t_f}{2}\right) \quad (2\text{-}5)$$

$$= (42.5 \text{ ksi})(½ \text{ in.})(24 \text{ in.})\left(4.34 \text{ in.} - \frac{½ \text{ in.}}{2}\right) + 2(42.5 \text{ ksi})(½ \text{ in.})\frac{(4.34 \text{ in.})^2}{2}$$

$$+ 0.85(12 \text{ ksi})(24 \text{ in.})\left[\frac{(4.34 \text{ in.} - ½ \text{ in.})^2}{2}\right]$$

$$+ (50 \text{ ksi})(½ \text{ in.})(24 \text{ in.})\left(25 \text{ in.} - 4.34 \text{ in.} - \frac{½ \text{ in.}}{2}\right)$$

$= 27{,}200$ kip-in.

$$M_n = 0.9M_{sd} \quad (2\text{-}6)$$

$= 0.9(27{,}200 \text{ kip-in.})$

$$= (24{,}500 \text{ kip-in.})\left(\frac{1 \text{ ft}}{12 \text{ in.}}\right)$$

$= 2{,}040$ kip-ft

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The available flexural strength is:

LRFD	ASD
$\phi_b = 0.90$ $M_{cx} = \phi_b M_n$ $= 0.90(2{,}040 \text{ kip-ft})$ $= 1{,}840 \text{ kip-ft}$	$\Omega_b = 1.67$ $M_{cx} = M_n/\Omega_b$ $= (2{,}040 \text{ kip-ft})/1.67$ $= 1{,}220 \text{ kip-ft}$

Check combined flexure and axial compression strength

The interaction strength of filled composite members constructed with high-strength materials must be evaluated according to AISC *Specification* Appendix Section 2.1.4. Specifically, interaction strength is evaluated using AISC *Specification* Equations I5-1a and I5-1b, with modified values for c_p and c_m.

$$c_p = 0.175 - \frac{0.075}{B/H} + \lambda\left(\frac{0.3}{P_n/P_{no}}\right)\left(\frac{f_c'}{F_y}\right) \qquad \text{(Spec. Eq. A-2-3)}$$

$$= 0.175 - \frac{0.075}{(25 \text{ in.})/(25 \text{ in.})} + 1.99\left[\frac{0.3}{(6{,}720 \text{ kips})/(7{,}960 \text{ kips})}\right]\left(\frac{12 \text{ ksi}}{50 \text{ ksi}}\right)$$

$$= 0.270$$

$$c_m = 0.6 + 0.3\left(\frac{P_n}{P_{no}}\right)^2 + 0.6\lambda\left(\frac{B}{H}\right)\left(\frac{F_{y,max}}{F_y}\right)\left(\frac{f_c'}{F_y}\right) \qquad \text{(Spec. Eq. A-2-4)}$$

$$= 0.6 + 0.3\left(\frac{6{,}720 \text{ kips}}{7{,}960 \text{ kips}}\right)^2 + 0.6(1.99)\left(\frac{25 \text{ in.}}{25 \text{ in.}}\right)\left(\frac{100 \text{ ksi}}{50 \text{ ksi}}\right)\left(\frac{12 \text{ ksi}}{50 \text{ ksi}}\right)$$

$$= 1.39$$

Evaluate the interaction equation.

LRFD	ASD
Compare P_u and P_c $\frac{P_u}{P_c} = \frac{2{,}750 \text{ kips}}{5{,}040 \text{ kips}}$ $= 0.546$ Because $P_u/P_c \geq c_p$, use AISC *Specification* Equation I5-1a $\frac{P_u}{P_c} + \frac{1-c_p}{c_m}\left(\frac{M_u}{M_c}\right) \leq 1.0$ $= \frac{2{,}750 \text{ kips}}{5{,}040 \text{ kips}} + \left(\frac{1-0.270}{1.39}\right)\left(\frac{740 \text{ kip-ft}}{1{,}840 \text{ kip-ft}}\right) \leq 1.0$ $= 0.757 < 1.0$ **o.k.**	Compare P_a and P_c $\frac{P_a}{P_c} = \frac{1{,}750 \text{ kips}}{3{,}360 \text{ kips}}$ $= 0.521$ Because $P_a/P_c \geq c_p$, use AISC *Specification* Equation I5-1a $\frac{P_a}{P_c} + \frac{1-c_p}{c_m}\left(\frac{M_a}{M_c}\right) \leq 1.0$ $= \frac{1{,}750 \text{ kips}}{3{,}360 \text{ kips}} + \left(\frac{1-0.270}{1.39}\right)\left(\frac{500 \text{ kip-ft}}{1{,}220 \text{ kip-ft}}\right) \leq 1.0$ $= 0.736 < 1.0$ **o.k.**

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Determine the available shear strength

Calculate the shear area of the steel section as the sum of the area of webs.

$$A_v = 2(25\text{ in.})(0.5\text{ in.})$$
$$= 25.0\text{ in.}^2$$

The span-to-depth ratio is computed as:

LRFD	ASD
$\frac{(M_u/V_u)}{d} = \frac{(740\text{ kip-ft})(12\text{ in./ft})/90\text{ kips}}{25\text{ in.}}$ $= 3.95$	$\frac{(M_a/V_a)}{d} = \frac{(500\text{ kip-ft})(12\text{ in./ft})/60\text{ kips}}{25\text{ in.}}$ $= 4.00$

Because $(M_r/V_r)/d \geq 0.7$, $K_c = 1$.

Calculate the nominal shear strength.

$$V_n = 0.6A_vF_y + 0.06K_cA_c\sqrt{f'_c} \qquad \text{(}Spec\text{. Eq. I4-1)}$$
$$= 0.6(25.0\text{ in.}^2)(50\text{ ksi}) + 0.06(1)(576\text{ in.}^2)\sqrt{12\text{ ksi}}$$
$$= 870\text{ kips}$$

The available shear strength is:

LRFD	ASD
$\phi_v = 0.90$ $\phi_v V_n = 0.90(870\text{ kips})$ $= 783\text{ kips} > 90\text{ kips}$ **o.k.**	$\Omega_v = 1.67$ $\frac{V_n}{\Omega_v} = \frac{870\text{ kips}}{1.67}$ $= 521\text{ kips} > 60\text{ kips}$ **o.k.**

Conclusion

The square filled composite column is found to be adequate for the required moment and forces.

2.7 AVAILABLE STRENGTH OF ROUND FILLED COMPOSITE MEMBERS

2.7.1 Limitations

The limitations for round filled composite members are the same as those for rectangular and square filled composite members. See Section 2.5.1 of this Design Guide.

2.7.2 Classification of Composite Sections

Filled composite members are required to be classified for local buckling according to AISC *Specification* Section I1.4. Sections are classified as compact composite, noncompact composite, or slender composite. For a round section to be compact composite, the diameter-to-thickness ratio, $\lambda = D/t$, must be less than or equal to λ_p as determined from Table 2-6 (based on AISC *Specification* Tables I1.1a and I1.1b). For the section to be noncompact composite, the diameter-to-thickness ratio must be greater than λ_p and less than or equal to λ_r as determined from Table 2-6. All other sections are classified as slender composite. However, the maximum permitted diameter-to-thickness ratio is $0.31E/F_y$. Any section with D/t greater than this value is not permitted. For members subjected to combined flexure and axial load, the classification can be different for flexure and axial load.

The use of the terms "compact composite," "noncompact composite," and "slender composite" in the AISC *Specification* is intended to distinguish the classification of composite members from that of structural steel members. The behavior assumed in design for a noncompact composite section,

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Table 2-6. Limiting Diameter-to-Thickness Ratios for Compression Steel Elements in Round Filled Composite Members				
Description of Element	**Diameter-to-Thickness Ratio**	**λ_p Compact Composite/ Noncompact Composite**	**λ_r Noncompact Composite/ Slender Composite**	**Maximum Permitted**
Round HSS subjected to axial compression	D/t	$\frac{0.15E}{F_y}$	$\frac{0.19E}{F_y}$	$\frac{0.31E}{F_y}$
Round HSS subjected to flexural bending		$\frac{0.09E}{F_y}$	$\frac{0.31E}{F_y}$	

for example, is different from that of a noncompact element of a structural steel member.

2.7.3 Axial Compressive Strength

The axial compressive strength of filled composite members is defined in AISC *Specification* Section I2.2b. The nominal axial compressive strength without consideration of length effects, P_{no}, is computed based on the classification of the section. For compact composite sections, P_{no} is equal to the plastic axial compressive strength, P_p.

$$P_{no} = P_p \qquad \text{(}Spec\text{. Eq. I2-9a)}$$

$$P_p = F_y A_s + 0.95 f_c' \left(A_c + A_{sr} \frac{E_s}{E_c} \right)$$

(from *Spec* Eq. I2-9b)

This equation is based on AISC *Specification* Equation I2-9b but with C_2 replaced with the value of 0.95 for round sections. The 0.95 factor on the concrete contribution is greater than the equivalent factor used for rectangular and square filled composite members because of the superior confinement provided by the round steel section.

For noncompact composite sections, P_{no} varies between P_p and P_y as shown in AISC *Specification* Equation I2-9c.

$$P_{no} = P_p - \frac{P_p - P_y}{(\lambda_r - \lambda_p)^2} (\lambda - \lambda_p)^2$$

(*Spec*. Eq. I2-9c)

where

P_y = yield axial compressive strength, kips, from AISC *Specification* Equation I2-9d

λ = the diameter-to-thickness ratio of the section

$\lambda_p = 0.15E/F_y$ from Table 2-6 and AISC *Specification* Table I1.1a

$\lambda_r = 0.19E/F_y$ from Table 2-6 and AISC *Specification* Table I1.1a

$$P_y = F_y A_s + 0.7 f_c' \left(A_c + A_{sr} \frac{E_s}{E_c} \right)$$

(*Spec*. Eq. I2-9d)

For slender composite sections, P_{no} is computed as:

$$P_{no} = F_n A_s + 0.7 f_c' \left(A_c + A_{sr} \frac{E_s}{E_c} \right)$$

(*Spec*. Eq. I2-9e)

where

$$F_n = \frac{0.72 F_y}{\left[\left(\frac{D}{t} \right) \frac{F_y}{E_s} \right]^{0.2}} \qquad \text{(}Spec\text{. Eq. I2-11)}$$

The nominal axial compressive strength (with consideration of member length effects) for the limit state of flexural buckling, P_n, is computed as shown in AISC *Specification* Equations I2-2 and I2-3. Note that the column curve defined by AISC *Specification* Equations I2-2 and I2-3 is identical to the column curve for structural steel members defined in AISC *Specification* Section E3 if $P_{no} = A_s F_y$ and $(EI)_{eff} = E_s I_s$.

When $\frac{P_{no}}{P_e} \le 2.25$

$$P_n = P_{no} \left(0.658^{\frac{P_{no}}{P_e}} \right) \qquad \text{(}Spec\text{. Eq. I2-2)}$$

When $\frac{P_{no}}{P_e} > 2.25$

$$P_n = 0.877 P_e \qquad \text{(}Spec\text{. Eq. I2-3)}$$

where P_e is the elastic critical buckling load determined from AISC *Specification* Equation I2-4; as specified in AISC *Specification* Appendix 7, Section 7.2.3(b), when using the effective length method; or through an elastic buckling analysis.

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$$P_e = \frac{\pi^2 (EI)_{eff}}{L_c^2} \qquad \text{(Spec. Eq. I2-4)}$$

where

$(EI)_{eff}$ = effective stiffness of the composite section, kip-in.^{2}, determined from AISC *Specification* Equation I2-12

L_c = effective length of the member, in.

The effective stiffness of the composite section, $(EI)_{eff}$, is computed by summing the contributions of each of the components of the cross section but with a reduction factor, C_3, applied to the concrete contribution to account for the level of cracking expected at the axial compressive strength of the member.

$$(EI)_{eff} = E_s I_s + E_s I_{sr} + C_3 E_c I_c \qquad \text{(Spec. Eq. I2-12)}$$

$$C_3 = 0.45 + 3\left(\frac{A_s + A_{sr}}{A_g}\right) \le 0.9 \qquad \text{(Spec. Eq. I2-13)}$$

Round filled composite members without internal reinforcing are axisymmetric. The critical buckling load, P_e, for these members should be computed for the axis with the largest effective length. For round filled composite members with internal reinforcing, the critical buckling load, P_e, should be computed for each coordinate axis and the least value used to compute P_n. Torsional buckling does not apply to filled composite members because of their high torsional stiffness.

The design compressive strength is determined using a resistance factor of $\phi_c = 0.75$ for LRFD. The allowable compressive strength is determined using a safety factor of $\Omega_c = 2.00$ for ASD. The design compressive strength need not be less than the design compressive strength of the bare steel member. The design compressive strength of the bare steel member can be greater than that of the composite member when the steel ratio is high and the contribution of the concrete is not enough to overcome the lower resistance factor or higher safety factor used with composite members.

2.7.4 Axial Tensile Strength

The calculation of axial tensile strength for round filled composite members is the same as that for rectangular and square filled composite members. See Section 2.5.4 of this Design Guide.

2.7.5 Flexural Strength

The flexural strength of filled composite members is defined in AISC *Specification* Section I3.4b. Unlike for encased composite members, direct bond interaction between the steel and concrete is sufficient to develop the composite action necessary to achieve the full moment capacity of the composite section; therefore, the determination of the flexural strength is not dependent on whether steel anchors or other supplementary shear transfer mechanisms are provided. However, due to the possibility of local buckling, the determination of flexural strength is dependent on the classification of the composite section. Classification of round filled composite members is discussed in Section 2.7.2 of this Design Guide.

For compact composite sections, the nominal flexural strength, M_n, is equal to the plastic moment strength, M_p, determined as the moment corresponding to a plastic stress distribution over the composite cross section. Closed-form equations for the flexural strength of round filled composite members are provided in AISC *Manual* Table 6-4, reproduced here as Figure 2-16. The flexural strength is denoted as M_B in this table. The table does not consider internal reinforcement. The composite column program that accompanies this Design Guide can compute the flexural strength considering internal reinforcement. The strain compatibility method and elastic stress distribution methods can conservatively be used in lieu of the plastic stress distribution moment for determining M_n. The effective stress-strain method may also be used with appropriately calibrated effective stress-strain relationships (e.g., Sakino et al., 2004; Han et al., 2005; Lai and Varma, 2016).

For noncompact composite sections, the nominal flexural strength, M_n, is determined as:

$$M_n = M_p - (M_p - M_y)\left(\frac{\lambda - \lambda_p}{\lambda_r - \lambda_p}\right) \qquad \text{(Spec. Eq. I3-5b)}$$

In AISC *Specification* Equation I3-5b, M_y is the yield moment corresponding to yielding of the tensile side of the section and first yield of the extreme compression fiber of the steel section. The capacity at first yield is calculated assuming a linear elastic stress distribution with the maximum concrete compressive stress limited to $0.7f_c'$ and the maximum steel stress limited to F_y. The width-to-thickness ratio, λ, and corresponding limits, λ_p and λ_r, are listed in AISC *Specification* Table I1.1b (Table 2-6 of this Design Guide).

For slender composite sections, the nominal flexural strength, M_n, is computed based on assumed stress distributions. The stress in the steel is assumed to vary linearly from F_n (AISC *Specification* Equation I2-11) at the extreme compression fiber of the steel section to F_y at the extreme tension fiber of the steel section. The stress in the concrete varies linearly from zero at the neutral axis (as defined by steel stress distribution) to $0.7f_c'$ at the extreme compression fiber of the concrete.

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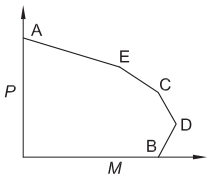
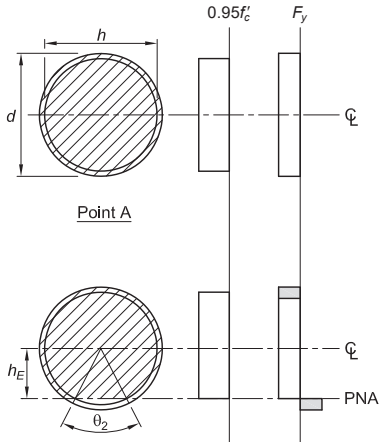
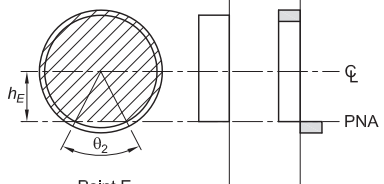
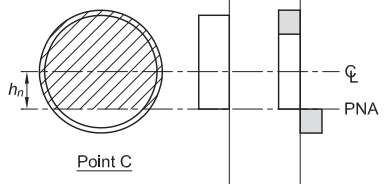
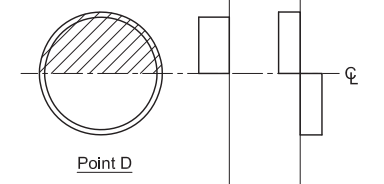
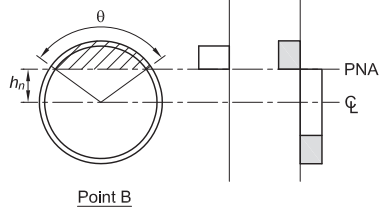


Table 6-4
Cross-Section Strength for Composite Filled Round HSS
Subject to Flexure about Any Axis

Section	Stress Distribution	Pt.	Defining Equation
		A	$P_A = F_y A_s + 0.95 f'_c A_c$ $M_A = 0$ $A_s = \pi(dt - t^2)$ $A_c = \frac{\pi h^2}{4}$
		E	$P_E = P_A - F_y\left(d^2 - h^2\right)\frac{\theta_2}{4} - 0.95 f'_c h^2\left(\frac{\theta_2 - \sin\theta_2}{8}\right)$ $M_E = F_y Z_{sE} + 0.95 f'_c\left(\frac{Z_{sE}}{2}\right)$ $Z_{cE} = \frac{h^3}{6}\sin^3\left(\frac{\theta_2}{2}\right)$ $Z_{sE} = \frac{d^3 - h^3}{6}\sin\left(\frac{\theta_2}{2}\right)$ $h_E = \frac{h_n}{2} + \frac{h}{4}$ $\theta_2 = \pi - 2\arcsin\left(\frac{2h_E}{h}\right)$
		C	$P_C = 0.95 f'_c A_c$ $M_C = M_B$
		D	$P_D = \frac{0.95 f'_c A_c}{2}$ $M_D = F_y Z_s + 0.95 f'_c\left(\frac{Z_c}{2}\right)$ Z_s = plastic section modulus of steel shape, in.^{3} $= \frac{d^3}{6} - Z_c$ $Z_c = \frac{h^3}{6}$
		B	$P_B = 0$ $M_B = F_y Z_{sB} + 0.95 f'_c\left(\frac{Z_{cB}}{2}\right)$ $Z_{sB} = \frac{d^3 - h^3}{6}\sin\left(\frac{\theta}{2}\right)$ $Z_{cB} = \frac{h^3}{6}\sin^3\left(\frac{\theta}{2}\right)$ $\theta = \frac{0.0260K_c - 2K_s}{0.0848K_c} + \frac{\sqrt{\left(0.0260K_c + 2K_s\right)^2 + 0.857K_cK_s}}{0.0848K_c}$ (rad) $K_c = f'_c h^2$ $K_s = F_y\left(\frac{d - t}{2}\right)t$ (thin HSS wall assumed) $h_n = \frac{h}{2}\sin\left(\frac{\pi - \theta}{2}\right)$
			

F_y = specified minimum yield stress of steel shape, ksi

Fig. 2-16. AISC Manual *Table 6-4—Axial and flexural interaction strength of round filled composite members based on the plastic stress distribution method.*

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These assumed stress distributions for calculating M_y or M_n for slender composite members cannot be determined from an elastic stress distribution and can only arise from an elastic stress distribution if the curvatures in the steel and concrete are different. No guidance is provided in the AISC *Specification* or Commentary on how to consider internal reinforcement in these calculations. A logical approach that is consistent with the axial strength provisions would be to back-calculate the curvature in the concrete assuming elastic behavior to determine the strains in each reinforcing bar and assign a stress to each bar using the transformed modulus, $0.7f_c'E_s/E_c$.

No closed-form expressions are available to compute M_y or M_n for slender composite sections; however, the composite column program that accompanies this Design Guide can compute the flexural strength, including these strengths.

The design flexural strength is determined using a resistance factor of $\phi_b = 0.90$ for LRFD. The allowable flexural strength is determined using a safety factor of $\Omega_b = 1.67$ for ASD.

2.7.6 Interaction Strength

The process of assessing interaction strength for round filled composite members is the same as that for rectangular and square filled composite members. See Section 2.5.6 of this Design Guide.

2.7.7 Shear Strength

New in the 2022 edition of the AISC *Specification* are provisions for the available shear strength of filled composite members that combine the contributions of steel and concrete. Based on the results of recent research, AISC *Specification* Section I4.2 permits the calculation of nominal strength of round filled composite members as:

$$V_n = 0.6A_vF_y + 0.06K_cA_c\sqrt{f_c'} \quad \text{(Spec. Eq. I4-1)}$$

where

A_c = area of concrete infill, in.^{2}

A_v = shear area of the steel portion of the composite member, in.^{2}

$= 2A_s/\pi$

The coefficient $K_c = 1$ if the cross section is not compact composite (AISC *Specification* Tables I1.1a and I1.1b) or if the shear span-to-depth ratio, $(M_r/V_r)/d$, is greater than or equal to 0.7. For the shear span-to-depth ratio, M_r and V_r are the maximum required flexural and shear strengths, respectively, along the member length (they are shown as M_u and V_u in the AISC *Specification* despite applying to both LRFD and ASD), and d is equal to the member depth in the direction of bending. If the cross section is compact composite and the shear span-to-depth ratio is less than 0.5, then $K_c = 9$. For members with compact composite cross sections and a shear span-to-depth ratio between 0.5 and 0.7, K_c can be determined from linear interpolation.

In most cases, $K_c = 1$. The greater value of K_c may be applicable for panel zones or other special situations where the shear span is small enough for a diagonal compression strut to form in the concrete.

Note that the coefficient 0.06 is used in AISC *Specification* Equation I4-1 instead of the more familiar coefficient 2 from ACI 318 because f_c' is in units of ksi instead of psi. Internal transverse reinforcement has been found to have minimal impact on the shear strength of filled composite members (Bruneau et al., 2018).

DESIGN EXAMPLE 2.7—Filled Round Composite Member Subjected to Combined Axial Compression and Flexure

Given:

Assess the adequacy of a filled composite column member constructed with the cross section shown in Figure 2-17 and having an effective length $L_{cx} = L_{cy} = 20$ ft for the following required strengths:

LRFD	ASD
$P_u = 680$ kips	$P_a = 450$ kips
$M_{ux} = 264$ kip-ft	$M_{ax} = 180$ kip-ft
$V_u = 20$ kips	$V_a = 15$ kips

The composite member consists of an ASTM A500/A500M Grade C HSS16.000×0.250 (ASTM, 2023) filled with normal weight concrete having a specified concrete compressive strength, $f_c' = 8$ ksi. The member includes four No. 7, ASTM A615/A615M Grade 60 longitudinal reinforcing bars arranged as shown in Figure 2-17.

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Solution:

From AISC *Manual* Table 2-4, the material properties of the steel shape are as follows:

ASTM A500/A500M Grade C
$F_y = 50$ ksi

From ASTM A615/A615M, the material properties of the reinforcement are as follows:

ASTM A615/A615M Grade 60
$F_{ysr} = 60$ ksi

From AISC *Manual* Table 1-13, the cross-sectional properties of the steel section are as follows:

$D = 16.0$ in. $\quad t = 0.233$ in. $\quad A_s = 11.5$ in.^{2} $\quad D/t = 68.7$ $\quad I_s = 359$ in.^{4}

From ACI 318, Appendix B, the area of a single No. 7 reinforcing bar is 0.60 in.^{2}

Calculate cross-sectional properties

The area and moment of inertia of the reinforcing are:

$$A_{sr} = 4(0.60 \text{ in.}^2)$$
$$= 2.40 \text{ in.}^2$$

$$I_{sr} = 2(0.60 \text{ in.}^2)\left(\frac{16.0 \text{ in.}}{2} - 2\text{½ in.}\right)^2$$
$$= 36.3 \text{ in.}^4$$

The cross-sectional properties of the concrete are:

$$A_c = \frac{\pi}{4}(D-2t)^2 - A_{sr}$$
$$= \frac{\pi}{4}\left[16.0 \text{ in.} - 2(0.233 \text{ in.})\right]^2 - 2.40 \text{ in.}^2$$
$$= 187 \text{ in.}^2$$

$$I_c = \frac{\pi}{64}(D-2t)^4 - I_{sr}$$
$$= \frac{\pi}{64}\left[16.0 \text{ in.} - 2(0.233 \text{ in.})\right]^4 - 36.3 \text{ in.}^4$$
$$= 2{,}820 \text{ in.}^4$$

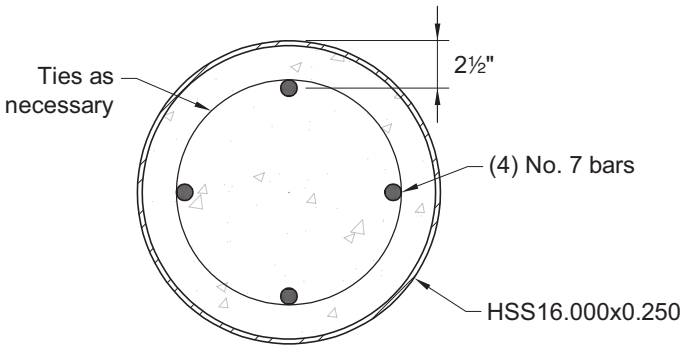


Fig. 2-17. Round filled composite member for Example 2.7.

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The gross area of the concrete is:

$$A_{gc} = \frac{\pi}{4}(D-2t)^2$$
$$= \frac{\pi}{4}\left[16.0 \text{ in.} - 2(0.233 \text{ in.})\right]^2$$
$$= 190 \text{ in.}^2$$

The gross area of the composite member is:

$$A_g = \frac{\pi}{4}D^2$$
$$= \frac{\pi}{4}(16.0 \text{ in.})^2$$
$$= 201 \text{ in.}^2$$

Assess the limitations of AISC Specification *Section I2.2a*

AISC *Specification* Section I2.2a(a) requires that the cross-sectional area of the structural steel section comprise at least 1% of the total composite cross section.

$$\frac{A_s}{A_g} = \frac{11.5 \text{ in.}^2}{201 \text{ in.}^2}$$
$$= 5.72\% > 1\% \quad \textbf{o.k.}$$

This satisfies the minimum required steel percentage of 1%.

AISC *Specification* Section I2.2a(b) requires that the filled composite section be classified for local buckling according to AISC *Specification* Section I1.4.

$$\lambda = D/t$$
$$= 68.7$$

From AISC *Specification* Table I1.1a:

$$\lambda_p = \frac{0.15E}{F_y}$$
$$= \frac{0.15(29{,}000 \text{ ksi})}{50 \text{ ksi}}$$
$$= 87.0$$

Because $\lambda \le \lambda_p$, the section is compact composite for compression.

AISC *Specification* Section I2.2a(c) requires that if longitudinal reinforcement is provided, the minimum internal transverse reinforcement consist of either a No. 3 bar spaced at a maximum of 12 in. on center, or a No. 4 bar or larger spaced at a maximum of 16 in. on center.

Ties are provided to meet these requirements.

AISC *Specification* Section I2.2a(d) requires that if longitudinal reinforcing steel is provided for strength, the maximum reinforcement ratio not exceed 8% of the gross area of the concrete in accordance with ACI 318, Section 10.6.1.1.

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$$\frac{A_{sr}}{A_{gc}} = \frac{2.40 \text{ in.}^2}{190 \text{ in.}^2}$$
$$= 1.26\% < 8\% \quad \textbf{o.k.}$$

The reinforcement satisfies the maximum permitted reinforcement ratio.

All the requirements of AISC *Specification* Section I2.2a have been met.

Determine the available axial compressive strength

Calculate the modulus of elasticity of concrete.

$$E_c = w_c^{1.5}\sqrt{f_c'}$$
$$= (145 \text{ lb/ft}^3)^{1.5}\sqrt{8 \text{ ksi}}$$
$$= 4{,}940 \text{ ksi}$$

Calculate the nominal axial compressive strength without consideration of length effects. For a compact composite section, $P_{no} = P_p$.

$$P_p = F_y A_s + 0.95 f_c' \left(A_c + A_{sr}\frac{E_s}{E_c}\right) \quad \text{(from } \textit{Spec.} \text{ Eq. I2.9b)}$$
$$= (50 \text{ ksi})(11.5 \text{ in.}^2) + 0.95(8 \text{ ksi})\left[187 \text{ in.}^2 + (2.40 \text{ in.}^2)\frac{29{,}000 \text{ ksi}}{4{,}940 \text{ ksi}}\right]$$
$$= 2{,}100 \text{ kips}$$

$$P_{no} = P_p \quad \text{(}\textit{Spec.}\text{ Eq. I2.9a)}$$
$$= 2{,}100 \text{ kips}$$

Calculate the effective stiffness of the composite section.

$$C_3 = 0.45 + 3\left(\frac{A_s + A_{sr}}{A_g}\right) \le 0.9 \quad \text{(}\textit{Spec.}\text{ Eq. I2-13)}$$
$$= 0.45 + 3\left(\frac{11.5 \text{ in.}^2 + 2.40 \text{ in.}^2}{201 \text{ in.}^2}\right) \le 0.9$$
$$= 0.657 < 0.9$$
$$= 0.657$$

$$(EI)_{eff} = E_s I_s + E_s I_{sr} + C_3 E_c I_c \quad \text{(}\textit{Spec.}\text{ Eq. I2-12)}$$
$$= (29{,}000 \text{ ksi})(359 \text{ in.}^4) + (29{,}000 \text{ ksi})(36.3 \text{ in.}^4) + 0.657(4{,}940 \text{ ksi})(2{,}820 \text{ in.}^4)$$
$$= 20{,}600{,}000 \text{ kip-in.}^2$$

Calculate the nominal compressive strength.

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$$P_e = \frac{\pi^2 (EI)_{eff}}{L_c^2} \qquad \text{(Spec. Eq. I2-4)}$$

$$= \frac{\pi^2 (20{,}600{,}000 \text{ kip-in.}^2)}{\left[(20 \text{ ft})(12 \text{ in./ft})\right]^2}$$

$$= 3{,}530 \text{ kips}$$

$$\frac{P_{no}}{P_e} = \frac{2{,}100 \text{ kips}}{3{,}530 \text{ kips}}$$

$$= 0.595$$

Because $P_{no}/P_e \le 2.25$, use AISC *Specification* Equation I2-2.

$$P_n = P_{no}\left(0.658^{\frac{P_{no}}{P_e}}\right) \qquad \text{(Spec. Eq. I2-2)}$$

$$= (2{,}100 \text{ kips})(0.658^{0.595})$$

$$= 1{,}640 \text{ kips}$$

The available compressive strength is:

LRFD	ASD
$\phi_c = 0.75$ $P_c = \phi_c P_n$ $= 0.75(1{,}640 \text{ kips})$ $= 1{,}230 \text{ kips}$	$\Omega_c = 2.00$ $P_c = \frac{P_n}{\Omega_c}$ $= \frac{1{,}640 \text{ kips}}{2.00}$ $= 820 \text{ kips}$

Assess the limitations of AISC Specification *Section I3.4a*

AISC *Specification* Section I3.4a(a) requires that filled composite sections be classified for local buckling according to AISC *Specification* Section I1.4.

$$\lambda = D/t$$

$$= 68.7$$

From AISC *Specification* Table I1.1b:

$$\lambda_p = \frac{0.09E}{F_y}$$

$$= \frac{0.09(29{,}000 \text{ ksi})}{50 \text{ ksi}}$$

$$= 52.2$$

$$\lambda_r = \frac{0.31E}{F_y}$$

$$= \frac{0.31(29{,}000 \text{ ksi})}{50 \text{ ksi}}$$

$$= 180$$

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Because $\lambda_p < \lambda \le \lambda_r$, the section is noncompact composite for flexure.

AISC *Specification* Section I3.4a(b) requires that the total cross-sectional area of the structural steel section comprise at least 1% of the total composite cross section.

The structural steel section was previously confirmed to comprise at least 1% of the total composite cross section.

AISC *Specification* Section I3.4a(c) requires that if longitudinal reinforcing steel is provided, the reinforcement ratio for continuous longitudinal reinforcement, ρ_{sr}, exceed 0.4% and the minimum internal transverse reinforcement consist of either a No. 3 bar spaced at a maximum of 12 in. on center or a No. 4 bar or larger spaced at a maximum of 16 in. on center.

$$\rho_{sr} = \frac{A_{sr}}{A_g}$$

$$= \frac{2.40 \text{ in.}^2}{201 \text{ in.}^2}$$

$$= 0.0119 > 0.004 \quad \textbf{o.k.}$$

This satisfies the minimum required reinforcement percentage of 0.4%.

Ties are provided to meet the minimum transverse reinforcing requirements.

AISC *Specification* Section I3.4a(d) does not apply to this member because it is a column.

The requirements of AISC *Specification* Section I3.4a have been met.

Determine the available flexural strength

Because the section is noncompact composite for flexural loading, the flexural strength is determined using AISC *Specification* Equation I3-3b, which requires M_p, the moment corresponding to plastic stress distribution over the composite cross section, and M_y, the yield moment corresponding to yielding of the tension flange and first yield of the compression flange. If the section did not have internal reinforcement, M_p could be determined as M_B from AISC *Manual* Table 6-4. However, no closed-form solutions are available for the flexural strength of round filled composite members with internal reinforcement. Therefore, the composite column program described in Appendix A will be used to determine M_p and M_y.

From the composite column program described in Appendix A:

$M_p = 364$ kip-ft

While M_p for this example is difficult to compute with hand calculations, it can be verified with hand calculations. In addition to the plastic moment, M_p, the program also determines the location of the PNA. For this example, the PNA is located 4.01 in. above the centerline of the cross section. The PNA divides the cross section into five stressed components: (1) steel in compression with a stress of $-F_y$ (negative for compression in this calculation); (2) steel in tension with a stress of F_y; (3) concrete in compression with a stress of $-0.95f'_c$; (4) the reinforcing in compression with a stress of $-F_{ysr}$; and (5) the reinforcing in tension with a stress of F_{ysr}. Concrete in tension is assumed to have no stress and thus is excluded from this calculation. The cross-sectional area and centroid of each of these components can be computed by geometry or using CAD software, as shown in Figure 2-18. With these values, the axial load and bending moment about the centroid of the cross section can be computed as shown in Table 2-7. The axial loads in the components sum to 4 kips. The expected value is zero. Deviation from the expected values is due to rounding of the PNA location, cross-sectional properties, etc. Nonetheless, the small value of the resulting axial load confirms that the program correctly identified the neutral axis. The moments in the components sum to 4,357 kip-in. = 363 kip-ft, which is near the value of M_p computed by the program. Again, the deviation is due to rounding. The close result confirms that the program correctly computed the plastic moment.

From the composite column program described in Appendix A:

$M_y = 281$ kip-ft

Like M_p, the value of M_y can be verified with hand calculations. However, unlike M_p, the components of the cross section are not all under uniform stress; therefore, the calculations are somewhat more difficult. For verification of M_y, it is helpful to divide

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Table 2-7. Verification of Plastic Moment, M_p					
Component	***A*, in.^{2}**	**$\bar{y}$, in.**	**Stress, ksi**	***P*, kips**	***M*, kip-in.**
Steel in compression	3.81	6.55	–50.0	–191	1,248
Steel in tension	7.73	–3.22	50.0	387	1,245
Concrete in compression	34.8	5.55	–7.6	–264	1,468
Reinforcing in compression	0.60	5.50	–60.0	–36	198
Reinforcing in tension	1.80	–1.83	60.0	108	198
Total	N/A	N/A	N/A	4	4,357

the cross section into four components: (1) the portion of the steel shape that remains elastic with stress varying linearly from $-F_y$ (negative for compression) at the top to F_y (positive for tension) at the bottom; (2) the portion of the steel shape that has yielded with a unform stress of F_y; (3) the concrete in compression with stress varying linearly from $-0.7f_c'$ at the top to zero at the neutral axis; and (4) the reinforcing with stress that varies based on distance from the neutral axis. The cross-sectional area, centroid, and moment of inertia of each of these components can be computed by geometry or using CAD software as shown in Figure 2-19. Note that the moments of inertia shown in this figure are about the centroid of the individual component, not the whole cross section.

The stress in the reinforcing should be computed from a distribution of strain based on what the concrete experiences. The strain at the extreme concrete compression fiber is:

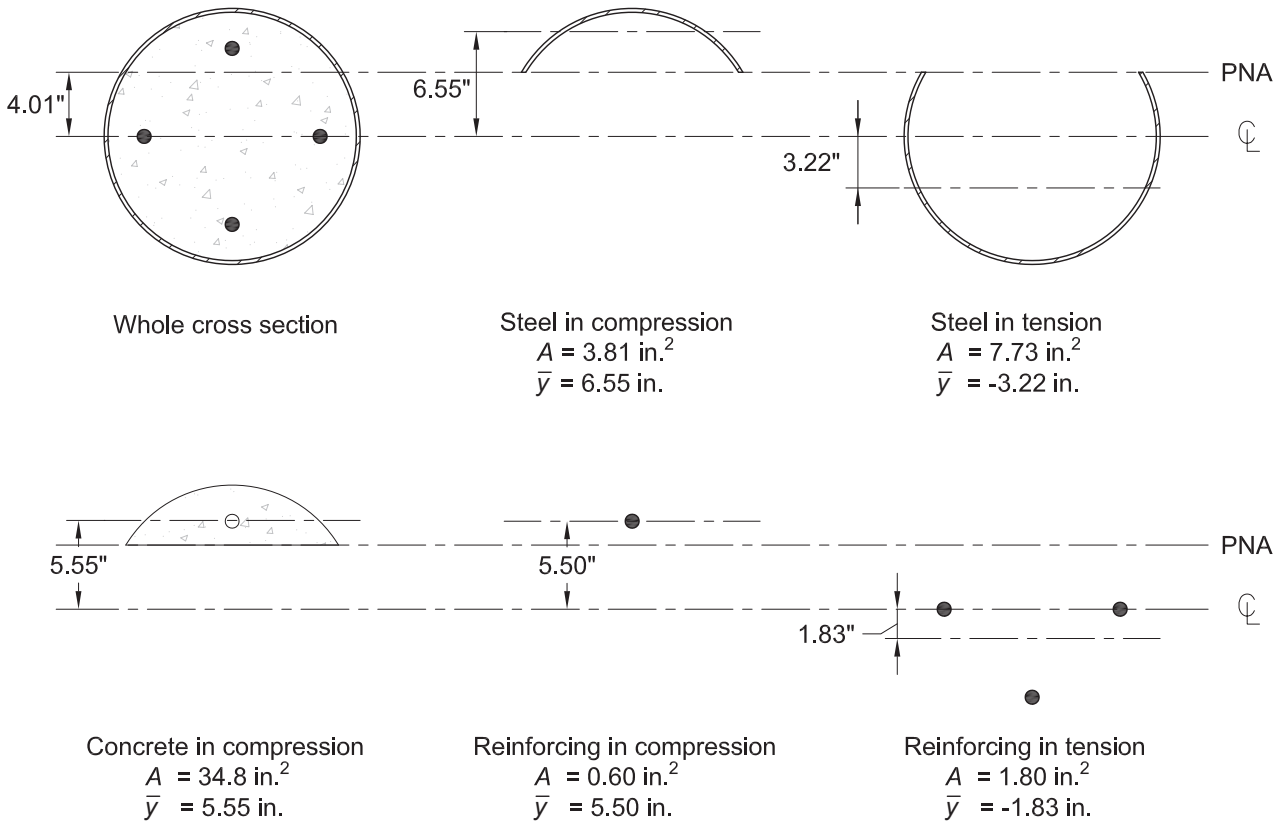


Fig. 2-18. Cross-sectional properties for verification of plastic moment, M_p.

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$$\varepsilon_c = -\frac{0.7 f_c'}{E_c}$$
$$= -\frac{0.7(8 \text{ ksi})}{4{,}940 \text{ ksi}}$$
$$= -0.00113$$

The strain in the reinforcing bar above the centerline is computed based on similar triangles as:

$$\varepsilon = \frac{5.50 \text{ in.} - 2.04 \text{ in.}}{8.00 \text{ in.} - 0.233 \text{ in.} - 2.04 \text{ in.}}(-0.00113)$$
$$= -0.000683$$

Thus, the stress in the reinforcing steel is (as long as the stress does not exceed the yield stress):

$$\sigma = E_s \varepsilon$$
$$= (29{,}000 \text{ ksi})(-0.000683)$$
$$= -19.8 \text{ ksi}$$

Similar calculations can be performed to determine the stress in the other reinforcing bars shown in Figure 2-19.

For the elastic portion of the steel and the concrete, the axial load and bending moment can be back-calculated from principles of structural mechanics. For example, for the steel that remains elastic, the stress is –50 ksi at y = 8.00 in. – 3.27 in. = 4.73 in. and 50 ksi at y = 2.04 in. – 5.96 in. – 3.27 in. = –7.19 in., where y is the distance from the centroid of the portion of steel that remains elastic. Noting that stress in an elastic cross section subjected to axial load and flexure is $P/A - My/I$, this results in a system of two equations and two unknowns:

$$-50 \text{ ksi} = \frac{P}{7.69 \text{ in.}^2} - \frac{M(4.73 \text{ in.})}{108 \text{ in.}^4}$$
$$50 \text{ ksi} = \frac{P}{7.69 \text{ in.}^2} - \frac{M(-7.19 \text{ in.})}{108 \text{ in.}^4}$$

Solving for P and M gives:

$$P = 79.3 \text{ kips}$$
$$M = 906 \text{ kip-in.}$$

However, these forces are about the centroid of the portion of steel that remains elastic and need to be converted to forces about the centroid of the whole cross section for combination with forces from the other components.

$$M = 906 \text{ kip-in.} + (79.3 \text{ kips})(3.27 \text{ in.})$$
$$= 1{,}165 \text{ kip-in.}$$

A summary of the calculations is shown in Table 2-8. The axial loads in the components sum to 4 kips. The expected value is zero. Deviation from the expected value is due to rounding of the PNA location, cross-sectional properties, etc. Nonetheless, the small value of the resulting axial load confirms that the program correctly identified the neutral axis. The moments in the components sum to 3,410 kip-in. = 284 kip-ft, which is different from M_y computed by the program, but only due to rounding in the calculations, confirming that the program correctly computed the yield moment.

Using the values of M_p and M_y from the composite column program, the nominal flexural strength is calculated as:

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Table 2-8. Verification of Yield Moment, M_y					
Component	***A*, in.^{2}**	**$\bar{y}$, in.**	**Stress, ksi**	***P*, kips**	***M*, kip-in.**
Steel that remains elastic	7.69	3.27	–50.0 to 50.0	–79	1,166
Steel yielded in tension	3.85	–6.51	50.0	193	1,253
Concrete in compression	62.9	4.41	–5.60 to 0.00	–146	783
Reinforcing bar above the centerline	0.60	5.50	–19.8	–12	65
Reinforcing bars at the centerline	1.20	0.00	11.7	14	0
Reinforcing bars below the centerline	0.60	–5.50	43.2	26	143
Total	N/A	N/A	N/A	–4	3,410

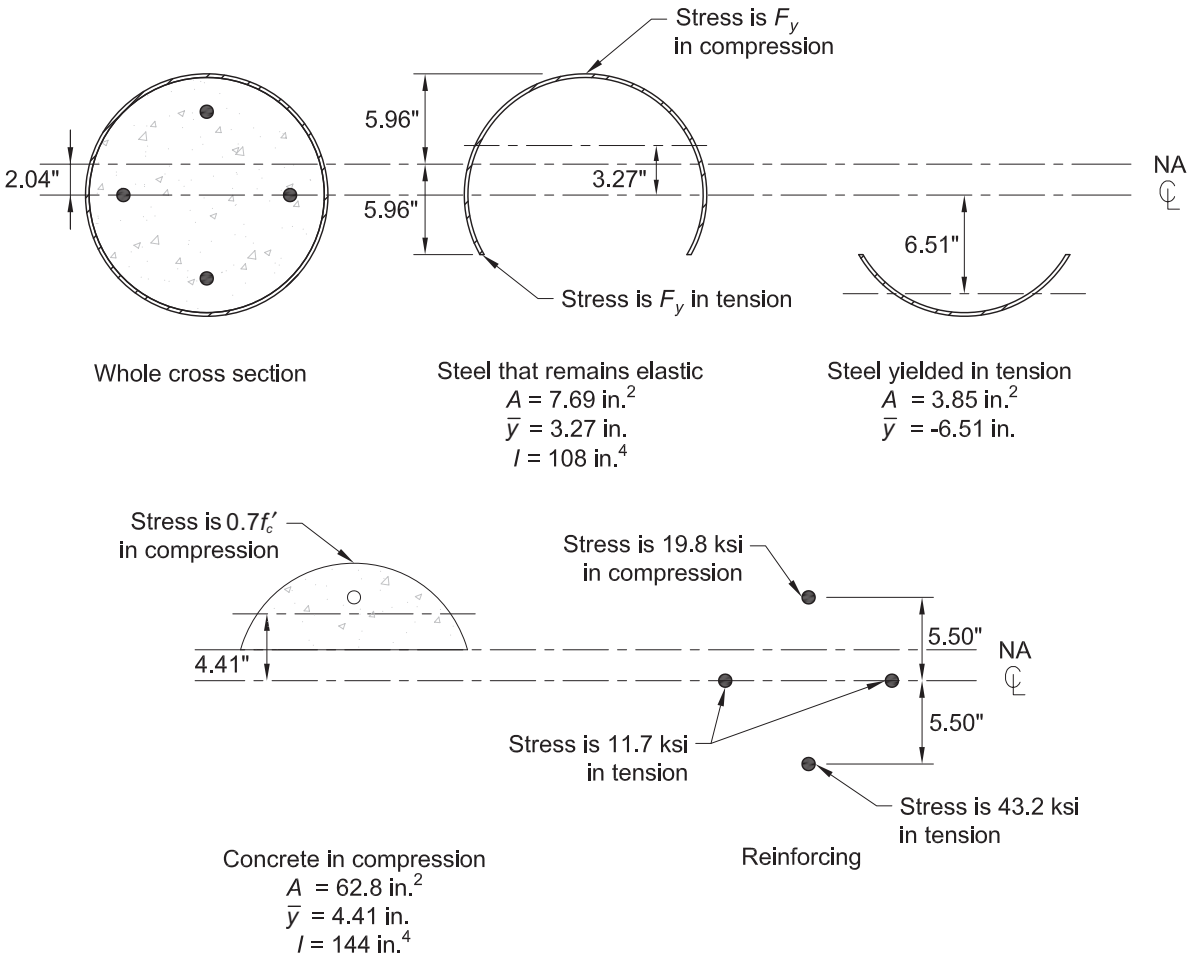


Fig. 2-19. Cross-sectional properties for verification of yield moment, M_y.

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$$M_n = M_p - (M_p - M_y)\left(\frac{\lambda - \lambda_p}{\lambda_r - \lambda_p}\right) \qquad \text{(Spec. Eq. I3-5b)}$$

$$= 364 \text{ kip-ft} - (364 \text{ kip-ft} - 281 \text{ kip-ft})\left(\frac{68.7 - 52.2}{180 - 52.2}\right)$$

$$= 353 \text{ kip-ft}$$

The available flexural strength is:

LRFD	ASD
$\phi_b = 0.90$ $M_c = \phi_b M_n$ $= 0.90(353 \text{ kip-ft})$ $= 318 \text{ kip-ft}$	$\Omega_b = 1.67$ $M_c = \frac{M_n}{\Omega_b}$ $= \frac{353 \text{ kip-ft}}{1.67}$ $= 211 \text{ kip-ft}$

Check combined flexure and axial compression strength

Because the section is noncompact composite for flexure, the interaction strength will be determined using AISC *Specification* Equations I5-1a and I5-1b.

Calculate the coefficients in the interaction equation according to AISC *Specification* Table I5.1.

$$c_{sr} = \frac{A_s F_y + A_{sr} F_{ysr}}{A_c f_c'} \qquad \text{(Spec. Eq. I5-2)}$$

$$= \frac{(11.5 \text{ in.}^2)(50 \text{ ksi}) + (2.40 \text{ in.}^2)(60 \text{ ksi})}{(187 \text{ in.}^2)(8 \text{ ksi})}$$

$$= 0.481$$

$$c_m = \frac{0.95}{c_{sr}^{0.32}} \le 1.67$$

$$= \frac{0.95}{(0.481)^{0.32}} \le 1.67$$

$$= 1.20 < 1.67$$

$$= 1.20$$

$$c_p = \frac{0.27}{c_{sr}^{0.4}}$$

$$= \frac{0.27}{(0.481)^{0.4}}$$

$$= 0.362$$

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Evaluate the interaction equation.

LRFD	ASD
Compare P_u and P_c $\frac{P_u}{P_c} = \frac{680 \text{ kips}}{1{,}230 \text{ kips}}$ $= 0.553 > c_p$ Because $P_u/P_c \geq c_p$, use AISC *Specification* Equation I5-1a: $\frac{P_u}{P_c} + \frac{1-c_p}{c_m}\left(\frac{M_u}{M_c}\right) \leq 1.0$ $= 0.553 + \left(\frac{1-0.362}{1.20}\right)\left(\frac{264 \text{ kip-ft}}{318 \text{ kip-ft}}\right) \leq 1.0$ $= 0.994 < 1.0$ **o.k.**	Compare P_a and P_c $\frac{P_a}{P_c} = \frac{450 \text{ kips}}{820 \text{ kips}}$ $= 0.549 > c_p$ Because $P_a/P_c \geq c_p$, use AISC *Specification* Equation I5-1a: $\frac{P_a}{P_c} + \frac{1-c_p}{c_m}\left(\frac{M_a}{M_c}\right) \leq 1.0$ $= 0.549 + \left(\frac{1-0.362}{1.20}\right)\left(\frac{180 \text{ kip-ft}}{211 \text{ kip-ft}}\right) \leq 1.0$ $= 1.00 \leq 1.0$ **o.k.**

Determine the available shear strength

Calculate the shear area of the steel section.

$$A_v = \frac{2A_s}{\pi}$$
$$= \frac{2(11.5 \text{ in.}^2)}{\pi}$$
$$= 7.32 \text{ in.}^2$$

Because the section is noncompact composite for flexure, $K_c = 1$.

Calculate the nominal shear strength.

$$V_n = 0.6A_vF_y + 0.06K_cA_c\sqrt{f_c'} \qquad \text{(}\textit{Spec.}\text{ Eq. I4-1)}$$
$$= 0.6(7.32 \text{ in.}^2)(50 \text{ ksi}) + 0.06(1)(187 \text{ in.}^2)\sqrt{8 \text{ ksi}}$$
$$= 251 \text{ kips}$$

The available shear strength is:

LRFD	ASD
$\phi_v = 0.90$ $\phi_v V_n = 0.90(251 \text{ kips})$ $= 226 \text{ kips} > 20 \text{ kips}$ **o.k.**	$\Omega_v = 1.67$ $\frac{V_n}{\Omega_v} = \frac{251 \text{ kips}}{1.67}$ $= 150 \text{ kips} > 15 \text{ kips}$ **o.k.**

Conclusion

The filled round composite column is found to be adequate for the required moment and forces.

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2.8 DRIFT AND SERVICEABILITY

In elastic analysis, the stiffness of frame elements is based on the stiffness of the cross section for various modes of deformation, including axial (where stiffness is denoted as *EA*), flexural (*EI*), shear (*GA*), and torsional (*GJ*). For moment-frame systems, the dominant mode of deformation is typically flexural; thus, the flexural stiffness, *EI*, is of prime importance.

Elastic analyses are used for many different purposes in building design, and the appropriate elastic section properties may differ depending on the purpose of the analysis. For strength design, appropriate elastic section properties typically reflect the level of inelasticity at the "ultimate" limit state but also depend on how available strengths are calculated. Requirements for the stiffness used in the determination of required strength are discussed in Section 2.2 of this Design Guide.

When computing wind drifts, appropriate elastic section properties typically reflect the level of inelasticity at a "service loading" level. Another common use of the elastic flexural stiffness is within an eigenvalue analysis to compute the fundamental period of the structure. Perea et al. (2017) compared the natural frequencies of a building with encased composite columns measured from an ambient vibration test to natural frequencies from an eigenvalue analysis of a model of the structure. They determined that modeling the columns with a flexural stiffness of $(EI)_{eff}$, as defined by AISC *Specification* Equation I2-5, resulted in a 10% underestimate of the natural frequency and that modeling the column using the gross flexural stiffness resulted in a 5% overestimate of the natural frequency.

The elastic flexural stiffness of both encased and filled composite members for the specific purpose of determination of lateral drifts under service loads was investigated by Denavit et al. (2018). This study quantified the stiffness that, when used in an elastic analysis of simple frame structures, matched the drift from an inelastic analysis. The resulting stiffness was seen to vary with both the type of loading (e.g., the relative magnitude of axial compressive and bending moment, which influences the location of the neutral axis) and the magnitude of loading. Specific design recommendations were developed for the elastic flexural stiffness for determination of deflections at service loads. The elastic flexural stiffness, $EI_{elastic}$, is proposed as the summation of the flexural stiffness for each of the components but with the concrete component factored down by some amount. That amount depends on the member type and level of axial compression, *P*.

For encased composite members:

When $P < 0.1P_{no}$

$$EI = E_s I_s + E_s I_{sr} + 0.4C_1 E_c I_c \tag{2-7}$$

where the coefficient C_1 is defined by AISC *Specification* Equation I2-6.

When $P \geq 0.1P_{no}$

$$EI = E_s I_s + E_s I_{sr} + 0.4E_c I_c \tag{2-8}$$

For filled composite members:

When $P < 0.1P_{no}$

$$EI = E_s I_s + E_s I_{sr} + 0.4C_3 E_c I_c \tag{2-9}$$

where the coefficient C_3 is defined by AISC *Specification* Equation I2-13.

When $P \geq 0.1P_{no}$

$$EI = E_s I_s + E_s I_{sr} + 0.6E_c I_c \tag{2-10}$$

In the AISC *Specification*, the axial stiffness of composite columns, *EA*, is taken as the summation of the elastic axial stiffnesses of each component, as shown in Equation 2-1. This agrees with most studies as summarized by Schiller et al. (1994).

$$EA = E_s A_s + E_s A_{sr} + E_c A_c \tag{2-1}$$

The shear stiffness, *GA*, is necessary when shear deformations are included in the elastic model (i.e., Timoshenko beam theory). Tomii and Sakino (1979) performed experiments on rectangular filled members and recommended an expression for *GA*.

$$GA = \frac{G_s A_s}{2} + \frac{G_c A_c}{1.2} \tag{2-11}$$

For round filled composite members and encased members, no suitable expressions for *GA* have been found in the literature, so it is recommended to use the shear stiffness of either the steel or concrete section alone.

The torsional stiffness, *GJ*, is necessary for three-dimensional analyses. For filled members, Perea (2010) developed expressions for the elastic torsional stiffness, *GJ*, based on a series of full-scale slender beam-column tests.

$$GJ = G_s J_s + \beta_t G_c J_c \tag{2-12}$$

where the coefficient β_t is given by Equation 2-13 for round members and Equation 2-14 for square and rectangular members.

$$\beta_t = 2\left(\frac{A_s}{A_g}\right) + \frac{2}{5} \leq 1 \tag{2-13}$$

$$\beta_t = \frac{2}{3}\left(\frac{A_s}{A_g}\right) + \frac{2}{15} \leq \frac{1}{3} \tag{2-14}$$

For encased composite members, no suitable expressions for *GJ* have been found in the literature, so it is recommended to use the torsional stiffness of either the steel or concrete section alone. The concrete section will typically provide greater torsional stiffness.

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Chapter 3
Connections

3.1 LOAD TRANSFER

When axial loads are applied to composite columns, the application of load to the various components is rarely in the same proportion that load is shared naturally based on the relative stiffnesses and strengths of the components. Thus, load transfer between the steel and concrete occurs near the connection to achieve a state of internal equilibrium, allowing the member to act in a composite manner (Jacobs and Hajjar, 2010). For example, if a beam is welded to the steel section of a filled composite member, all the load from the beam is applied initially to the steel section. However, stresses at the steel-concrete interface will form that transfer a portion of the applied load to the concrete fill. Design requirements for load transfer are presented in AISC *Specification* Section I6.

3.1.1 Force Allocation

Jacobs and Hajjar (2010) illustrate a design procedure for analyzing force transfer requirements for composite columns. This procedure is summarized in Figure 3-1. The various force allocation methods are illustrated in Table 3-1. The first step of the procedure is to determine the required external force applied to the composite member, P_r. The next step is to determine the required transfer force, V_r'. Equations for V_r' are provided in the AISC *Specification* for cases where the external force is applied to the steel only or to the concrete only.

When the external force is applied entirely to the steel member, the required longitudinal shear force that is transferred to the concrete member can be calculated as:

$$V_r' = P_r\left(1-\frac{F_yA_s}{P_{no}}\right) \qquad (Spec.\ \text{Eq. I6-1})$$

where

A_s = area of steel section, in.^{2}

F_y = specified minimum yield stress of the steel elements, ksi

P_{no} = nominal axial compressive strength without consideration of length effects, kips

P_r = required external force applied to the composite member, kips

V_r' = required transfer force, kips

When the external axial force is applied entirely to the concrete member, the required longitudinal shear force that is transferred to the steel member can be calculated using AISC *Specification* Equations I6-2a and I6-2b.

For encased or filled composite members that are compact composite or noncompact composite:

$$V_r' = P_r\left(\frac{F_yA_s}{P_{no}}\right) \qquad (Spec.\ \text{Eq. I6-2a})$$

For slender filled composite members:

$$V_r' = P_r\left(\frac{F_nA_s}{P_{no}}\right) \qquad (Spec.\ \text{Eq. I6-2b})$$

where

F_n = critical buckling stress of the structural steel section of filled composite members, ksi, determined from AISC *Specification* Equation I2-10 or I2-11

These equations for the required transfer force assume that once in equilibrium, the force in the composite member is shared by individual components in proportion to their strength. While no equations are provided for the case where external force is applied concurrently to both the steel and the concrete, the required transfer force should be computed based on the same assumption. In this case, the portion of the external force applied directly to the concrete, P_{rc}, and the portion of the external force applied directly to the steel section, P_{rs}, are determined separately. Note that $P_r = P_{rc} + P_{rs}$. The required transfer force is computed from one of the following equations.

Equation 3-1 is based on the force in the concrete. The required transfer force is the absolute value of the difference between the portion of external force applied directly to the concrete and that required by AISC *Specification* Equation I6-1.

$$V_r' = \left|P_{rc} - P_r\left(1-\frac{F_yA_s}{P_{no}}\right)\right| \qquad (3\text{-}1)$$

Equations 3-2 and 3-3 are based on the force in the steel. The required transfer force is the absolute value of the difference between the portion of external force applied directly to the steel and that required by AISC *Specification* Equations I6-2a or I6-2b.

For encased or filled composite members that are compact composite or noncompact composite:

$$V_r' = \left|P_{rs} - P_r\frac{F_yA_s}{P_{no}}\right| \qquad (3\text{-}2)$$

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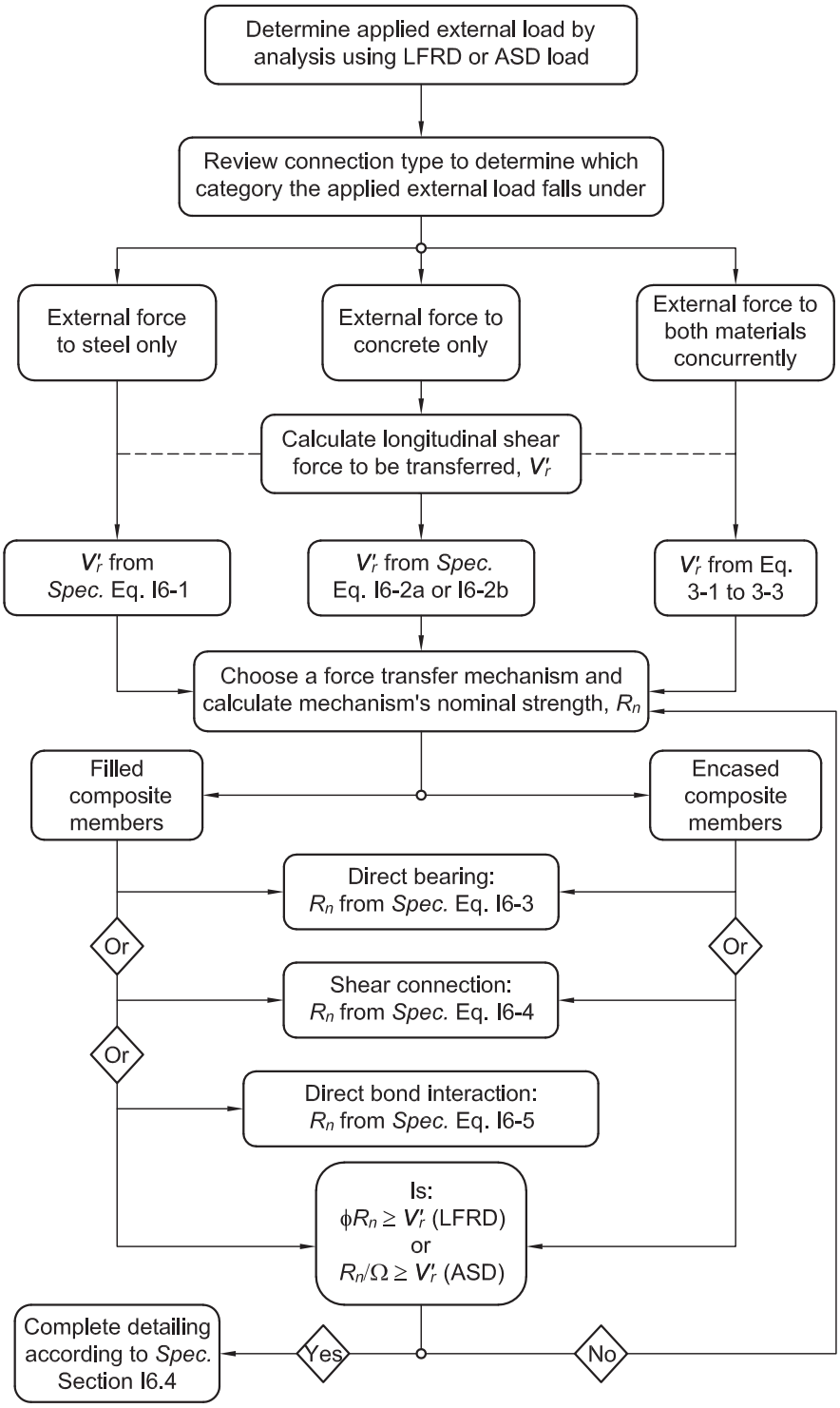
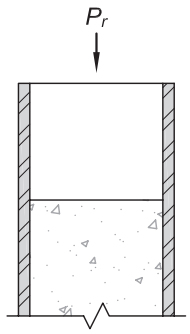
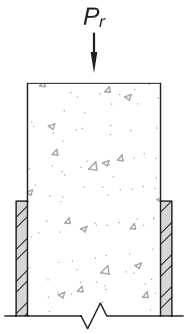
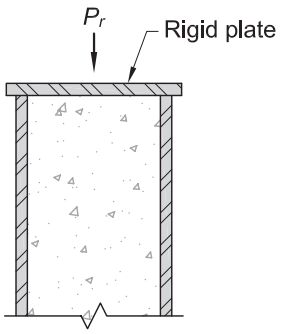
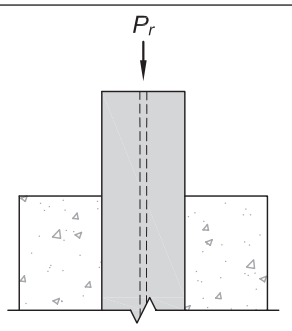
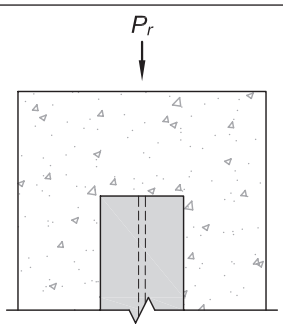
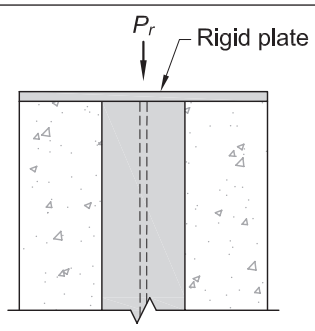
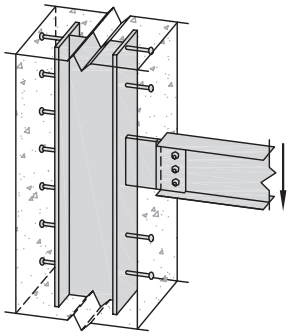
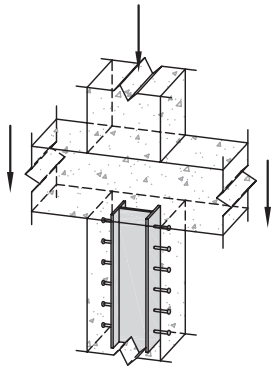
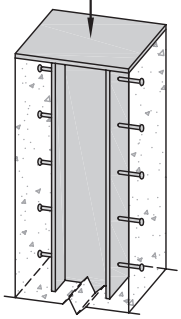


Fig. 3-1. Load transfer flow chart (adapted from Jacobs and Hajjar, 2010).

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Table 3-1. Force Allocation Methods (adapted from Jacobs and Hajjar, 2010)

	External force applied to steel column	External force applied to concrete column	External force applied to both steel and concrete columns
Filled			
Encased			
			

For slender filled composite members:

$$V_r' = \left| P_{rs} - P_r \frac{F_n A_s}{P_{no}} \right| \qquad (3\text{-}3)$$

3.1.2 Force Transfer Mechanisms

Once the required transfer force, V_r', is determined, the next step is to choose a force transfer mechanism such that the available strength of the mechanism (i.e., ϕR_n or R_n/Ω) equals or exceeds the required transfer force. The permitted mechanisms are direct bearing, shear connection, and direct bond interaction. Evaluating multiple force transfer mechanisms and using the one that provides the largest available strength is permitted, but the force transfer mechanisms should not be combined. Table 3-2 provides a summary of the force transfer mechanisms.

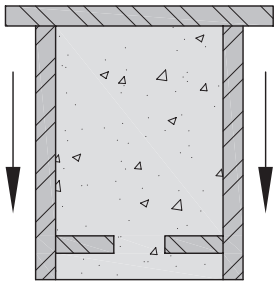
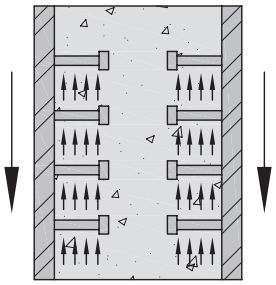
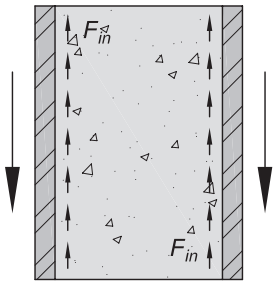
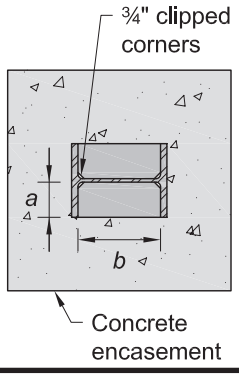
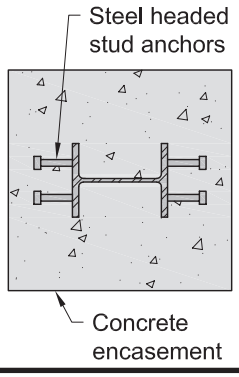
Direct Bearing

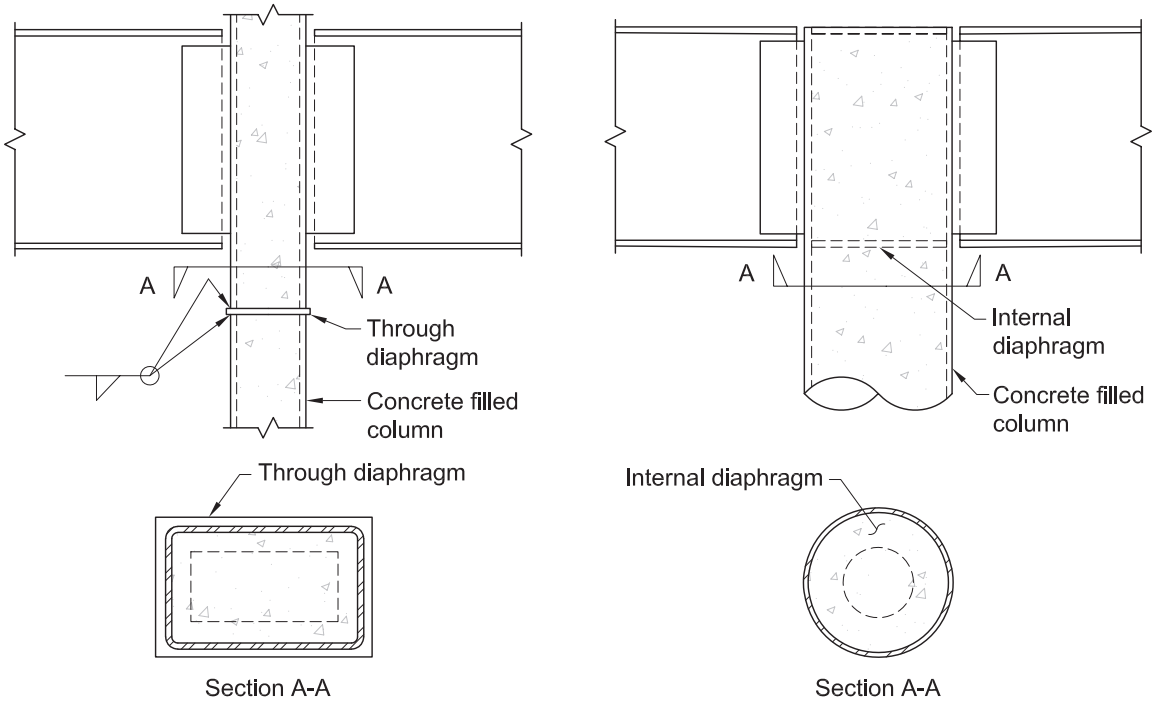
Direct bearing is perhaps the most effective means of force transfer, but it typically requires the installation of components that can add cost. Examples of force transfer via an internal bearing mechanism include internal steel plates within a filled composite member, as shown in Figure 3-2; continuity plates welded to a steel wide-flange column that is encased within concrete; and a stub beam connected to a steel wide-flange column that is encased within concrete.

Strength for direct bearing is based on provisions for bearing on concrete in AISC *Specification* Section J8, but with

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Table 3-2. Force Transfer Mechanisms

	Direct bearing	Shear connection	Direct bond interaction
Filled			
Encased			Not Allowed



(a) Through bearing plate *(b) Internal bearing plate*

Fig. 3-2. Filled member with a direct bearing mechanism within the load introduction length.

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the term $\sqrt{A_2/A_1}$ set equal to 2.0 because of the high level of confinement typical within composite members. The nominal strength for direct bearing is given by AISC *Specification* Equation I6-3.

$$R_n = 1.7 f_c' A_1 \qquad (Spec.\ \text{Eq. I6-3})$$

where

A_1 = loaded area of concrete, in.^{2}

The design bearing strength is determined using a resistance factor of $\phi_B = 0.65$ for LRFD. The allowable bearing strength is determined using a safety factor of $\Omega_B = 2.31$ for ASD.

Shear Connection

Use of steel headed stud anchors is also an efficient and effective means of force transfer if they can be installed. Only anchors installed within the load introduction length, L_{in}, as defined in AISC *Specification* Section I6.4 and shown in Figure 3-3 can be considered to contribute to the force transfer mechanism. The available force transfer strength is simply the summation of the individual strengths of the anchors within the load transfer length as described in AISC *Specification* Equation I6-4.

$$R_c = \sum Q_{cv} \qquad (Spec.\ \text{Eq. I6-4})$$

where

Q_{cv} = available shear strength of a steel headed stud anchor, placed within the load introduction length, kips

The available strengths of the individual anchors are determined in accordance with AISC *Specification* Section I8.3a for steel headed stud anchors or Section I8.3d for steel channel anchors. The resistance factor or safety factor defined in those sections should be applied.

Direct Bond Interaction

Direct bond interaction is only applicable to filled composite members (not encased composite members). The strength for this transfer mechanism is limited but does not require the installation of any component, which can be highly beneficial, especially for smaller sections. Despite the name, direct bond interaction is largely a frictional response. Experiments have shown large variability and strong dependence on the dimensions of the steel section (Zhang et al., 2012).

The nominal strength for direct bond interaction is given by AISC *Specification* Equation I6-5.

$$R_n = p_b L_{in} F_{in} \qquad (Spec.\ \text{Eq. I6-5})$$

where

F_{in} = nominal bond stress, ksi

L_{in} = load introduction length, in., determined in accordance with AISC *Specification* Section I6.4

R_n = nominal bond strength, kips

p_b = perimeter of the steel-concrete bond interface within the composite cross section, in.

The equations for F_{in} in the AISC *Specification* were derived by Zhang et al. (2012).

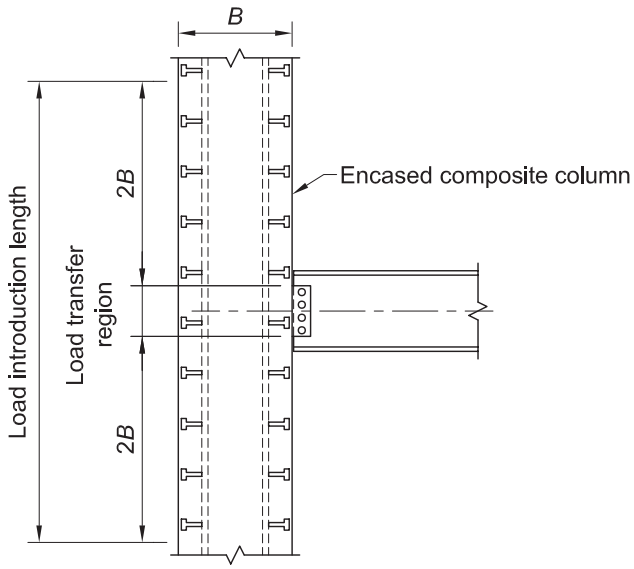


Fig. 3-3. Load introduction length.

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For rectangular cross sections

$$F_{in} = \frac{12t}{H^2} \leq 0.1 \quad (3\text{-}4)$$

For round cross sections

$$F_{in} = \frac{30t}{D^2} \leq 0.2 \quad (3\text{-}5)$$

where

D = outside diameter of round HSS, in.

H = maximum transverse dimension of rectangular steel member, in.

t = design wall thickness of HSS member as defined in AISC *Specification* Section B4.2, in.

The design bond strength is determined using a resistance factor of $\phi_d = 0.50$ for LRFD. The allowable bond strength is determined using a safety factor of $\Omega_d = 3.00$ for ASD.

3.2 SIMPLE CONNECTIONS

3.2.1 Beam to Filled Composite Column Simple Connections

In general, simple connections between a beam and filled composite member are similar in form and design to simple connections between a beam and an HSS or built-up box column. Suitable connections include the single-plate shear connection, as shown in Figure 3-4; single- or double-angle connections; seated connections; and shear end-plate connections. AISC *Manual* Part 10 provides guidance on these connections. Connections that are welded to the column avoid the need for blind bolts or other special techniques or attachments to overcome access issues. Welding to the face of the column typically occurs prior to the column being filled with concrete. However, in the case where the shear connection is welded to the column after the composite column is filled with concrete, consideration should be given to the effect of the heat from the welding operation on the hardened concrete. AISC Design Guide 21 (Miller, 2017) discusses related considerations for welding on in-place embed plates. The constructability issues related to bolted connections exist for HSS or built-up box columns and can be exacerbated in filled composite members because any bolted connections to a filled composite column would need to be made before the concrete is placed. When a beam is connected directly to the steel section as described in this section, the transfer of load from the steel section needs to be considered as described in Section 3.1 of this Design Guide.

3.2.2 Beam to Encased Composite Column Simple Connections

Simple connections between a beam and an encased composite member can be made directly to the structural steel section within the composite member using conventional simple connections such as single-plate shear connections, single- or double-angle connections, or shear end-plate connections. AISC *Manual* Part 10 provides guidance on these connections. Rotational restraint of the beam provided by the concrete is typically neglected, and the beam is designed conservatively as a simple span member. Enclosure face plates can be shop welded to the floor beam at the face of the composite columns to minimize surface cracks in the column concrete at the beam connection. When a beam is connected directly to the steel section as described in this section, transfer of load from the steel section needs to be considered as described in Section 3.1 of this Design Guide.

3.3 NONSEISMIC MOMENT CONNECTIONS

This section focuses on nonseismic moment connections because not all the connections presented have been subjected to the rigorous experimental testing necessary to confirm the ductility requirements for use in special and intermediate moment frames. Some of the connections have been shown to provide sufficient ductility for use in seismic applications.

3.3.1 Beam to Filled Composite Column Moment Connections

In general, steel beam to filled composite column moment connections are similar to those using a steel column with no concrete infill. The majority of research conducted on these types of connections is based on seismic applications for composite special moment frames. However, they can and are often used in nonseismic applications. Although any configuration of moment connection that satisfies the laws of mechanics and equilibrium for transferring forces from the beam to the column can be used, this section will be limited to the following configurations: (1) connections using diaphragms, (2) double split-tee connections, (3) through beam connections, and (4) proprietary moment connections.

Connections Using Diaphragm Plates

Diaphragm plate connections can be utilized with either rectangular or round filled composite members and include either internal diaphragm, external diaphragm, or through-diaphragm plates, as shown in Figure 3-5. Diaphragm plates are aligned with the beam flanges and welded to the steel shape of the composite column. They are used to transfer the moment forces from the beam to or through the column. The flexural strength is provided by the shear resistance of the tension/compression couple provided by the webs of the steel section and formation of a diagonal concrete compression strut in the panel zone. Connections using internal or through diaphragms require openings in the diaphragm plates to allow for proper filling of the steel section with concrete (Lai et al., 2019).

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Zhao et al. (2010) developed design examples for connections using external and through diaphragms. This reference is not based on U.S. codes and standards, but the concepts can be applied to U.S. practice with appropriate judgment.

Internal diaphragm connections—Moment connections to filled HSS or built-up sections using internal diaphragm connections can be utilized to transfer moment to or through the column. The internal diaphragms consist of plates welded to the inside of the HSS or built-up column similar to continuity plates in wide-flange columns. These diaphragm plates should align with the flanges of the moment connected beam. However, aligning these plates with the beam flanges within an HSS column has complications that are not a problem in columns with an open section where the diaphragms are placed as continuity plates or in built-up box columns where the diaphragms are placed during the assembly of the closed section.

The placement of internal diaphragms in HSS cross sections requires one cut of the column at each level so the internal diaphragm plates can be welded inside the column. The two pieces of the column are subsequently rejoined. The internal diaphragm plates in concrete-filled columns also require cutting a hole in the diaphragm plate that allows the pouring of concrete inside the HSS cross section, as illustrated in Figure 3-5(a). These internal diaphragm plates can only be welded to the column on one side requiring either a partial-joint-penetration (PJP) or complete-joint-penetration (CJP) groove weld. The column splice is CJP groove welded with an internal backing plate around the HSS. Welding defects have been observed at the corners when the internal diaphragm is welded all around (Kurobane et al., 2004), but these defects are eliminated when the diaphragm plates have corner clips or when the fillet welds are placed on the flat sides only.

External diaphragm connections—An example of an external diaphragm connection is shown in Figure 3-5(b). These connections utilize diaphragm plates that wrap around the HSS column and are welded to the column walls with CJP groove welds or fillet welds. The plates can be either bolted or welded to the beam. AISC Design Guide 24 (Packer and Olson, 2024) provides additional discussion and a design example for this connection type.

Through diaphragm connections—An example of a through diaphragm connection is shown in Figure 3-5(c). The filled HSS or built-up column in this connection is required to be cut and beveled at the position of the beam flanges so that the diaphragm plates can be placed through and welded to the HSS column with CJP groove welds with an internal backing plate around the HSS. As is the case for internal diaphragms connections, the through diaphragm connection requires a hole in the through plates that allows the pouring of concrete inside the HSS cross section. AISC

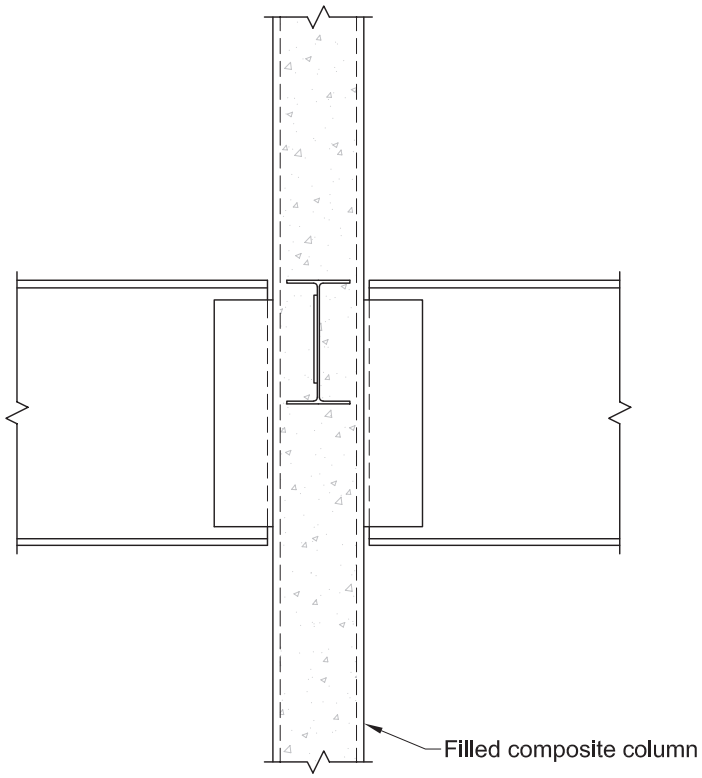
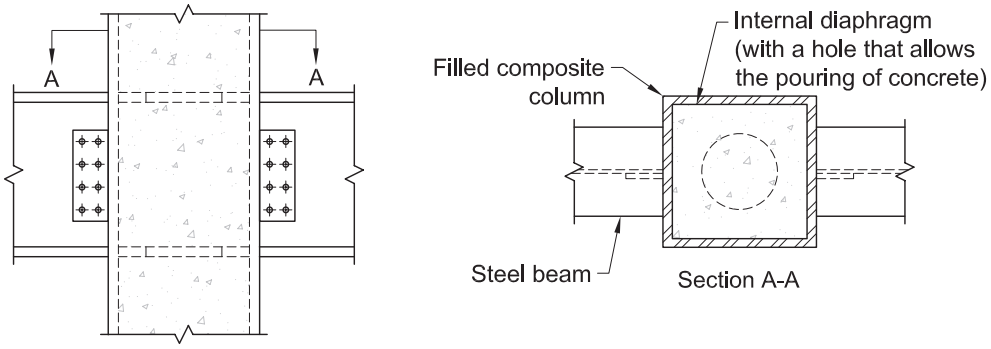
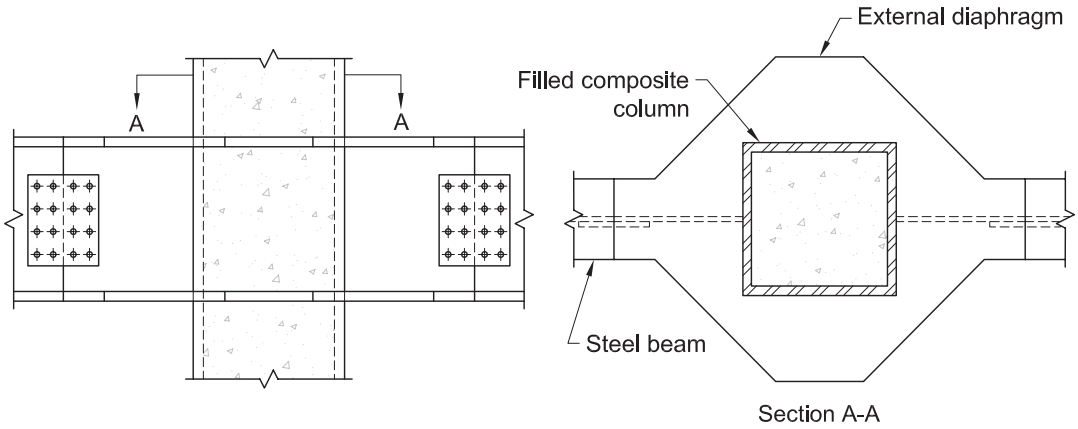


Fig. 3-4. Simple connections of beams to a filled composite column.

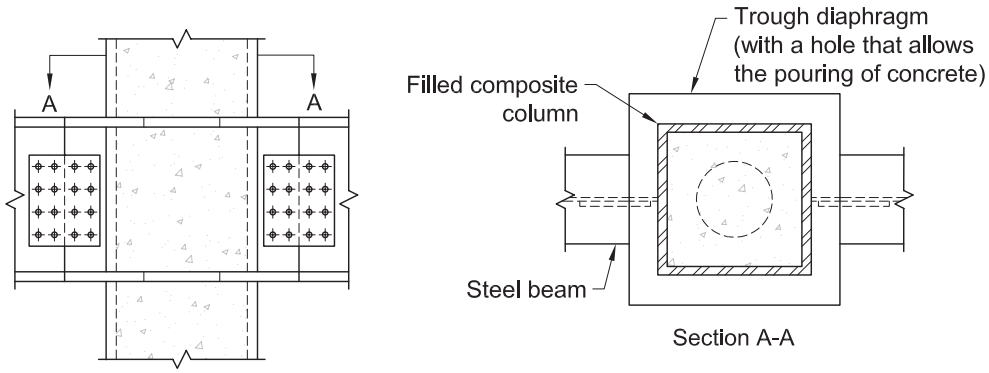
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(a) Connection using internal diaphragm plates



(b) Connection using external diaphragm plates



(c) Connection using through diaphragm plates

Fig. 3-5. Connections using diaphragm plates.

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Design Guide 24 provides additional discussion and a design example for this connection type.

Double Split-Tee Connections

Double split-tee (DST) connections use T-stubs in the form of WT members or built-up plates to connect a wide-flange beam to a square or rectangular filled composite member using high-strength through bolts, as shown in Figure 3-6. The T-stub can then be either bolted or welded to the beam flange. Similar to diaphragm plate moment connections, the flexural strength of the connection is a function of the shear resistance to the tension/compression force couple provided by the webs of the steel section and a concrete compression strut in the panel zone.

Lai et al. (2019) and Lai (2020) provide a design procedure and examples for DST connections; however, the procedure is not entirely consistent with the provisions in AISC *Prequalified Connections for Special and Intermediate Steel Moment Frames for Seismic Applications*, ANSI/AISC 358 (AISC, 2022b), hereafter referred to as AISC 358, for the double-tee moment connection for structural steel columns.

The flexibility of the connection must be carefully considered to ensure that the connections behave as fully restrained moment connections. Elongation of the through bolts and bending of the T-stub flange can be major sources of flexibility.

DST connections cannot be used with round filled composite members.

Through Beam Connections

Through beam connections to filled HSS or built-up members, as shown in Figure 3-7, are often fabricated by cutting I-shaped slots in the wall of the HSS, sliding the beam through the slot, and welding the beam to the HSS with either CJP groove welds or fillet welds (Elremaily and Azizinamini, 2001). The moment is transferred to the column by shear in the beam web, shear in the webs of the HSS, and a compression strut in the concrete core within the panel zone. Experimental tests conducted on these types of connections have indicated that the full flexural strength of the beam can be developed if the strong column/weak beam criteria are used.

Lai et al. (2019) and Lai (2020) provide a design procedure for through beam connections to filled HSS or built-up columns.

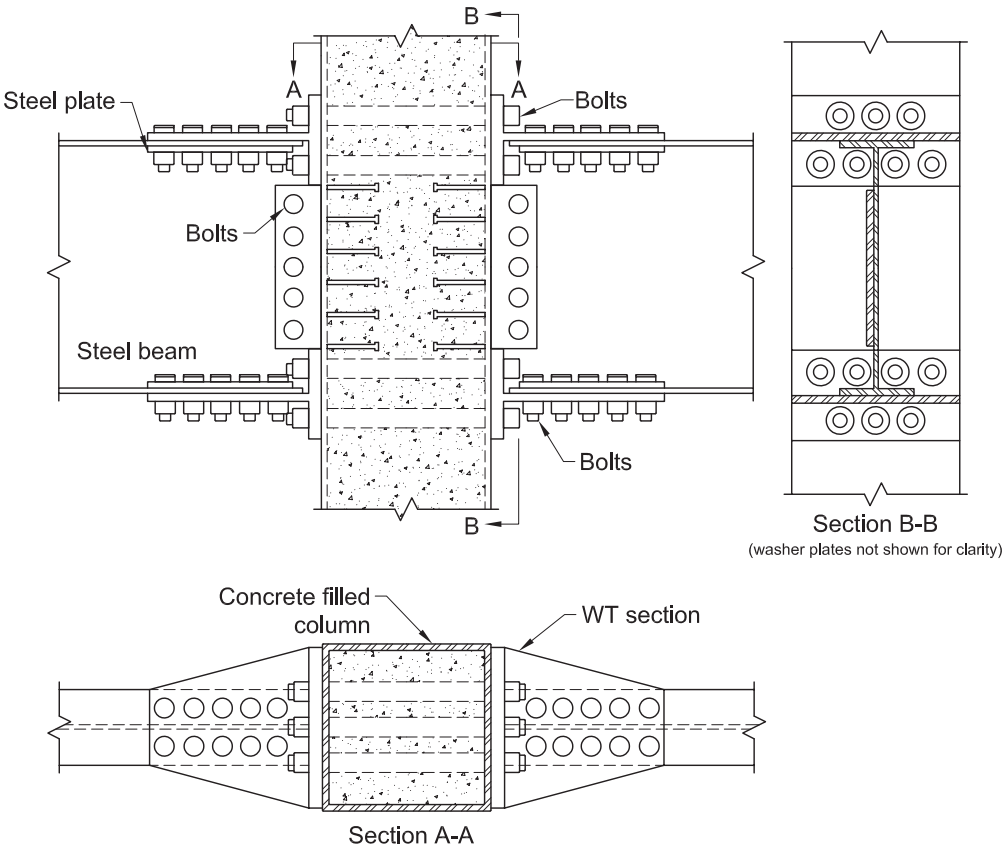


Fig. 3-6. Double split-tee moment connection.

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Proprietary Moment Connections

The ConXL moment connection, as shown in Figure 3-8(a), is a proprietary assembly for the development of a moment connection for a beam-to-filled composite column joint. The connection uses a field-bolted collar flange assembly with either forged or cast steel column fittings that have been shop welded to the columns. Due to the nature of the connection, all moment connected beams must be of the same nominal beam depth. The SidePlate moment connection, as shown in Figure 3-8(b), utilizes interconnecting plates to connect beams to columns including filled composite columns. Field-welded and field-bolted options are available.

Both the ConXL and SidePlate moment connections were developed for seismic applications and are included as prequalified connections for special and intermediate moment frames in AISC 358. However, these connections can be used more broadly, including for nonseismic applications. Both connections will develop the full flexural strength of the connected wide-flange beam to the concrete filled HSS or built-up box column.

Additional information related to the design of these connections can be found on the manufacturers' web sites (ConXtech, 2025; SidePlate, 2025) and in AISC 358.

3.3.2 Beam to Encased Composite Column Moment Connections

In composite frame construction, steel beams often frame into the composite column at the floor level. Sometimes these beams will be simply supported floor beams where conventional double-angle framed beam connections or single-plate shear connections may be utilized, and the resulting axial load in the steel column will be transferred to the concrete. More often, however, the steel beams will be part of the lateral load-resisting system of the building and require a moment connection to the composite column. When the beam is significantly larger than the steel shape of the composite column, it is more practical for the beam to be continuous through the connection and the steel shape of the composite column to be interrupted and CJP groove welded to the flanges of the beam. To increase the speed of erection and minimize field welding, the beam and column are often prefabricated in the shop to form "tree columns" or "tree beams" with field connections at the mid-height of column and midspan of beam using high-strength bolts as shown in Figure 3-9. With proper detailing and design, the connection panel will deform as a monolithic unit fully utilizing both the steel and concrete strengths (Deierlein et al., 1989).

The design of the connection is limited by the vertical bearing strength and panel shear strength to resist the horizontal and vertical couples resulting from the applied beam moments. When the vertical bearing strength of the concrete is of concern, vertical reinforcing in the form of reinforcing bars, structural steel rods, or Dywidag (threaded) bars can be attached to the beam flange to resist compression where the beam flange bears on the concrete and tension at the opposite flange where gaps form between the steel beam and concrete column. Panel shear is resisted by the web of the steel column, concrete compression strut, and concrete compression field. The steel web resists the shear force in a manner consistent with the web panel-zone shear strength defined in the AISC *Specification* Section J10.6. The concrete strut is engaged by the addition of vertical stiffener plates within the web of the steel beam. The design of these stiffener plates determines the contribution of the concrete strut in resisting the horizontal shear forces and they can be extended above and below the beam to engage concrete beyond the beam flanges (Deierlein et al., 1989). The concrete compression field is developed in the region outside the steel beam by horizontal struts that form through bearing against (1) the embedded steel column, (2) stiffener plates extended above and below the beam, or (3) shear studs welded to the beam

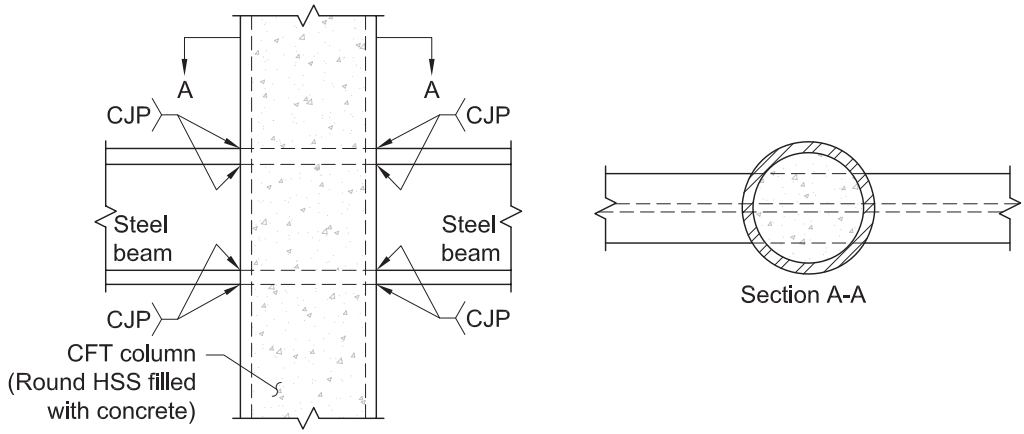


Fig. 3-7. Through beam connection.

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flanges (Sheikh et al., 1989). Deierlein et al. provide design equations and a design example for designing a moment connection to an encased composite column using the above approach.

Figure 3-10 shows an elevation view of an encased composite column that highlights suggested details for the distribution of ties, steel headed stud anchors, and splices along the column. The details of the column joint are shown in Figure 3-11. This detail shows face-bearing plates (confinement stiffener plates) and steel angles with optional studs welded to the beam flanges. These components develop the concrete compression strut that resists the horizontal shear force within the panel zone. In Figure 3-11, the confinement of the concrete within the joint is provided by ties that cross the beam web through holes that are drilled in the web. In addition to hooked ties for the longitudinal reinforcing bars, an alternative to provide shear strength and concrete confinement within the joint is to use band plates that are welded to the steel beams, as shown in Figure 3-12.

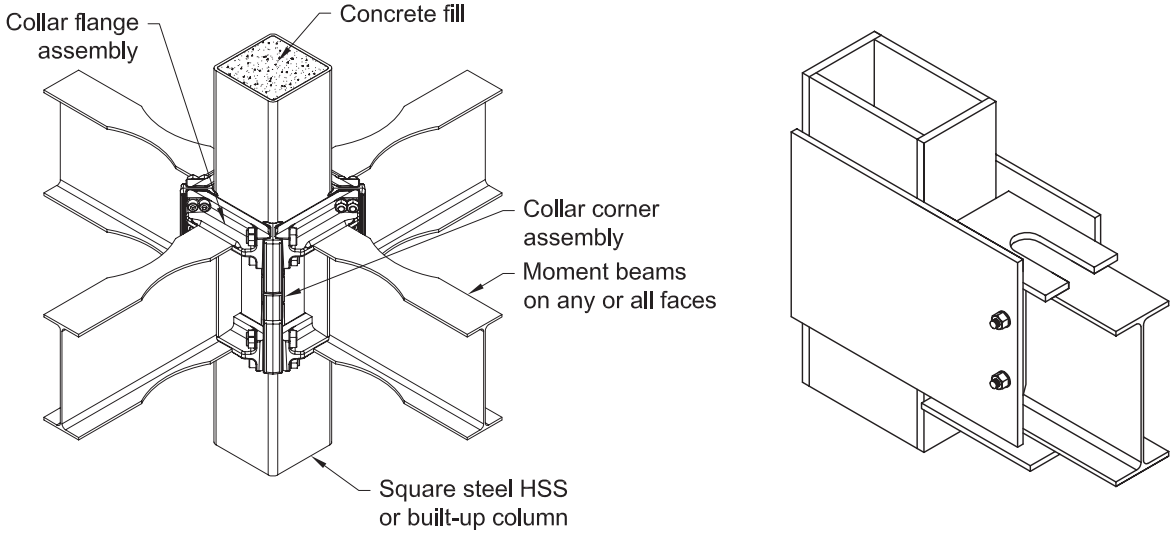
3.4 LOAD TRANSFER FROM COMPOSITE COLUMN TO BASE PLATE AND FOUNDATION

3.4.1 Encased Composite Columns

Typically the smallest possible base plate for the encased steel column in a composite column is specified to accommodate the anchor bolts anchoring it to the foundation during the erection phase. This will cause the least possible amount of interference between the base plate and the dowels coming up from the foundation to splice with the longitudinal vertical bars of the composite column. The design engineer must provide dowels from the composite column to the foundation to transmit the column load in excess of the allowable bearing stress on the foundation concrete, $\phi 0.85 f_c'$ times the effective bearing area—the total composite column area less the area of the encased wide-flange column base plate. In some cases, depending on the base plate size, it may be necessary to add additional foundation dowels to adequately transmit the load carried by the concrete of the composite column. A typical base plate detail is shown in Figure 3-13. Design Example 3.4 illustrates the design of a base plate for an encased composite column.

3.4.2 Filled Composite Columns

Load transfer to base plates and foundations of filled composite columns is not specifically addressed in the AISC *Specification*; however, the AISC *Specification* Commentary references Hitaka et al. (2003) and Stephens et al. (2016), which describe several configurations of filled composite member base connections and their practical limitations.



(a) Assembled ConXL moment connection

(b) Assembled SidePlate connection—uniaxial field-welded configuration

Fig. 3-8. Assembled proprietary moment connections (AISC, 2022b).

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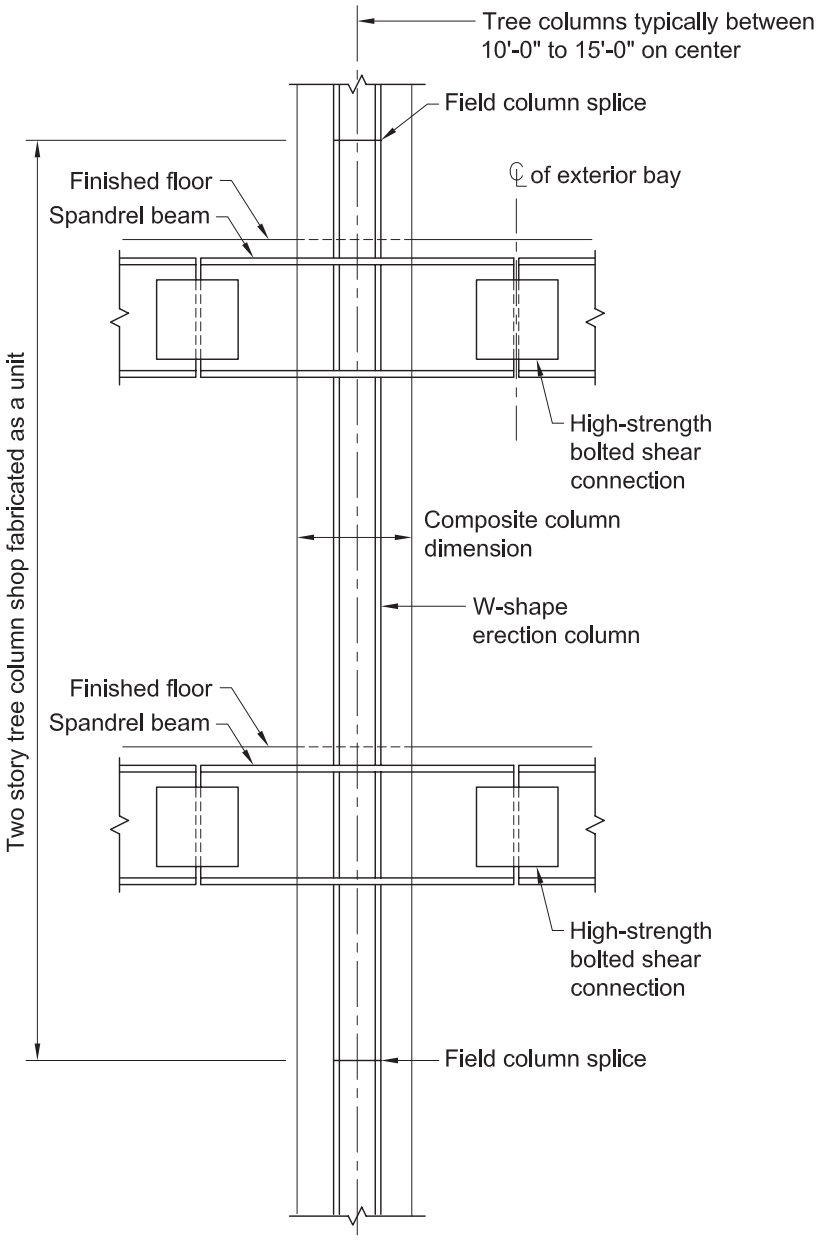


Fig. 3-9. Tree column in an encased composite column frame.

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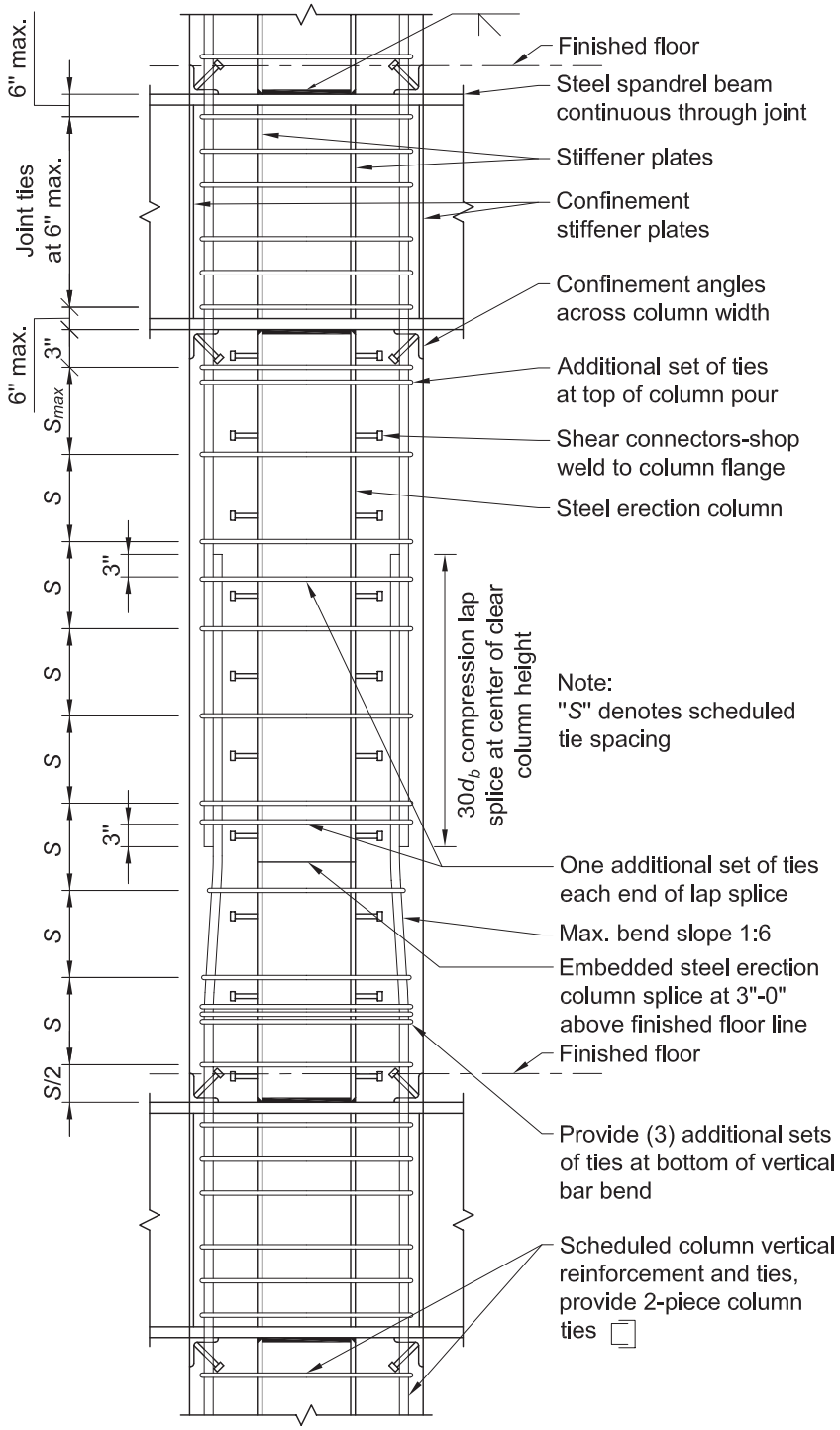


Fig. 3-10. Composite column elevation.

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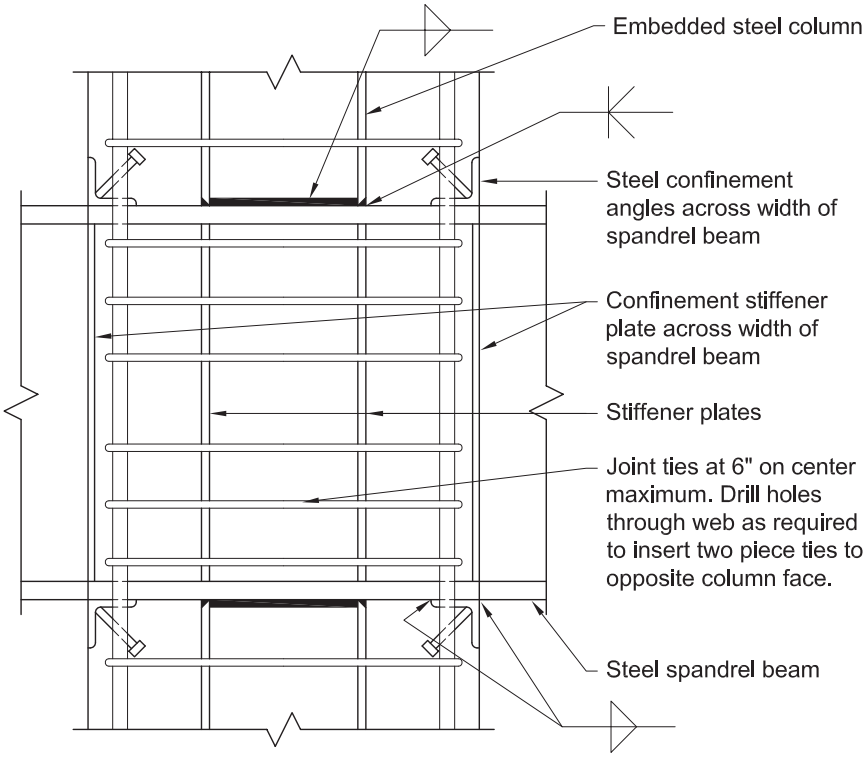


Fig. 3-11. Composite column detail.



Fig. 3-12. Through beam connection.

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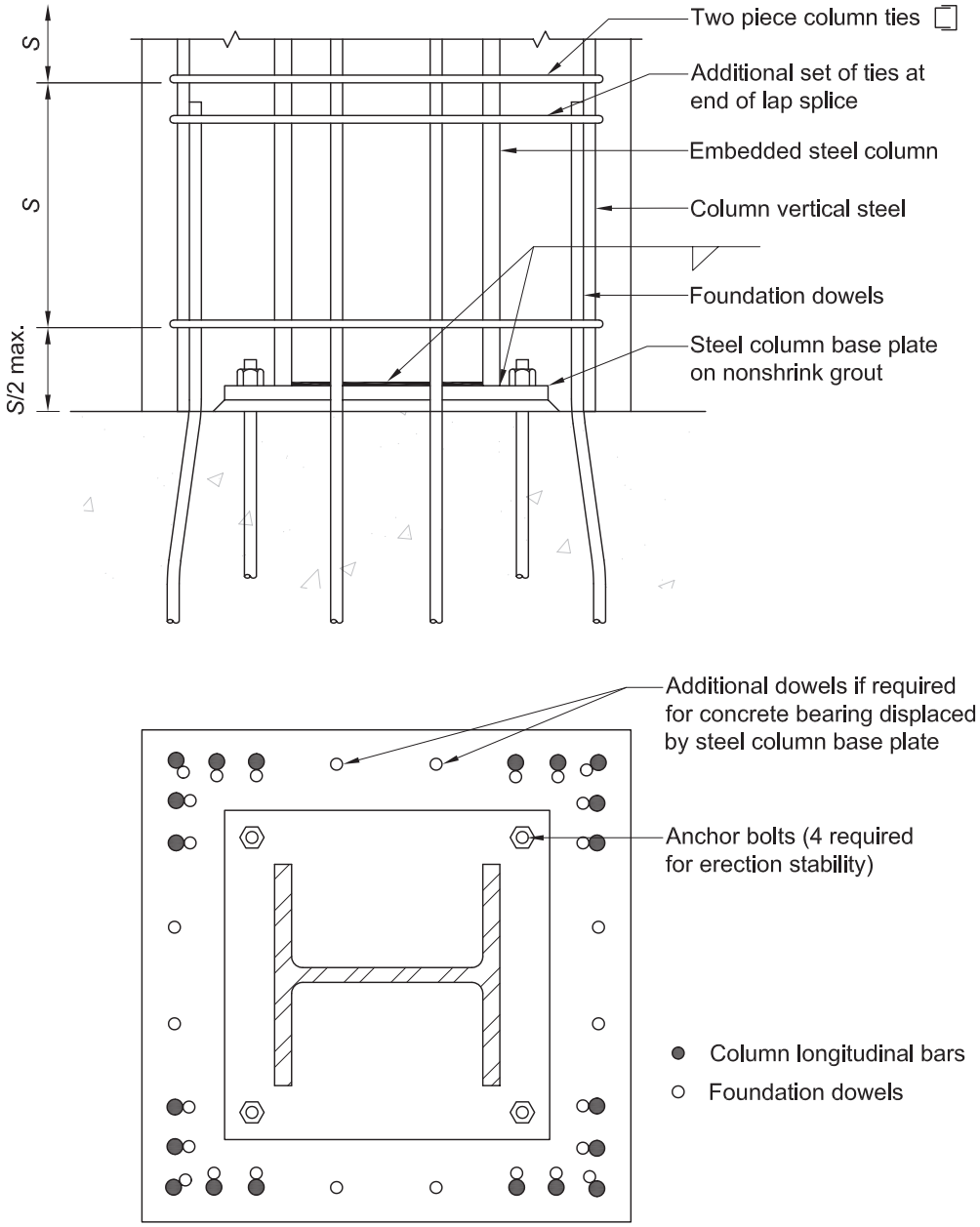


Fig. 3-13. Base plate detail.

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DESIGN EXAMPLE 3.4—Encased Composite Column Base Plate

Design the base plate for the 18 in. × 18 in. composite column shown in Figure 3-14. The column consists of a W10×54 ASTM A992/A992M steel section that is encased in normal-weight concrete (w_c = 145 lb/ft^3) with a specified compressive strength of f'_c = 8 ksi and reinforced with four No. 8 ASTM A615/A615M Grade 60 longitudinal bars. The base plate is ASTM A572/A572M Grade 50 steel, and the foundation is a 48 in. × 48 in. normal-weight concrete footing with a specified compressive strength of f'_c = 3 ksi.

The required strengths are as follows:

LRFD	ASD
P_u = 1,000 kips	P_a = 667 kips

Solution:

From AISC *Manual* Tables 2-4 and 2-5, the material properties are as follows:

Column
A992/A992M
F_y = 50 ksi

Base plate
A572/A572M Grade 50
F_y = 50 ksi

From ASTM A615/A615M, the reinforcement material properties are as follows:

Longitudinal reinforcement
F_{ysr} = 60 ksi

From AISC *Manual* Table 1-1, the geometric properties are as follows:

W10×54 Column
A = 15.8 in.^{2} d = 10.1 in. b_f = 10.0 in. t_f = 0.615 in.

From ACI 318, Appendix B, the geometric properties of single reinforcing bars are as follows:

No. 8 Bar
A = 0.79 in.^{2}

No. 9 Bar
d_b = 1.128 in.
A = 1.00 in.^{2}

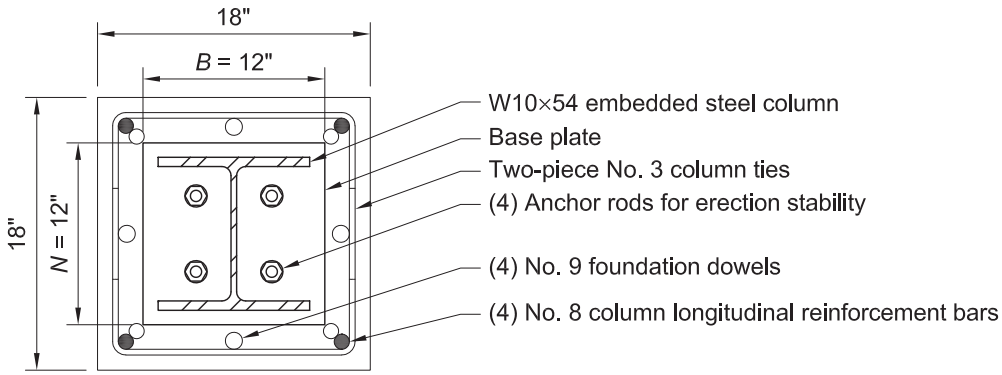


Fig. 3-14. Encased composite column base plate for Example 3.4.

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Calculate cross-sectional properties

The area of the longitudinal reinforcing is:

$$A_{sr} = 4(0.79 \text{ in.}^2)$$
$$= 3.16 \text{ in.}^2$$

The gross area of the composite member is:

$$A_g = (18 \text{ in.})(18 \text{ in.})$$
$$= 324 \text{ in.}^2$$

The area of concrete is:

$$A_c = A_g - A_s - A_{sr}$$
$$= 324 \text{ in.}^2 - 15.8 \text{ in.}^2 - 3.16 \text{ in.}^2$$
$$= 305 \text{ in.}^2$$

The modulus of elasticity of concrete is:

$$E_c = w_c^{1.5}\sqrt{f'_c}$$
$$= (145 \text{ lb/ft}^3)^{1.5}\sqrt{8 \text{ ksi}}$$
$$= 4{,}940 \text{ ksi}$$

Calculate the portion of the axial load carried by the steel member

Using strain compatibility, the axial force on the steel member can be estimated as:

$$P_{rs} = \frac{P_r E_s A_s}{E_s A_s + E_s A_{sr} + E_c A_c}$$

LRFD	ASD
$P_{us} = \dfrac{P_u E_s A_s}{E_s A_s + E_s A_{sr} + E_c A_c}$	$P_{as} = \dfrac{P_a E_s A_s}{E_s A_s + E_s A_{sr} + E_c A_c}$
$= \dfrac{(1{,}000 \text{ kips})(29{,}000 \text{ ksi})(15.8 \text{ in.}^2)}{\left[\begin{array}{l}(29{,}000 \text{ ksi})(15.8 \text{ in.}^2) \\ +(29{,}000 \text{ ksi})(3.16 \text{ in.}^2) \\ +(4{,}940 \text{ ksi})(305 \text{ in.}^2)\end{array}\right]}$	$= \dfrac{(667 \text{ kips})(29{,}000 \text{ ksi})(15.8 \text{ in.}^2)}{\left[\begin{array}{l}(29{,}000 \text{ ksi})(15.8 \text{ in.}^2) \\ +(29{,}000 \text{ ksi})(3.16 \text{ in.}^2) \\ +(4{,}940 \text{ ksi})(305 \text{ in.}^2)\end{array}\right]}$
$= 223 \text{ kips}$	$= 149 \text{ kips}$

Try a 12 in. × 12 in. base plate.

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Check concrete bearing strength of the foundation beneath the base plate

$$A_1 = (12 \text{ in.})(12 \text{ in.})$$
$$= 144 \text{ in.}^2$$

$$A_2 = (48 \text{ in.})(48 \text{ in.})$$
$$= 2{,}300 \text{ in.}^2$$

$$P_p = 0.85 f_c' A_1 \sqrt{A_2/A_1} \le 1.7 f_c' A_1 \quad \text{(Spec. Eq. J8-2)}$$
$$= 0.85(3 \text{ ksi})(144 \text{ in.}^2)\sqrt{(2{,}300 \text{ in.}^2)/(144 \text{ in.}^2)} \le 1.7(3 \text{ ksi})(144 \text{ in.}^2)$$
$$= 1{,}470 \text{ kips} > 734 \text{ kips}$$
$$= 734 \text{ kips}$$

The available bearing strength is:

LRFD	ASD
$\phi_c = 0.65$ $\phi_c P_p = 0.65(734 \text{ kips})$ $= 477 \text{ kips} > 223 \text{ kips}$ **o.k.**	$\Omega_c = 2.31$ $\frac{P_p}{\Omega_c} = \frac{734 \text{ kips}}{2.31}$ $= 318 \text{ kips} > 149 \text{ kips}$ **o.k.**

Determine the base plate thickness

Use the base plate design procedure in AISC *Manual* Part 14.

$$m = \frac{N - 0.95d}{2} \quad \text{(Manual Eq. 14-2)}$$
$$= \frac{12 \text{ in.} - 0.95(10.1 \text{ in.})}{2}$$
$$= 1.20 \text{ in.}$$

$$n = \frac{B - 0.8 b_f}{2} \quad \text{(Manual Eq. 14-3)}$$
$$= \frac{12 \text{ in.} - 0.8(10.0 \text{ in.})}{2}$$
$$= 2.00 \text{ in.}$$

$$n' = \frac{\sqrt{db_f}}{4} \quad \text{(Manual Eq. 14-4)}$$
$$= \frac{\sqrt{(10.1 \text{ in.})(10.0 \text{ in.})}}{4}$$
$$= 2.51 \text{ in.}$$

Conservatively assume $\lambda = 1$.

$$\lambda n' = 1(2.51 \text{ in.})$$
$$= 2.51 \text{ in.}$$

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The critical base plate cantilever dimension, l, is determined as the largest of m, n, and $\lambda n'$.

$$l = \max(m,\ n,\ \lambda n')$$
$$= \max(1.20 \text{ in.},\ 2.00 \text{ in.},\ 2.51 \text{ in.})$$
$$= 2.51 \text{ in.}$$

Calculate the minimum base plate thickness using AISC *Manual* Equations 14-7a and 14-7b. Note that the required strengths are taken as the portion of the axial load caried by the steel member, P_{rs}.

LRFD	ASD
$t_{min} = l\sqrt{\dfrac{2P_{us}}{0.90F_yBN}}$	$t_{min} = l\sqrt{\dfrac{1.67(2P_{as})}{F_yBN}}$
$= (2.51 \text{ in.})\sqrt{\dfrac{2(223 \text{ kips})}{0.90(50 \text{ ksi})(12 \text{ in.})(12 \text{ in.})}}$	$= (2.51 \text{ in.})\sqrt{\dfrac{1.67(2)(149 \text{ kips})}{(50 \text{ ksi})(12 \text{ in.})(12 \text{ in.})}}$
$= 0.658 \text{ in.}$	$= 0.660 \text{ in.}$

Use a ¾-in.-thick base plate.

Check compression transfer from the concrete encasement to the foundation

The portion of the required axial load not carried by the steel member is:

LRFD	ASD
$P_{uc} = P_u - P_{us}$	$P_{ac} = P_a - P_{as}$
$= 1{,}000 \text{ kips} - 223 \text{ kips}$	$= 667 \text{ kips} - 149 \text{ kips}$
$= 777 \text{ kips}$	$= 518 \text{ kips}$

For the concrete encasement, A_1 is taken as the gross area of the column less the area of the base plate.

$$A_1 = A_g - BN$$
$$= 324 \text{ in.}^2 - (12 \text{ in.})(12 \text{ in.})$$
$$= 180 \text{ in.}^2$$

The nominal bearing strength is:

$$P_p = 0.85f'_cA_1\sqrt{A_2/A_1} \le 1.7f'_cA_1 \qquad \text{(}\textit{Spec.}\text{ Eq. J8-2)}$$
$$= 0.85(3 \text{ ksi})(180 \text{ in.}^2)\sqrt{(2{,}300 \text{ in.}^2)/(180 \text{ in.}^2)} \le 1.7(3 \text{ ksi})(180 \text{ in.}^2)$$
$$= 1{,}640 \text{ kips} > 918 \text{ kips}$$
$$= 918 \text{ kips}$$

The available bearing strength is:

LRFD	ASD
$\phi_c = 0.65$	$\Omega_c = 2.31$
$\phi_cP_p = 0.65(918 \text{ kips})$	$\dfrac{P_p}{\Omega_c} = \dfrac{918 \text{ kips}}{2.31}$
$= 597 \text{ kips} < 777 \text{ kips}$	$= 397 \text{ kips} < 518 \text{ kips}$

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Because the available bearing strength is less than the required concrete compressive transfer force, dowels are required to transfer the remaining compressive force.

Design the column-to-foundation dowels

Calculate the required dowel area as the difference between the required concrete compressive transfer force and the available bearing strength divided by the dowel available stress.

LRFD	ASD
$A_{d,req} = \dfrac{P_{uc} - \phi_c P_p}{\phi_c F_{ysr}}$ $= \dfrac{777\,\text{kips} - 597\ \text{kips}}{0.90(60\ \text{ksi})}$ $= 3.33\ \text{in.}^2$	$A_{d,req} = \dfrac{P_{ac} - P_p/\Omega_c}{F_{ysr}/\Omega_c}$ $\dfrac{518\ \text{kips} - 397\ \text{kips}}{(60\ \text{ksi})/1.67}$ $= 3.37\ \text{in.}^2$

Try four No. 9 dowels.

$$A_d = 4(1.00\ \text{in.}^2)$$
$$= 4.00\ \text{in.}^2$$

LRFD	ASD
$A_d = 4.00\ \text{in.}^2 > A_{d,req} = 3.33\ \text{in.}^2$ **o.k.**	$A_d = 4.00\ \text{in.}^2 > A_{d,req} = 3.37\ \text{in.}^2$ **o.k.**

From ACI 318, Section 16.3.4.1, the minimum area of reinforcement crossing the column-to-foundation interface must be at least $0.005A_g$.

$$A_{d,min} = 0.005A_g$$
$$= 0.005(324\ \text{in.}^2)$$
$$= 1.62\ \text{in.}^2 < 4.00\,\text{in.}^2 \quad \textbf{o.k.}$$

From ACI 318, Section 25.4.9, the development length, l_{dc}, for deformed bars in compression is calculated as follows for the foundation. Note, $\lambda = 1.0$ and $\psi_r = 1.0$.

$$l_{dc} = \max\left[\left(\frac{F_{ysr}\psi_r}{50\lambda\sqrt{f_c'}}\right)d_b,\ 0.0003F_{ysr}\psi_r d_b,\ 8\ \text{in.}\right]$$
$$= \max\left[\left(\frac{(60{,}000\ \text{lb/in.}^2)(1.0)}{50(1.0)\sqrt{3{,}000\ \text{lb/in.}^2}}\right)(1.13\ \text{in.}),\ 0.0003(60{,}000\ \text{lb/in.}^2)(1.0)(1.13\ \text{in.}),\ 8\ \text{in.}\right]$$
$$= \max(24.8\ \text{in.}, 20.3\ \text{in.}, 8\ \text{in.})$$
$$= 24.8\ \text{in.}$$

From ACI 318, Section 25.4.9, the development length, l_{dc}, for deformed bars in compression is calculated as follows for the column. Note, $\lambda = 1.0$ and $\psi_r = 1.0$.

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$$l_{dc} = \max\left[\left(\frac{F_{ysr}\psi_r}{50\lambda\sqrt{f_c'}}\right)d_b,\ 0.0003F_{ysr}\psi_r d_b,\ 8\text{ in.}\right]$$

$$= \max\left[\left(\frac{(60{,}000\text{ lb/in.}^2)(1.0)}{50(1.0)\sqrt{8{,}000\text{ lb/in.}^2}}\right)(1.13\text{ in.}),\ 0.0003(60{,}000\text{ lb/in.}^2)(1.0)(1.13\text{ in.}),\ 8\text{ in.}\right]$$

$$= \max(15.2\text{ in.}, 20.3\text{ in.}, 8\text{ in.})$$

$$= 20.3\text{ in.}$$

Use four No. 9 dowels embedded 25 in. into the foundation and embedded 21 in. into the column.

Conclusion

A PL¾ in. × 12 in. × 12 in. base plate with four No. 9 dowels embedded 25 in. into the foundation and embedded 21 in. into the column is found to be adequate to transfer the required forces from the encased composite column to the foundation.

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Chapter 4
Practical Design Considerations

4.1 FIRE RESISTANCE

International Building Code (ICC, 2024), hereafter referred to as the IBC, Section 704 requires that all structural members achieve a fire resistance rating as specified in IBC Table 601. Further, IBC Section 703 requires that the fire resistance rating be determined in accordance with the test procedures set forth in ASTM E119 (ASTM, 2024b), UL263 (UL, 2022), or IBC Section 703.3. A 3 hour rating is the greatest required for building structures.

The concrete in filled composite members not only increases strength and stiffness but also increases fire resistance. Often, unfilled built-up box or HSS members will require external fire protection in the form of spray-on fireproofing or encasement with gypsum board to meet the fire resistance requirements of the IBC. However, filled composite members can, in some circumstances, achieve the same fire resistance as protected built-up members without the need for external fire protection. As a result, properly designed filled composite members can economically enable architectural and structural design with visible steel and increased usable space in the building (Kodur and Fike, 2009). AISC Design Guide 19, *Fire Resistance of Structural Steel Framing*, discusses the need to provide vent holes at the top and bottom of filled composite members to relieve steam pressure from accumulating inside the steel section resulting from the exposure to fire (Ruddy et al., 2003).

Encased composite columns have an inherent resistance to the elevated temperatures produced in a fire by virtue of the minimum required concrete cover to the reinforcing steel and structural steel, much like reinforced concrete columns. It is standard practice to provide a minimum of 1½ in. of concrete cover to the reinforcing steel of a composite column. Concrete cover is specified in ACI 318, Section 20.5.1.3.

AISC *Specification* Appendix Section 4.3.2b includes criteria for the design and evaluation of both filled composite columns and encased composite columns and provides four limitations to the application of the included equations. One of these, for example, limits the material and geometric properties to those shown in AISC *Specification* Appendix Tables A-4.3.5 or A-4.3.5M. This table does not allow the use of high strength concrete or columns with an effective length that exceeds 15 ft. Often, a significant benefit to using filled composite columns is for those conditions where high strength concrete and large effective lengths are necessary. In those cases, and others where the limits of Appendix Section 4.3.2a are not met, the AISC *Specification* provides an alternative performance-based design approach in Appendix Section 4.2 defined as Structural Design for Fire Conditions by Analysis that are not subject to the limitations of Tables A-4.3.5 and A-4.3.5M. The specifics of this approach, however, are beyond the scope of this Design Guide. AISC Design Guide 19 includes additional information and design examples.

4.2 REINFORCEMENT DETAILS

4.2.1 Encased Composite Columns

Longitudinal Reinforcing Bar Arrangement

Encased composite columns can take on just about any shape for which formwork can be made and stripped. They can be square, rectangular, round, triangular, or any other configuration, with just about any corresponding reinforcing bar arrangement common to concrete columns. For use in composite frame construction, however, square or rectangular columns are the most practical shape, with bar arrangements tending to place the vertical reinforcing bars at or near the four corners of the cross section. AISC *Specification* Section I2.1b only addresses the case of doubly symmetric sections, so bar layouts for such designs need to be doubly symmetric. Although, any vertical and transverse reinforcing layout that conforms to the requirements of ACI 318 is acceptable; Figure 4-1 shows preferred arrangements which allow beams to frame into the encased steel shape without interrupting the continuous vertical bars, while also generating the maximum design strength for the column. Cross ties meeting the requirements of ACI 318 must be used, including when the vertical reinforcing is concentrated at the corners of the concrete section. Figure 4-2 shows an example detail with restraining bars that are placed to meet the requirements of ACI 318. These restraining bars are discontinuous to allow steel beams to pass through and connect directly to the steel shape.

AISC *Specification* Section I2.1a(b) requires that the detailing and placement of longitudinal reinforcing, including bar spacing and concrete cover requirements, must conform to ACI 318. The area of continuous longitudinal reinforcing must be at least 0.4% of the gross area of the composite member per AISC *Specification* Section I2.1a(c). These spacing and cover requirements are listed here and shown diagrammatically in Figure 4-3:

- Minimum concrete cover to longitudinal and transverse reinforcement must comply with ACI 318, Table 20.5.1.3.1.

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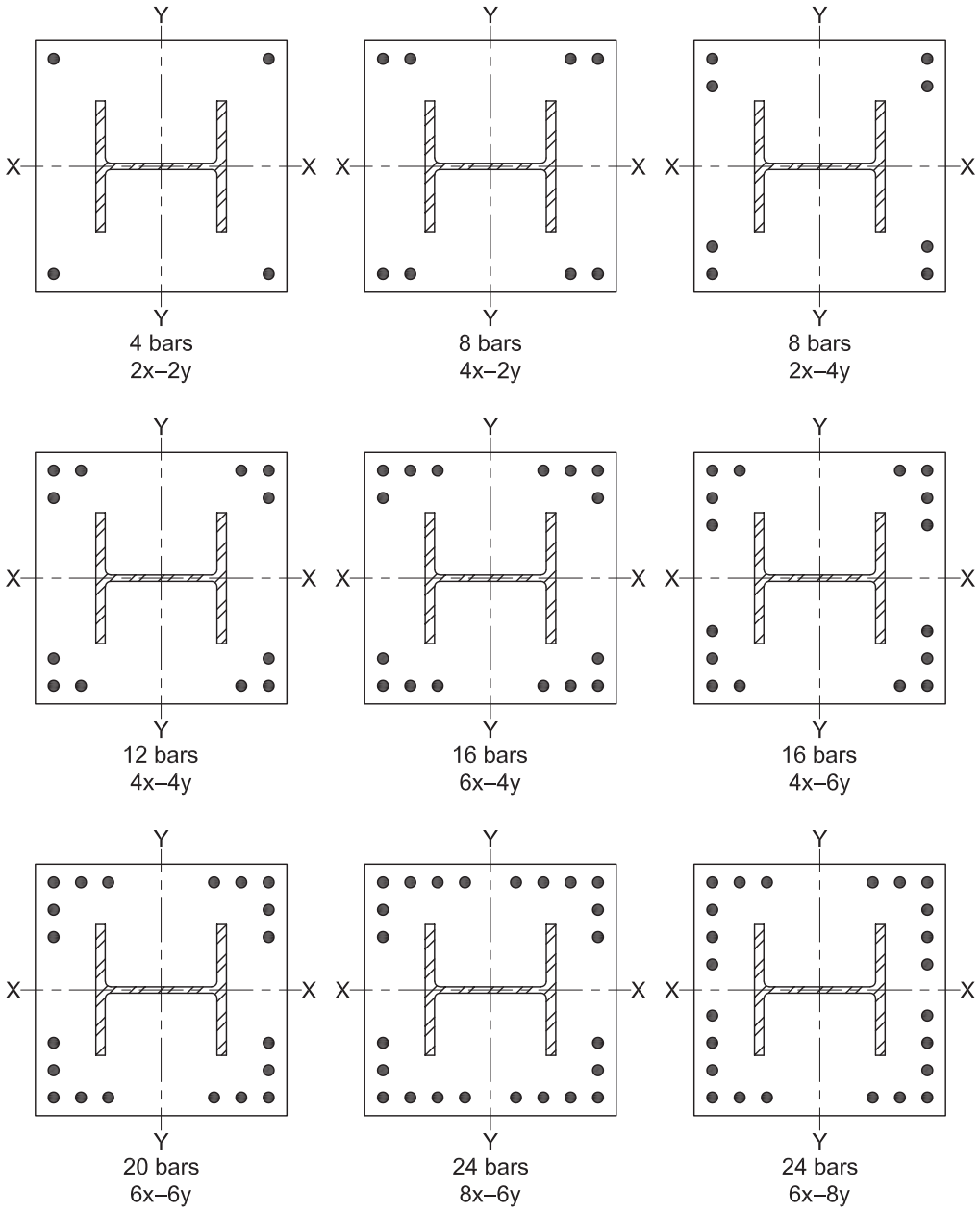


Fig. 4-1. Longitudinal bar arrangement in encased composite columns.

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- The clear distance between longitudinal bars must not be less than the greater of 1½ bar diameters, 1½ in., and ⁴⁄₃ of the nominal maximum coarse aggregate size (ACI 318, Section 25.2.3). This distance is shown as s_1 in Figure 4-3. The clear distance limitations apply also to contact lap splices and adjacent bars (ACI 318, Section 25.5.1.2).
- The clear distance between longitudinal bars and the steel shape must not be less than the greater of 1½ bar diameters and 1½ in. according to AISC *Specification* Section I2.1e and I2.2e. This distance is shown as s_2 in Figure 4-3.

Transverse Reinforcing

Reinforcing steel cages (longitudinal bars and ties) are usually set around the steel column after it has been erected. Because the steel column is erected in an earlier erection sequence, open U-shaped or single leg ties are suitable for composite columns. For round members, circular ties or hoops are usually more effective. Ties are used to provide lateral stability of the longitudinal bars and confinement of the concrete. The requirements of AISC *Specification* Section I2.1a(b) are as follows:

- Minimum tie bar size and spacing must be a No. 3 bar spaced at a maximum of 12 in. or a No. 4 bar spaced at a maximum of 16 in.
- The spacing of the transverse reinforcing must not be greater than 0.5 times the least dimensions of the cross section.

Detailing of transverse reinforcement must conform to the following requirements from ACI 318:

- The clear spacing of ties must be at least ⁴⁄₃ times the nominal maximum aggregate size per ACI 318, Section 25.7.2.1(a).
- The center-to-center spacing of the transverse reinforcing must not be greater than 16 longitudinal bar diameters or 48 tie bar diameters per ACI 318, Section 25.7.2.1(b).
- Ties must be at least No. 4 in size for No. 11, No. 14, No. 18, and bundled longitudinal bars and No. 3 in size for No. 10 bars and smaller per ACI 318, Section 25.7.2.2.
- Rectilinear ties must be arranged such that every corner and alternate longitudinal bar should have lateral support provided by a corner of a tie, with an included angle of not more than 135°, and no bar should be farther than 6 in. clear on each side along the tie from a laterally supported bar per ACI 318, Section 25.7.2.3.
- A lap splice of two pieces of an open U-shaped tie must be at least equal to 1.3 times the tensile development length, Class B Lap Splice, for the specified yield strength per ACI 318, Section 25.7.1.7.

Suggested details for composite column ties are shown in Figure 4-2.

Longitudinal Reinforcing Bar Splices

The requirements for splicing vertical longitudinal reinforcing bars for encased composite columns should follow the

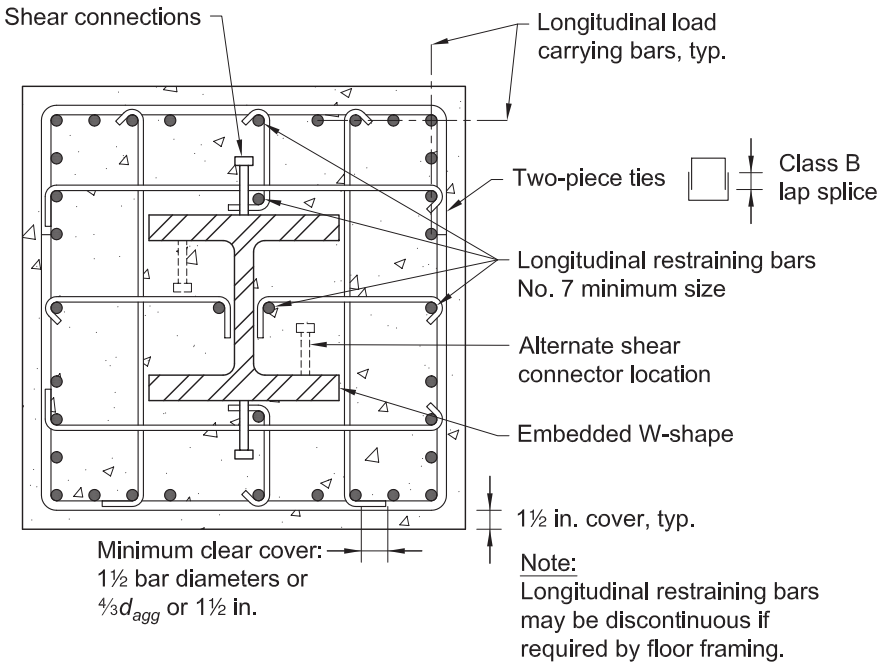


Fig. 4-2. Encased composite member vertical and transverse reinforcement.

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same rules as applied for conventional reinforced concrete columns as specified in ACI 318, Section 25.5. Several additional comments should be made regarding composite columns. First, additional vertical longitudinal "restraining bars" are often required with encased columns to limit the maximum clear spacing between laterally unsupported bars to 6 in. as limited by ACI 318, Section 25.7.2.3(b). These bars would be interrupted at the floor levels where the floor framing member must extend and connect to the steel column. Because they are not continuous, they should not be considered to contribute to the strength of the composite section. However, because they are provided to satisfy the maximum spacing between laterally supported longitudinal reinforcing bars, they should be enclosed with ties. A No. 7 bar is the minimum size recommended for these restraining bars. Second, it is suggested that, in high-rise composite frame construction, the vertical bar splices be located at the middle clear height of the composite column. This point is usually near the inflection point (point of zero moment) of the column where the more economical compression lap splices or compression butt splices may be used. The more expensive tension lap or tension butt splices may be required if splices are made at the floor line.

Encased Composite Columns with Built-Up Shapes

In a composite section that consists of two or more built-up shapes encased in concrete, the built-up shapes must be interconnected with lacing, tie plates, or comparable components to prevent buckling of the individual shapes due to construction loading applied prior to hardening of the concrete.

4.2.2 Filled Composite Members

Filled composite members do not require longitudinal reinforcement, although it can be used for additional strength. When reinforcement is included, it should conform to the following:

- Transverse reinforcement in the form of ties or hoops should consist of either No. 3 bars spaced at a maximum of 12 in. or No. 4 bars spaced at 16 in.
- The amount of longitudinal reinforcement should not be less than 0.4% nor exceed 8% of the gross area of the concrete section.
- The clear distance between longitudinal bars and the structural steel section should not be less than 1½ bar diameters or 1½ in.
- The requirements for splicing longitudinal reinforcing bars for filled composite columns should follow the same rules as applied for conventional reinforced concrete columns as specified in ACI 318, Section 25.5.

4.3 CONCRETE PLACEMENT IN FILLED COMPOSITE MEMBERS

Casting the concrete in filled composite members is a critical task in the construction sequence. Poorly placed concrete can have detrimental effects on the strength and stiffness of the composite member. Thus, care must be taken to ensure good quality and integrity of the concrete within the hollow member after the concrete has hardened.

Common practice is to place the concrete in the HSS or box section by either pouring the concrete through the open section at the top or pumping the concrete from a cut opening close to the bottom. Pumping from the bottom requires a repair (typically welded) to the pump opening once the concrete hardens. Pouring concrete from the top avoids the need to cut and repair the steel shape, but it may require careful concrete placement to avoid segregation and voids as the concrete falls into place. These issues can be avoided by introducing the pumping hose as deep as possible into the steel shape to minimize the dropping distance, using special concrete—for example, self-consolidating concrete—that minimizes segregation, or both. When using conventional

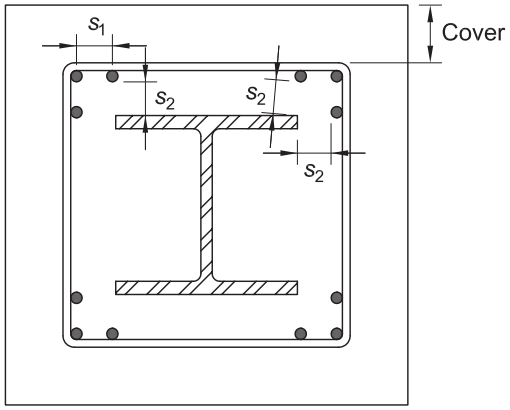


Fig. 4-3. Composite column cover and bar spacing requirements.

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concrete, care must be taken to ensure consolidation regardless of whether the concrete is poured or pumped. Self-consolidating concrete may be particularly advantageous for filled composite members.

When the concrete is placed in the steel shape—with either technique—the concrete introduces hydrostatic pressures on the walls that deform the steel shape. In high-rise buildings, concrete pouring for several stories is common to speed up the construction process; however, the magnitude of hydrostatic pressure resulting from the wet concrete increases with pour height. The hydrostatic pressure is not critical for round HSS due to shape efficiency (Bergmann et al., 1995; Uy and Das, 1999; Perea, 2010; Leon et al., 2011). However, for square and rectangular shapes, the outward pressure can cause bulging of the walls of the steel shape. The bulging of the walls will be greatest near the bottom of the pour height as shown in Figure 4-4. Temporary braces or lateral reinforcement, such as those shown in Figure 4-5, can be added to minimize bulging.

Analytical investigations of the wet concrete effects have been previously reported by Uy and Das (1997, 1999) through a folded plate finite element approach for thin-walled steel box sections with and without temporary lateral bracing. The variables investigated by these authors in their parametric study included number of simultaneous stories being poured, number of equally distributed braces between floor stories, h/t ratios, and boundary conditions. As expected, their results for the estimated lateral deflection are reduced with closer brace spacing and are increased for higher brace spacing when simultaneously cast with upper stories.

Perea (2010) found that local buckling of filled composite members will develop earlier if significant initial out-of-straightness of the walls due to concrete casting exists.

To avoid the deleterious effects of bulging, it is recommended that stress in the steel section be limited to half the steel yield stress, $0.5F_y$, and the maximum deformation in any wall of the steel section be limited to the column unbraced length divided by 2,000 (i.e., $L/2{,}000$) when the steel section is subject to hydrostatic pressure from wet concrete. The deformation limit is half of the tolerance on out-of-straightness defined in the AISC *Code of Standard Practice* (AISC, 2022a).

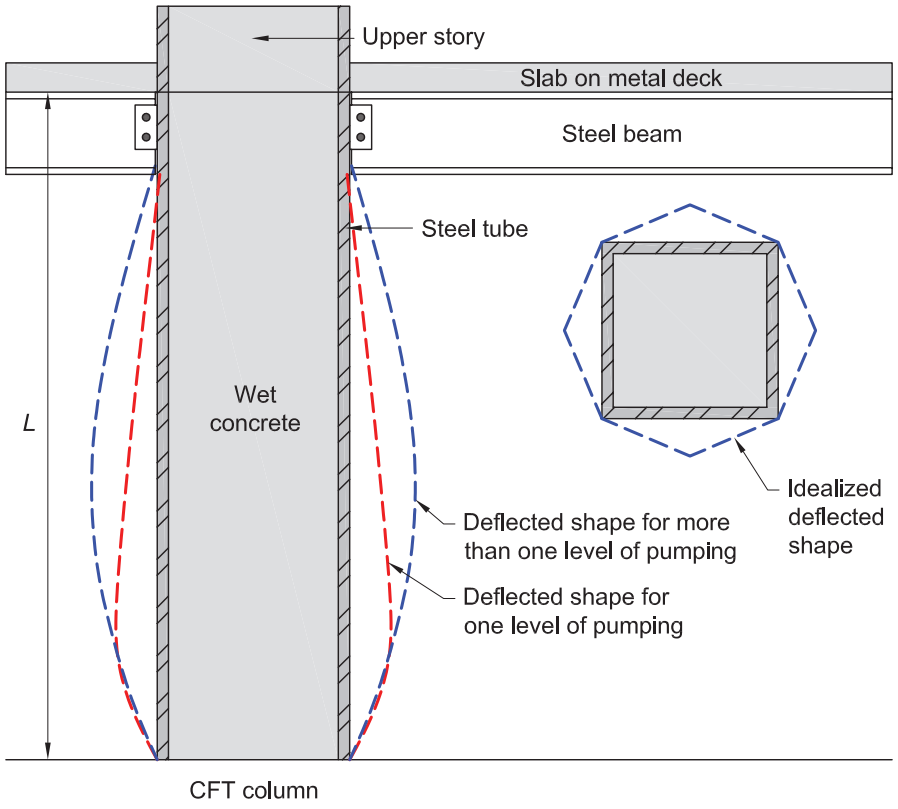
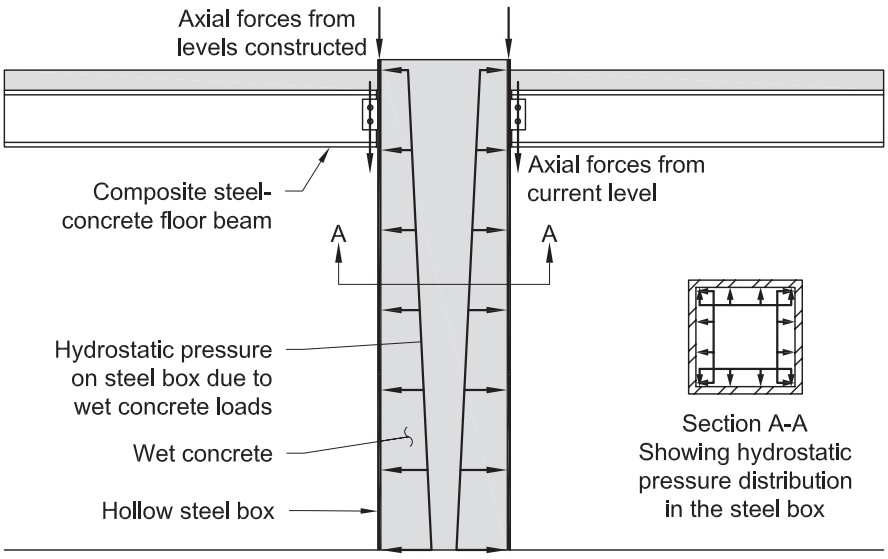
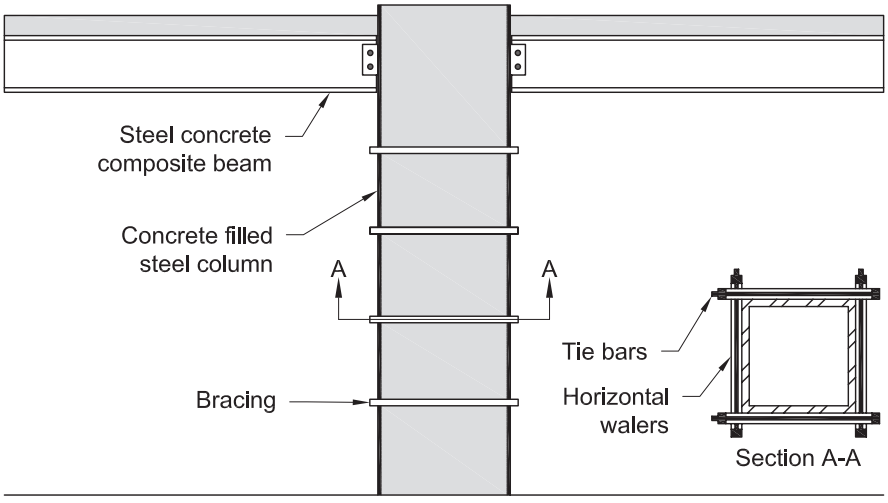


Fig. 4-4. Deflected shape tendency in rectangular filled composite members (adapted from Uy and Das, 1999).

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(a) Hydrostatic pressure from wet concrete



(b) Temporary stiffeners on rectangular filled composite columns to reduce outward expansion

Fig. 4-5. Hydrostatic pressure and temporary stiffeners to reduce outward expansion (adapted from Uy and Das, 1999).

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The stress and deformation can be evaluated with a relatively simple finite element model using shell elements for the steel section. Using shell element models is beneficial because they can consider explicitly the beneficial effect of internal diaphragms or stiffener plates that may be included. For basic cases, the AISC *Specification* Commentary provides equations C-I2-1 and C-I2-2, which were derived based on equations for rectangular pressure vessels (Perea, 2010; Leon et al., 2011).

$$\sigma_{max} = \max\left[\begin{array}{l}\left(\frac{2h_c}{b_c+4h_c}\right)\frac{ph_c^2}{t^2} \\ \frac{1}{3}\left(\frac{3b_c+4h_c}{b_c+4h_c}\right)\frac{ph_c^2}{t^2}\end{array}\right] \le 0.5F_y$$

(*Spec.* Eq. C-I2-1)

$$\delta_{max} = \frac{1}{32}\left(\frac{5b_c+4h_c}{b_c+4h_c}\right)\frac{ph_c^4}{E_s t^3} \le \frac{L}{2{,}000}$$

(*Spec.* Eq. C-I2-2)

where

E_s = modulus of elasticity of steel, ksi
= 29,000 ksi

L = unbraced member length of the steel encasement, in.

b_c = the shorter inner width of the rectangular cross section, in.
$= B - 2t$

h_c = the longer inner width of the rectangular cross section, in.
$= H - 2t$

p = hydrostatic pressure at the base of the wet concrete, ksi

t = wall thickness of HSS or built-up shape, in.

The hydrostatic pressure at the base of the wet concrete is computed as:

$$p = \frac{L_p w_c}{144{,}000} \tag{4-1}$$

where

L_p = height of wet concrete pour, ft

w_c = weight of concrete per unit volume, lb/ft^3

Note that AISC *Specification* Equations C-I2-1 and C-I2-2 were validated by Perea (2010) for filled composite members built with HSS shapes that have inherent continuity at the corners. The same level of continuity does not necessarily exist in the corners of built-up box sections where the plates are connected by welds. The wet weight of concrete places fillet welds at corners of built-up box shapes into a complex state of stress, and continuity at the corner of the box section may not exist.

If the stress or deformation limits are exceeded, the pour heights can be made shorter, the steel shape can be made thicker, or stiffeners can be added. Added temporary stiffeners should be placed before filling the steel shape with concrete and are recommended to stay in place for 7 days after the concrete is placed or until the concrete reaches 80% of the specified compressive strength. The spacing of stiffeners can be determined by analysis. In most cases, using temporary stiffeners with a spacing equal to the maximum transverse dimension of the steel member and only in the bottom third of the pour height will be sufficient.

4.4 ERECTION STABILITY

The steel components of a composite frame are erected before placement and hardening of concrete. In some construction methods, such as that shown in Figure 1-2, several stories of steel framing are erected before concrete is placed. Stability of the bare steel framing is critical for safety during construction and must be assessed. The steel components alone may not be stable nor fully able to carry lateral loads—such as wind—and thus require temporary erection bracing. The engineer of record should state the assumptions of bare steel frame stability in the contract documents, and they should design and detail the necessary temporary bracing on the drawings or require that the erector hire an erection engineer to develop an engineered erection plan that ensures stability during construction. Griffis (1986) presents case studies and describes in more detail an efficient sequence of erection and design responsibility during erection for composite frame structures. The AISC *Code of Standard Practice* and ASCE/SEI 37, *Design Loads on Structures during Construction* (ASCE, 2014), include requirements for erection stability.

4.5 SHORTENING OF COMPOSITE COLUMNS IN TALL STRUCTURES

The effects of long-term creep, shrinkage, and column shortening in composite columns are not specifically addressed in the AISC *Specification*. However, in tall structures, these must be considered during design. The shortening is caused by creep and shrinkage that are affected by differing stress levels, loading histories, ratios of reinforcement, volume-to-surface ratios, and environmental conditions. The concrete is subjected to elastic, creep, and shrinkage shortening, while the steel is subjected to elastic shortening only (Fintel et al., 1987). This phenomenon is most prominent in encased composite columns. Lehman et al. (2015) suggest that drying shrinkage in filled composite members is negligible for most applications. The primary purpose for accounting for this shortening is to minimize the distortion of the floor slabs,

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cladding, finishes, partitions, and other components that could lead to impaired serviceability. Fintel et al. developed an analytical procedure to predict the anticipated elastic and inelastic shortening that will occur in the columns so that it can be compensated for during construction.

The steel components of composite columns are typically detailed to lengths consistent with the finished floor elevations that do not inherently consider such shortening. To compensate for initial elastic shortening, shims can be installed between two column lifts with a thickness corresponding to the expected shortening. Alternatively, to eliminate the need to install shims, the column lengths can be detailed to slightly longer lengths to account for the shortening (Fintel et al., 1987).

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Chapter 5
Future Perspective

Composite construction has been a pivotal construction approach for the last several decades, harnessing the power of combining steel and concrete to best effect. Numerous innovations have been put forward over those years, with new approaches for analysis and design enabling an expanding range of composite structural systems to be used in both nonseismic and seismic zones. Filled and encased composite members, composite beams, composite wall systems, composite foundation systems, and other related composite systems are a testament to the research and design innovation that has taken place over the years.

Looking forward, the architecture, engineering, and construction industry is on the cusp of a major shift in focus toward developing structural systems that are more sustainable and resilient. Composite construction can and should be at the core of these innovations, and a future edition of this Design Guide can provide critical information for this transformation. Specifically, innovative systems that ensure the use of sustainable materials are needed. Opportunities are ripe for incorporating sustainable materials into composite construction—concrete and steel materials coupled with fabrication and construction practices that dramatically decrease the amount of greenhouse gas emissions, material flow, pollution, and waste associated with the structure. Monolithic concrete pours could be replaced by efficient, shape-optimized precast components connected in ways that facilitate design for deconstruction. Incorporation of replaceable energy-dissipating components in composite systems can limit irreparable damage and better position composite structures to bounce back quickly from extreme events. Composite construction can embrace these new materials and systems.

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Appendix A
Composite Column Program

The first edition of this Design Guide included numerous tables of the available strength of encased composite members. With an expanded scope, including filled composite members and newer provisions, the tables are replaced in this edition of the Design Guide with a companion composite column program. The program is implemented in a Microsoft Excel macro-enabled workbook and is available on the AISC website at **www.aisc.org/dg**. The workbook contains worksheets for round filled composite members, square and rectangular filled composite members, square and rectangular filled composite members constructed with high-strength materials, and encased composite members. The workbook computes axial compressive strength, axial tensile strength, flexural strength, shear strength, as well as interaction diagrams for combined axial compression and flexure. While not intended to be a full design solution, this spreadsheet allows for easy calculation of available strengths, several of which cannot be efficiently computed by hand. In particular, the flexural strength of many composite sections is difficult to compute by hand. Flexural strength is computed in the workbook with macros that discretize the composite cross section into small fibers and determine the neutral axis and flexural strength numerically. Refer to the notes in the "Information" worksheet for assumptions used by the program.

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Symbols

A	Area of cross-sectional element, in.^{2}
A_c	Cross-sectional area of concrete, in.^{2}
A_d	Cross-sectional area of dowel bars, in.^{2}
$A_{d,min}$	Minimum cross-sectional area of dowel bars, in.^{2}
$A_{d,req}$	Required cross-sectional area of dowel bars, in.^{2}
A_g	Gross cross-sectional area, in.^{2}
A_{gc}	Gross cross-sectional area of concrete, in.^{2}
A_s	Cross-sectional area of steel, in.^{2}
A_{sr}	Cross-sectional area of longitudinal reinforcing bars, in.^{2}
A_v	Shear area of the steel portion of the composite cross section, in.^{2}
A_1	Loaded area of concrete, in.^{2}
A_2	Maximum area of the portion of the supporting surface that is geometrically similar to and concentric with the loaded area, in.^{2}
B	Flange width of rectangular cross section, in.
B	Base plate width, in.
B_1, B_2	Second-order effect multipliers
C_d	Deflection amplification factor
C_m	Equivalent uniform moment factor
C_{v1}	Web shear strength coefficient
C_1, C_2, C_3	Reduction factors
D	Dead load, kips
D	Outside diameter of HSS, in.
E_c	Modulus of elasticity of concrete = $w_c^{1.5}\sqrt{f'_c}$, ksi
E_s	Modulus of elasticity of steel = 29,000 ksi
EA	Axial stiffness, kips
EI	Flexural stiffness, kip-in.^{2}
EI*	Flexural stiffness required to be used in the analysis, kip-in.^{2}
$(EI)_{eff}$	Effective stiffness of the composite section, kip-in.^{2}
F_{in}	Nominal bond stress, ksi
F_n	Nominal stress, ksi
F_y	Specified minimum yield stress, ksi
$F_{y,max}$	Maximum permitted yield stress of steel, ksi
F_{ysr}	Specified minimum yield stress of reinforcing steel, ksi
G_c	Shear modulus of elasticity of concrete, ksi
G_s	Shear modulus of elasticity of steel = 11,200 ksi
GA	Shear stiffness, kips
GJ	Torsional stiffness, kip-in.^{2}
H	Horizontal point load, kips
H	Web depth of rectangular cross section, in.
I	Moment of inertia of portion of cross section, in.^{4}
I_c	Moment of inertia of concrete, in.^{4}
I_s	Moment of inertia of steel, in.^{4}
I_{sr}	Moment of inertia of longitudinal reinforcing steel, in.^{4}
J_c	Torsional constant of concrete, in.^{4}
J_s	Torsional constant of steel, in.^{4}
K	Effective length factor
K_c	Coefficient based on width-to-thickness ratio of the shear element
L	Live load, kips
L	Unbraced length, in.
L_c	Effective length of member, in.
L_{c1}	Effective length of member in the plane of bending, in.
L_{in}	Load introduction length, in.
L_p	Height of wet concrete pour, ft
L_r	Roof live load, kips
M	Moment in cross-sectional element, kip-in.
M_B	Nominal flexural strength at point B on the interaction diagram, kip-in.
M_D	Nominal flexural strength at point D on the interaction diagram, kip-in.
M_c	Available flexural strength, kip-in.

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M_{lt} First-order moment due to lateral translation, kip-in.

M_n Nominal flexural strength, kip-in.

M_{nt} First-order moment with the structure restrained against lateral translation, kip-in.

M_p Plastic moment, kip-in.

M_r Required flexural strength, kip-in.

M_{sd} Moment corresponding to the stress distribution in a rectangular filled composite member, kip-in.

M_1, M_2 Moments at the top and bottom of the column, kip-in.

N Base plate depth, in.

N_i Notional load applied at level i, kips

P Vertical point load, kips

P Axial force in cross-sectional element, kips

P_C Nominal axial compressive strength at point C without member length effects, kips

P_c Available axial compressive strength, kips

P_{cc} Available axial compressive strength at point C, kips

P_e Elastic critical buckling load, kips

$P_{e\ story}$ Elastic critical buckling strength of the story in the direction of translation being considered, kips

P_{e1} Elastic critical buckling strength of member in the plane of bending, kips

P_{lt} First-order axial compressive force due to lateral translation, kip-in.

P_{mf} Total vertical load in columns in the story that are part of moment frames, kips

P_n Nominal axial compressive strength, kips

P_{no} Nominal axial compressive strength without consideration of length effects, kips

P_{nt} First-order axial compressive force with the structure restrained against lateral translation, kips

P_p Plastic axial compressive strength, kips

P_r Required axial compressive strength, kips

P_{rc} Portion of external force directly applied to concrete, kips

P_{rs} Portion of external force directly applied to steel, kips

P_{story} Total vertical load supported by the story, kips

P_u Required axial compressive strength, kips

P_y Yield axial compressive strength, kips

Q_{cv} Available shear strength of a steel headed stud anchor, placed within the load introduction length, kips

R Response modification coefficient

R_a Required strength determined using ASD load combinations, kips

R_c Available force transfer strength, kips

R_n Nominal strength, kips

R_u Required strength determined using LRFD load combinations, kips

S Scheduled tie spacing, in.

V_n Nominal shear strength, kips

V_r Required shear strength, kips

V_r' Required transfer force, kips

Y_i Gravity load applied at level i, kips

Z_s Plastic section modulus of steel, in.^{3}

a_p Location of the neutral axis of high-strength rectangular filled composite member, in.

b Clear distance between webs less the inside corner radii, in.

b_c Shorter inner width of the rectangular cross section, in.

b_f Width of flange, in.

b_i Internal flange width of rectangular cross section, in.

c_m, c_p, c_{sr} Interaction coefficients for noncompact composite and slender composite filled composite members

d Depth of member, in.

d_{agg} Diameter of aggregate, in.

d_b Reinforcing bar diameter, in.

f_c' Specified compressive strength of concrete, ksi

h Clear distance between flanges less inside corner radii, in.

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h_c Longer inner width of the rectangular cross section, in.

h_i Internal web depth of rectangular cross section, in.

l Critical base plate cantilever dimension, in.

l_{dc} Development length for deformed bars in compression, in.

l_{sc} Lap splice length for deformed bars in compression, in.

m, n, n' Base plate cantilever dimensions, in.

p Hydrostatic pressure at the base of the wet concrete, ksi

p_b Perimeter of the steel-concrete bond interface within the composite cross section, in.

t Thickness of cross-sectional element, in.

t_f Flange thickness, in.

t_{min} Minimum base plate thickness, in.

t_w Web thickness, in.

w_c Density of concrete, lb/ft^3

x Subscript relating symbol to major-axis bending

y Subscript relating symbol to minor-axis bending

y Distance from centroid of cross section to centroid of cross-sectional element, in.

$\overline{y}$ Distance to the plastic neutral axis from the centerline of the gross section, in.

Δ_H First-order interstory drift, in.

Ω Safety factor

Ω_0 Overstrength factor

α Biaxial bending shape factor

α LRFD/ASD adjustment factor

β_t Coefficient for the contribution of concrete to torsional stiffness

χ Stability reduction factor

δ_{max} Maximum displacement in the steel section due to hydrostatic pressure of wet concrete, in.

ε Strain of cross-sectional element, in./in.

ε_c Strain at the extreme concrete compression fiber, in./in.

ε_t Strain at the centroid of the extreme tensile reinforcement bar, in./in.

ε_{ty} Strain of reinforcing steel at yield stress, in./in.

ϕ Resistance factor

λ Cross-sectional element width-to-thickness ratio

λ_p Limiting compact composite/noncompact composite width-to-thickness ratio

λ_r Limiting noncompact composite/slender composite width-to-thickness ratio

ρ_{sr} Continuous longitudinal reinforcement ratio

σ Stress of cross-sectional element, ksi

σ_{max} Maximum stress in the steel section due to hydrostatic pressure of wet concrete, ksi

τ_b Stiffness reduction parameter

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